PHINMA UNIVERSITY OF ILOILO
COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

DIGITRACK: AN INSTRUCTOR SCHEDULING AND PERFORMANCE MONITORING SYSTEM FOR DRIVING SCHOOL

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Abrot, Carl Edciel R.

Geroy, Blessed Rose P.

Madrid, Roseanne P.

Valdez, Jerry Mae R.

Villar, Crisamae M.

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CHAPTER I

INTRODUCTION

Driving schools play an important role in preparing individuals for the responsibilities associated with operating motor vehicles. Driver education generally involves the development of theoretical knowledge, practical driving skills, awareness of traffic regulations, and appropriate attitudes toward road safety. In the Philippines, the regulation of driving institutions and driver education is governed by requirements established by the Land Transportation Office (LTO). Memorandum Circular No. 2019-2176 provides rules and regulations concerning the accreditation, supervision, and control of driving institutions and the standardization of driver education. The circular emphasizes compliance with prescribed driver education standards, instructor qualifications, facilities, equipment, course components, and minimum training requirements [1]. These requirements demonstrate the importance of maintaining organized records and processes within institutions that provide driver education.

As driving schools manage multiple students, instructors, training schedules, and records, administrative activities become an important part of daily operations. Scheduling a training session requires consideration of the student's availability, the instructor's availability, the type of training, the duration of the session, and other existing appointments. When these activities are handled through separate records or manual coordination, maintaining an updated and consistent schedule can become more difficult. Previous research on timetabling and resource utilization has also identified the importance of systematic scheduling approaches in managing limited resources efficiently [2].

Information management is another important consideration in the operation of educational and training institutions. Digital information systems can provide centralized access to records and support administrative processes by organizing information that would otherwise be distributed across separate documents or storage methods. Grepon et al. demonstrated how an information system approach can support school management by integrating administrative information and improving access to organizational records [3]. Similar principles can be applied to driving school operations where information concerning students, instructors, schedules, training sessions, and feedback must be maintained and retrieved when needed.

Motosafe Driving School, located in Miagao, Iloilo, provides driver education and training services that require coordination between students, instructors, and school administrators. Based on the operational context examined in this study, several activities related to scheduling, instructor management, training monitoring, and feedback are handled through manual or distributed processes. These processes may include paper-based records, spreadsheets, verbal coordination, and other forms of communication. While such methods can be used for daily operations, they may become difficult to manage when the number of appointments and records increases.

The proposed system, DigiTrack: An Instructor Scheduling and Performance Monitoring System for Motosafe Driving School, is designed to address these operational concerns through a centralized web-based platform. The system focuses on instructor scheduling, instructor availability, workload monitoring, student training session records, student feedback, performance monitoring, and administrative reporting. Rather than replacing administrative decision-making, DigiTrack is intended to provide organized information and system supported processes that can assist administrators, instructors, and students in carrying out their respective responsibilities.

The development of DigiTrack is therefore focused on a specific organizational need: providing Motosafe Driving School with a centralized system for managing information related to scheduling and instructor performance. The study does not claim that the system will directly reduce road accidents, guarantee driver competency after licensing, or replace government regulatory systems. Instead, it concentrates on the development, implementation, testing, and evaluation of a system intended to improve the organization and management of the school's internal operations.

1.1 BACKGROUND OF THE STUDY

Driving education is an important component of preparing individuals to operate vehicles responsibly. It combines theoretical instruction with practical training so that learners can acquire knowledge of traffic rules, road safety principles, vehicle operation, and appropriate driving practices. The quality of driver education depends not only on the content of instruction but also on the organization of the training process. Driving

institutions must coordinate instructors, students, schedules, training sessions, records, and other administrative requirements while complying with applicable regulations.

The importance of organized driver education can be understood within the broader context of road-safety policy and international transportation research. The International Transport Forum's Road Safety Annual Report 2023 reviews road-safety indicators and policy developments across countries [4]. This report provides broader road-safety context, but it should not be treated as direct evidence that a particular driving-school management feature improves road safety.

In the Philippine setting, driving institutions operate within regulatory requirements established by the Land Transportation Office. LTO Memorandum Circular No. 2019-2176 establishes rules concerning the accreditation, supervision, and control of driving institutions and standardization of driver education [1]. The memorandum identifies requirements related to driving institutions, instructors, facilities, equipment, course components, and driver education standards. Compliance with such requirements requires institutions to maintain appropriate records and organize training activities in a manner that allows information to be monitored and retrieved when necessary.

The operational management of a driving school involves several interconnected activities. A student may need to be registered, assigned to a course, scheduled for training, assigned to an instructor, and monitored throughout the completion of required sessions. Instructors, meanwhile, must manage their assigned schedules and conduct training sessions according to the school's procedures. Administrators must coordinate these activities, maintain records, respond to scheduling changes, review feedback, and monitor the overall operation of the institution.

These activities are interdependent rather than separate administrative tasks. A change in an instructor's availability may affect a student's appointment, the sequence of a training course, and the school's ability to use its available training hours. Similarly, a cancellation or rescheduling request can require staff to check several records before confirming a replacement schedule. When the information needed for these decisions is spread across paper forms, spreadsheets, messages, or separate files, staff may need to reconcile the records manually. The issue is not simply the number of records maintained; it is the effort required to keep related records consistent as circumstances change.

The operational challenge becomes more noticeable when several instructors and students require training during the same period. Each instructor has a finite working schedule, and each student may have availability constraints related to school, employment, transportation, or other responsibilities. Training sessions may also differ in duration or course requirements. These conditions mean that a usable schedule must consider more than an open time slot. It must also reflect whether the instructor and student are available, whether either person already has another appointment, and whether the assignment is consistent with the school's operating procedures. Automated conflict checking can assist with these checks, while the final assignment remains subject to authorized administrative decisions.

A centralized information system can provide a common location for maintaining instructor, student, course, appointment, training, attendance, and feedback records. This does not automatically guarantee that every record will be accurate. Accuracy still depends on correct data entry, appropriate permissions, timely updates, and clear responsibilities among users. However, a centralized system can reduce the need to reconcile separate copies of the same information and can make it easier for authorized users to review related records together. This role is consistent with the broader use of information systems to organize operational data and support management decisions [3].

For DigiTrack, the practical objective is to support routine administrative work rather than replace the judgment of the driving-school administrator. The system can present schedules, identify conflicting assignments, organize training progress, and make performance-related information easier to review. It should not be described as guaranteeing perfectly balanced workloads or automatically determining the best possible schedule. Workload comparisons are useful indicators for administrators, but differences in instructor assignments may reflect legitimate factors such as course type, availability, training duration, or specific student needs. The system therefore provides organized information that supports decisions instead of making unsupported claims about optimal outcomes.

The local regulatory context also shapes the information that a driving school needs to maintain. LTO Memorandum Circular No. 2019-2176 establishes rules concerning the accreditation, supervision, and control of driving institutions and the standardization of driver education [1]. DigiTrack is not intended to replace these regulations or serve as an official government compliance platform. Its purpose is narrower: to help the client organize operational records and schedules that are relevant to its internal management. Any regulatory interpretation, accreditation decision, or official submission remains the responsibility of the driving school and the appropriate authority.

The study is also relevant because digitalization should be evaluated in relation to actual users and tasks. Administrators need to review schedules and records efficiently; instructors need access to information relevant to their assigned training; and students need clear information about their schedules and training progress. A system that centralizes data but presents it in a confusing way may fail to provide practical value. Likewise, an interface that is easy to use but does not prevent unauthorized access would be insufficient for a system containing personal information. DigiTrack therefore treats operational usefulness, usability, and appropriate access control as connected design concerns.

Scheduling is particularly important because instructor availability and student availability must be considered at the same time. A schedule must also account for existing appointments and the type of training being conducted. Research on timetabling has shown that scheduling approaches can affect the utilization of available resources and the organization of activities [2]. Although the context of previous timetabling research may differ from driving schools, the general scheduling principle is relevant to DigiTrack because instructors represent a limited resource whose availability must be coordinated with student demand.

In addition to scheduling, information management plays an important role in school administration. Grepon et al. [3] demonstrated the use of an information system approach to support school management by organizing administrative information and providing a more structured method for managing records. This illustrates the potential value of centralized information systems in educational environments. For a driving school, a similar approach can be used to organize instructor information, student information, appointments, training sessions, and feedback.

Motosafe Driving School is the specific organizational setting of this study. The school offers driver education services involving theoretical and practical training. The existing operational processes examined by the researchers indicate that scheduling, instructor coordination, and monitoring activities are handled through manual or distributed methods. Information may be recorded in paper-based documents, spreadsheets, or through verbal communication between staff members. When information is distributed among several sources, employees may need to check multiple records before confirming an appointment or retrieving a particular piece of information.

Manual scheduling does not necessarily mean that the process is ineffective in every situation. For a small number of appointments, staff members may be able to coordinate schedules using calendars, spreadsheets, or direct communication. However, the process becomes more demanding as the number of students, instructors, and training sessions increases. Each new appointment requires the administrator to consider existing schedules, instructor availability, student availability, and other operational constraints.

One concern identified in the current process is the possibility of scheduling conflicts. A scheduling conflict occurs when two or more activities are assigned to the same instructor, student, vehicle, or time period in a manner that prevents the activities from being conducted as planned. When schedules are maintained manually, administrators must crosscheck existing appointments before creating or changing a booking. Errors may occur when an existing appointment is overlooked or when a schedule is not updated consistently across different records.

Another concern is instructor workload monitoring. Instructor workload refers to the amount of training activity assigned to an instructor during a particular period. Without organized records, it can be difficult for administrators to compare assignments among instructors or determine whether an instructor has a relatively high or low number of scheduled sessions. Research examining workload and teacher performance has found that workload can be relevant to professional factors associated with performance evaluation [5]. Although the cited research concerns teachers rather than driving instructors, the finding provides a relevant basis for considering workload as an administrative factor that may be monitored through an information system.

For DigiTrack, workload monitoring does not mean that the system independently determines the ideal workload of every instructor. Instead, the system is intended to provide administrators with organized information about instructor schedules and assignments so that they can make informed decisions. This distinction is important because final assignment decisions may depend on factors that cannot be fully represented by an automated calculation, including instructor qualifications, course requirements, vehicle availability, and school policies.

Performance monitoring is another area addressed by the study. Instructor performance can involve several indicators, including student feedback, ratings, training session records, and other information used by the school for internal evaluation. Without a centralized feedback mechanism, information about student experiences may be difficult to consolidate and review over time. A digital feedback feature can provide a structured means for collecting student ratings and comments and storing them with the relevant training records.

Existing research and systems demonstrate that technology can be applied to driving school management. Kannan et al. presented an advanced management system for car driving schools that addresses management related processes through a digital system [6]. Davanhipprage and Jagtap presented a web-based driving-school management project described as AI-driven, with features related to booking, scheduling, student management, and administration [7]. The paper should be treated as a system description rather than independent proof that advanced AI methods were implemented or validated. Mithrasena's TrainMe project likewise explored a web-based management system for a driving school, demonstrating that web-based information systems can be applied to the management needs of driving training institutions [8].

Although these studies and systems provide useful references, DigiTrack is developed for the specific operational context of Motosafe Driving School. The proposed system focuses on instructor scheduling, workload monitoring, student feedback, performance monitoring, training session records, and centralized administrative information. Its purpose is not to reproduce every feature of existing driving school systems but to address the operational requirements identified during the analysis of Motosafe's current processes.

A centralized database is particularly relevant to these requirements. Instead of keeping student information, instructor information, schedules, and feedback in separate locations, the proposed system stores related information in a structured database. This allows authorized users to retrieve information according to their role. For example, administrators may need access to overall scheduling and performance information, instructors may need access to their assigned schedules, and students may need access to their own training information and feedback functions.

The proposed web-based approach also provides a common interface through which authorized users can access the system. A web-based application can be accessed using a compatible browser without requiring a separate application installation on every client device. This approach is appropriate for a system intended for multiple user roles because the same centralized application can provide different functions depending on the user's role and permissions.

However, accessibility through a web browser should not be interpreted as a guarantee that the system is available under all conditions. System availability depends on factors such as server availability, network connectivity, device compatibility, and maintenance. Therefore, the study describes DigiTrack as a web-based system accessible to authorized users through supported devices rather than claiming unlimited or guaranteed 24hour availability.

DigiTrack is intended to support both administrative and operational activities. Administrators can use the system to manage instructors, students, schedules, training records, feedback, and reports. Instructors can use the system to view relevant schedules and update information assigned to them. Students can use the system to view trainingrelated information and submit feedback where applicable. Rolebased access control is used to ensure that users interact only with the functions and information permitted for their role.

The system also provides a basis for monitoring training activities. Scheduled sessions can be recorded and their completion status can be updated by authorized users. This creates a structured record of training activities that can be used by administrators when reviewing student progress. The system does not automatically determine whether a physical driving session has occurred; session completion remains dependent on authorized users entering the appropriate information.

Student feedback is another component of the proposed system. Feedback collected through DigiTrack is intended primarily for internal quality monitoring. Students can provide ratings and comments regarding their training experience, while administrators can review the collected information to identify patterns and areas that may require attention. The feedback mechanism does not constitute an official government evaluation or certification.

The use of a centralized system can also support administrative reporting. Rather than manually combining information from several records, administrators can retrieve information stored in the database and view summaries through the administrative dashboard. Such reports can assist management in reviewing schedules, instructor assignments, session completion, and feedback information.

The development of DigiTrack therefore responds to an operational requirement identified at Motosafe Driving School. The system is designed not merely as a general school management application but as a localized solution for driving school scheduling and instructor performance monitoring. Its development takes into consideration the school's users, processes, and information requirements.

The study also recognizes the boundaries of the proposed system. DigiTrack does not directly connect to the LTO or other government databases. It does not replace government accreditation or licensing procedures. It does not use GPS or vehicle sensors to automatically monitor physical driving activities. It also does not include payment processing as part of the current system scope.

Consequently, the intended contribution of DigiTrack is primarily organizational and technological. It seeks to provide Motosafe Driving School with a centralized platform that can organize scheduling information, instructor assignments, student training records, and feedback. The system is evaluated based on its functionality, usability, accuracy, and suitability for the intended users rather than on long-term road safety outcomes.

Through the development of DigiTrack, the researchers aim to demonstrate how a web-based information system can be applied to a localized operational problem in a driving school environment. The project provides an opportunity to apply principles of database management, web development, scheduling, role-based access, user interface design, and system evaluation to a real organizational context.

The requirements of a small driving school should be translated into system behavior that can be observed and verified. In DigiTrack, this means that the administrator should be able to maintain instructor and student records, create and review appointments, identify conflicts covered by the system's rules, and review relevant training and feedback information. Each function should have a clear purpose and should connect to the information used by another function where appropriate. For example, an appointment should be associated with the correct student and instructor, and a training record should be traceable to the activity it represents. This relationship between records is one of the reasons a structured database and consistent workflows are important.

The development problem also includes deciding which processes should remain under human control. Scheduling rules can identify invalid overlaps, but some decisions require contextual judgment. An instructor may be unavailable for reasons that are not yet recorded, a student may request a change, or a course may require a particular assignment. The system should support the administrator by making relevant information visible and applying defined checks, while allowing authorized staff to manage legitimate exceptions. This approach is more suitable for the stated project than claiming that the application autonomously optimizes every assignment. It also gives the evaluation a concrete focus: whether the system applies its stated rules correctly and helps users complete the intended tasks.

1.2 STATEMENT OF THE PROBLEM

This study aims to develop DigiTrack, a webbased scheduling and performance monitoring system for Motosafe Driving School, in order to facilitate its scheduling, instructor management, and operation processes. In particular, the development of such a solution would help to achieve greater efficiency, avoid human mistakes, and make better decisions using ystemsupported scheduling and workload monitoring. The development of this system represents a strategic response to the operational challenges that have become increasingly apparent as the school has grown and as student expectations for service quality have risen.

DigiTrack will enable the school to manage key processes such as automated instructor scheduling, equitable workload distribution, systematic student feedback collection, and real time performance tracking. By

integrating these functions into a centralized platform, the system intends to increase operational efficiency and provide Motosafe with a more professional and effective tool for delivering driver education services. The integration of these functions eliminates the need for multiple disconnected systems and manual workarounds, creating a streamlined workflow that reduces redundancy and improves data consistency. All stakeholders, administrators, instructors, and students will benefit from improved access to information and more efficient processes.

1.2.1 General Problem Statement

Motosafe Driving School faces significant operational inefficiencies due to its reliance on manual scheduling practices and the lack of a centralized instructor monitoring system. These challenges result in uneven instructor workloads, frequent scheduling conflicts, difficulty maintaining consistent training quality, and limited ability to track student progress and instructor performance in real time. The cumulative effect of these problems is reduced operational efficiency, increased administrative costs, lower student satisfaction, and constrained capacity for growth. Without intervention, these problems will likely worsen as student enrollment continues to increase and competitive pressures intensify.

The operational inefficiencies at Motosafe extend beyond simple inconvenience to affect the school's core mission and longterm viability. Scheduling conflicts and recordkeeping errors undermine student trust in the school's professionalism and reliability. Workload imbalance among instructors affects teaching quality and contributes to staff turnover, making it difficult to maintain consistent training standards. The inability to monitor performance systematically prevents the school from identifying and addressing quality issues promptly. These problems create a cycle of declining service quality and student satisfaction that threatens the school's reputation and competitive position in the market.

1.2.2 Specific Problems

1. Manual Scheduling Conflicts

At Motosafe Driving School, most procedures for booking training sessions and assigning instructors are handled through manual systems. Appointments are recorded on paper logs, in basic spreadsheets, or by verbal agreement. Although manual systems can function adequately for smallscale operations, they frequently cause problems such as overlapping appointments, double bookings, missed sessions, and human error in instructor assignment. These issues become particularly severe when the school handles multiple student requests simultaneously, as administrators may inadvertently assign the same instructor to two different students or schedule a student with an instructor who is already unavailable.

The manual scheduling process at Motosafe involves multiple steps that are all vulnerable to human error. When a student requests a training session, administrative staff must check instructor availability through paper calendars or Excel spreadsheets that may not be uptodate. Staff must then verify vehicle availability, as certain instructors only have access to specific vehicle types. Next, staff must ensure that the requested time slot doesn't conflict with existing commitments for the student themselves, as students often have multiple sessions scheduled. Each of these verification steps relies on manual crossreferencing and memory, creating multiple opportunities for error.

Finding the correct record of scheduled sessions can also become challenging for employees when managing several training events at once. Manual systems similarly cause problems such as a lack of accuracy in tracking completed sessions, missed followups, and inconsistent recordkeeping, which may lower the overall level of service quality offered to students. Paper logs can be lost, damaged, or misplaced, resulting in permanent loss of important scheduling information. Excel spreadsheets may become corrupted or contain formula errors that go undetected. Verbal agreements made over the phone may not be recorded accurately or may be forgotten entirely. These inconsistencies create gaps in the scheduling record that can lead to conflicts and confusion.

In this respect, it is advisable to develop an automated, computerbased scheduling system that would enhance performance, eliminate scheduling conflicts, and improve operational efficiency within Motosafe Driving School. An automated system would provide a single source of truth for all scheduling information, with realtime updates and automatic conflict detection. The system would eliminate the need for manual crossreferencing and reduce the cognitive load on administrative staff, allowing them to focus on exception handling and customer service rather than routine data entry and verification.

Consequences of Manual Scheduling Conflicts

The downstream effects of scheduling conflicts extend beyond administrative inconvenience. Student dissatisfaction increases when sessions are missed or rescheduled at the last minute, leading to negative wordofmouth and reduced enrollment. Each missed session represents lost revenue for the school, as well as wasted time for both instructors and students. Additionally, the time spent resolving conflicts reduces the time administrators can dedicate to strategic planning and quality improvement initiatives. Operational inefficiencies in manual scheduling can lead to unutilized time slots and reduced capacity, directly affecting a driving school's overall operational efficiency and revenue potential.

Student frustration with scheduling conflicts often leads to complaints that damage the school's reputation. In the age of social media and online reviews, negative experiences can spread quickly and affect enrollment from prospective students. Parents and guardians who are paying for training services expect reliability and professionalism, and scheduling errors undermine confidence in the school's ability to deliver on its promises. Over time, repeated scheduling problems can lead to student attrition, with students transferring to competing schools that offer more reliable service. This attrition not only represents lost revenue but also increases marketing costs needed to replace departing students.

Scheduling conflicts can create additional administrative work. Staff may need to contact students and instructors, identify alternative time slots, and update records when appointments overlap or change. The sources reviewed for this manuscript do not establish a verified weekly time estimate specific to Motosafe Driving School; therefore, this study does not assign a numerical estimate to the time spent resolving conflicts. This time could instead be used for other administrative activities, such as improving training materials, developing programs, or enhancing customer service.

2. Workload Imbalance Among Instructors

There is currently no systematic or objective way to ensure that instructors at Motosafe Driving School are assigned training tasks fairly and equitably. Assignments are often made based on convenience, habit, or informal communication rather than on realtime availability, current workload, or utilization rates. As a result, some instructors become overworked with excessive training sessions, leading to fatigue, reduced teaching quality, and potential burnout, while other instructors remain underutilized with very few or no assignments during certain periods.

This workload imbalance not only affects instructor morale and job satisfaction but also creates inefficiencies in resource utilization. Overworked instructors may rush through lessons, compromise on training quality, or become unavailable for additional sessions, while underutilized instructors represent wasted human resources that could otherwise contribute to meeting student demand. The school pays for the same number of instructor hours regardless of utilization rate, meaning that underutilization represents a direct financial loss. At the same time, overworked instructors may make more mistakes, provide lowerquality instruction, and burn out more quickly, all of which have financial and operational consequences.

Without a centralized scheduling and workload-monitoring mechanism, administrators may find it difficult to compare instructor assignments, identify potential schedule conflicts, and review the distribution of training sessions. These limitations call for a system that presents instructor availability, existing appointments, and workload-related information in one place. DigiTrack is intended to support administrative decisions by organizing this information and checking for scheduling conflicts; it does not independently guarantee equitable assignments or eliminate the need for administrator judgment.

Impact of Workload Imbalance on Instructors

Workload imbalance directly affects instructor morale, job satisfaction, and retention rates. Overworked instructors may experience burnout, leading to decreased teaching quality, increased absenteeism, and eventual resignation. Underutilized instructors represent wasted human resources and may feel undervalued, also leading to turnover. High instructor turnover increases recruitment and training costs for the school and disrupts continuity for students. The cost of replacing an instructor can be substantial, including recruitment advertising, screening and interviewing time, background checks, training costs, and the reduced productivity of new instructors during their learning curve.

Burnout among overworked instructors manifests in several ways that affect both the instructors themselves and the school's operations. Instructors experiencing burnout may show signs of fatigue, irritability, and decreased motivation. Their teaching quality may decline as they rush through lessons or become less engaged with students. They may be more likely to call in sick or take unexpected time off, disrupting scheduled sessions. Over time, burnout can lead to resignation, forcing the school to find and train replacement instructors.

The cumulative effect of instructor burnout is a cycle of declining quality and increasing turnover that is difficult to break.

Underutilized instructors face different but equally serious challenges. Instructors who receive few assignments may feel undervalued and irrelevant, questioning their role in the organization. They may become disengaged and less motivated to maintain their skills and knowledge. Financial stress may result if their income is commissionbased or tied to the number of sessions conducted. These instructors are more likely to seek employment elsewhere, taking their experience and skills with them. The school loses the investment made in training and developing these instructors, and students lose the benefit of their expertise.

3. Lack of Centralized Performance Monitoring

Motosafe Driving School currently lacks a structured, centralized mechanism to track and monitor instructor performance and training quality in real time. Student feedback is either not collected systematically, stored in disparate locations (such as paper forms, personal notebooks, or separate chat messages), or not analyzed for actionable insights. As a result, administrators have difficulty identifying which instructors are performing well, which ones need additional training or coaching, and where systemic gaps exist in the training program. This lack of visibility into performance makes it impossible to manage quality proactively or to reward high performers appropriately.

The current approach to collecting student feedback at Motosafe is inconsistent and fragmented. Some instructors may collect feedback through paper forms that students complete at the end of their training, but these forms may not be returned or may be lost. Other instructors may solicit feedback through informal conversations or messaging apps, but this feedback is not recorded systematically. Some feedback may be shared verbally with administrators during casual conversations, but it is not documented or tracked over time. This fragmented approach means that valuable information about instructor performance is lost or never captured in the first place.

The absence of a reliable feedback and performance monitoring system makes it challenging to assess service quality consistently, detect underperforming instructors early, and implement targeted improvements. Without systematic data, administrators must rely on anecdotal evidence and occasional complaints to identify performance issues. By the time a problem becomes evident through student complaints or attrition, it may have been ongoing for months, affecting many students and damaging the school's reputation. Early intervention is impossible without timely performance data, allowing problems to escalate before they are addressed.

Furthermore, without historical performance data, administrators cannot identify trends such as declining satisfaction scores, repeated complaints about specific instructors, or patterns in student completion rates. This calls for an efficient, centralized feedback system that allows students to rate and comment on their training experiences immediately after sessions, stores this data securely, and provides administrators with realtime performance metrics and visual dashboards for informed decisionmaking. A systematic feedback system would enable the school to track performance over time, identify patterns and trends, and make datadriven decisions about instructor development and program improvements.

Impact of Lack of Performance Monitoring on Training Quality

Without systematic performance monitoring, the school cannot ensure consistent training quality across all instructors. Underperforming instructors may continue teaching without improvement, directly impacting student learning outcomes and safety. The school also misses opportunities to recognize and reward highperforming instructors, which affects motivation and staff retention. Over time, inconsistent training quality can damage the school's reputation and reduce its competitiveness in the market. In the driving education industry, where student safety is paramount, inconsistent training quality poses both ethical and legal risks.

Student safety is the most critical concern affected by inconsistent training quality. Instructors who provide substandard instruction may fail to teach essential safety skills or may model poor driving behaviors that students then adopt. This can result in new drivers who are illprepared for realworld driving conditions, increasing their risk of accidents and endangering other road users. The school bears responsibility for the quality of instruction it provides, and inconsistent monitoring makes it impossible to ensure that all instructors meet minimum quality standards. This liability risk is particularly significant in an industry where poor training can have lifeordeath consequences.

The school also loses competitive advantage when it cannot demonstrate consistent training quality. In a market where students have multiple driving school options, those that can demonstrate high and consistent quality through systematic monitoring and improvement are more likely to attract and retain students. The inability to track and showcase performance metrics puts Motosafe at a disadvantage compared to competitors

that use data to drive quality improvement and marketing efforts. This competitive disadvantage can manifest in lower enrollment, reduced pricing power, and difficulty attracting highquality instructors.

4. Unavailability of an Organized Database for Operational Planning

There is no central, digital database at Motosafe Driving School where student records, instructor profiles, session schedules, completion statuses, and feedback data are stored in an organized, easily accessible manner. Information is scattered across paper logs, multiple spreadsheets, and informal communication channels, making the process of collecting, retrieving, and analyzing records extremely tedious and timeconsuming. It is also difficult for employees to conduct meaningful analysis on operational data, such as identifying peak training periods, commonly requested time slots, instructors with the highest utilization, or students who are lagging in progress. This fragmentation of data creates significant barriers to effective operational management and strategic planning.

The scattered nature of data at Motosafe means that staff members must search through multiple sources to find information they need. Student records might be in one spreadsheet, session schedules in paper calendars, instructor profiles in another document, and feedback in yet another location. This fragmentation makes simple tasks like generating a report on student completion rates timeconsuming and errorprone. Staff members may need to manually consolidate data from multiple sources, increasing the risk of errors and omissions. The time spent searching for and consolidating data reduces the time available for analysis and decisionmaking.

Decisionmaking at the organization becomes limited because of the lack of a data management system that helps gain insights into the school's performance, student demand patterns, and instructor effectiveness. Administrators cannot easily generate reports on key metrics such as instructor utilization rates, student completion rates, average feedback scores, or scheduling efficiency. This project seeks to develop a webbased scheduling and performance monitoring system that uses a centralized database to collect, store, organize, and analyze all operational information, enabling datadriven planning and optimization of driving school operations. A centralized database would provide a single source of truth for all operational data, making it easy to retrieve, analyze, and report on key metrics.

The inability to analyze operational data prevents the school from identifying opportunities for improvement and optimization. For example, the school cannot easily determine which time slots are most popular with students, making it difficult to optimize instructor scheduling for peak demand periods. The school cannot identify patterns in student completion rates that might indicate problems with specific instructors or training methods. The school cannot track trends in student enrollment over time to plan for capacity needs. These analytical gaps limit the school's ability to make informed strategic decisions and optimize its operations for efficiency and quality.

Impact of Disorganized Data on DecisionMaking

The absence of an organized database limits the school's ability to make informed strategic decisions. Administrators cannot identify trends such as peak enrollment periods, most popular training times, or instructors with the highest student satisfaction. This lack of insight prevents the school from optimizing resource allocation, planning for growth, or responding effectively to changes in student demand. Decisionmaking becomes reactive rather than proactive, limiting the school's longterm sustainability. Strategic planning becomes guesswork rather than datadriven decisionmaking.

Without data insights, the school cannot optimize its operations for maximum efficiency. For example, the school cannot determine the optimal number of instructors to maintain based on historical demand patterns, potentially leading to overstaffing during slow periods or understaffing during peak periods. The school cannot identify which training times are most popular, making it difficult to schedule classes when students are most likely to attend. The school cannot track which instructors have the highest completion rates or satisfaction scores, making it difficult to allocate students optimally or identify training needs. These inefficiencies accumulate over time, reducing the school's competitive position and profitability.

The school also faces compliance risks when data is disorganized and difficult to access. LTO accreditation requires accurate recordkeeping and the ability to produce reports on student training hours, instructor qualifications, and completion rates. Disorganized data makes it difficult to demonstrate compliance during audits or inspections, potentially jeopardizing the school's accreditation status. The time and effort required to assemble compliance reports from scattered sources is substantial, diverting resources from other important activities. A centralized database would ensure that compliance records are always available and accurate, reducing regulatory risk.

Taken together, these problems describe a need for consistent operational information rather than a need for an advanced optimization engine. The system must help authorized users view and maintain records, coordinate assignments, and identify obvious conflicts before a schedule is confirmed. A conflict check can indicate that an instructor or student has another assignment during the proposed period, but it does not independently decide which of several valid schedules is best. This distinction defines the intended level of automation and keeps the project aligned with the actual functions implemented for the client.

The research questions are addressed through the design and implementation of the system and through evaluation activities appropriate to the project's stage. Questions about the current process are addressed by gathering information from the client and reviewing existing workflows. Questions about system functionality are addressed by checking whether the developed modules support the stated requirements. Questions about usability and operational usefulness require feedback from intended users and should be interpreted only in light of the evaluation instrument, respondents, and data actually collected. This separation helps prevent technical implementation claims from being mistaken for evidence of long-term organizational impact.

1.2.3 Research Questions

To guide the development and evaluation of DigiTrack, this study addresses the following research questions:

1. How effectively can an automated scheduling module prevent conflicting appointments and overlapping bookings at Motosafe Driving School?

2. To what extent can DigiTrack support administrators in reviewing instructor assignments and availability and identifying potential scheduling conflicts?

3. How effective is a digital student feedback mechanism in improving instructor performance monitoring at Motosafe Driving School compared to manual feedback collection?

4. Can a centralized database and administrative dashboard improve data-driven decision-making for driving school operations?

5. How functional, accurate, and usable is the DigiTrack system for administrators, instructors, and students?

These questions will be answered through system development, testing, and user evaluation in the subsequent chapters of this study.

1.3 OBJECTIVES OF THE STUDY

The objectives of this study are the specific goals the researcher intends to accomplish through the development of the proposed system. They represent the expected outcomes of creating a system that will address the problems identified in the current operational processes at Motosafe Driving School, especially in scheduling, workload distribution, performance monitoring, and datadriven decisionmaking. These objectives are divided into two parts: the General Objective and the Specific Objectives.

1.3.1 General Objective

The main purpose of this study is to develop and implement DigiTrack, a webbased scheduling and performance monitoring system designed to streamline instructor management and optimize workload distribution for Motosafe Driving School. Specifically, the system will help the school organize the process of assigning instructors to students, monitor instructor availability and utilization, balance workloads equitably, collect and analyze student feedback, and generate realtime reports on operational performance. The general objective encompasses the overall goal of the project, which is to transform Motosafe's operational processes from manual and inefficient to automated and efficient.

The development of this webbased system will also make Motosafe Driving School’s operations more efficient and technologically advanced. It is expected to help the school better organize its training services, reduce administrative burdens, minimize scheduling conflicts, and provide higherquality driver education services to its students. Beyond the immediate operational improvements, the system will position the school as a modern, professional institution committed to quality and continuous improvement. This technological advancement will enhance the school's competitive position in the market and support longterm growth and sustainability.

The general objective also includes the broader goal of contributing to road safety in the community by improving the quality and consistency of driver education. By enabling better instructor management and performance monitoring, the system will help ensure that all students receive highquality instruction that prepares them for safe driving. This aligns with the school's mission and contributes to public safety goals. The system represents an investment not only in operational efficiency but also in the school's core mission of producing safe, competent drivers.

1.3.2 Specific Objectives

To meet the general objective of the research, the specific objectives are outlined below. These objectives are aimed at addressing the analysis, design, development, implementation, and evaluation of the proposed system. They have been formulated to solve the problems associated with Motosafe Driving School regarding manual scheduling, workload imbalance, lack of performance monitoring, and absence of centralized data management. These objectives form the foundation for creating an efficient webbased scheduling and performance monitoring system and will guide the research to ensure that the final system effectively addresses the existing problems.

To analyze the current situation and requirements of Motosafe Driving School and its stakeholders.

The study will examine and evaluate the processes through which current operations are undertaken, management practices, and the challenges faced by Motosafe Driving School administrators, instructors, and students. The processes for scheduling training sessions, assigning instructors, tracking student progress, and recording information will be carefully examined. An assessment of the current situation is essential in identifying areas where the manual process is inefficient, challenging, and in need of technological support. In addition, the evaluation of the current situation is critical in determining the functional and nonfunctional requirements and the needs of users concerning the proposed system. Understanding the current workflow will facilitate a better understanding of the driving school’s needs and enable the development of a system suited to meet those needs. The examination of the requirements of instructors and students is also critical in identifying the features and functionalities required from a webbased information system. This objective will guide the development of a system that not only meets the operational objectives of the driving school but also addresses the needs of its instructors and students.

To design and develop an automated scheduling module that prevents conflicting appointments and overlaps.

This study aims to design and develop an automated scheduling module using a webbased interface to prevent scheduling conflicts and eliminate overlapping appointments at Motosafe Driving School. The scheduling module will provide a digital environment where all lesson booking and instructor assignment operations can be effectively performed on one platform. A webbased scheduling system enables users to interact with schedule data and perform tasks related to lesson booking via internetaccessible devices such as computers, tablets, and smartphones. Compared to manual scheduling systems, a webbased automated scheduler offers better accuracy, faster performance, and realtime conflict detection. The module will incorporate userfriendly interfaces and proper validation functionalities to facilitate effective utilization by administrators and instructors. Key functionalities of the scheduling module include realtime availability checking, automatic conflict detection, automated instructor assignment suggestions, and calendar visualization for easy schedule overview. During the development of the scheduling module, appropriate software development practices and database management principles will be applied to increase the efficiency, security, and reliability of the system. The module will ensure that no two students are assigned to the same instructor at the same time and that instructors are not scheduled beyond their available hours.

To develop a web-based scheduling and workload-monitoring system that helps administrators review instructor availability, existing appointments, and assignment information.

The study aims to develop a centralized scheduling and workload-monitoring system that organizes instructor availability, appointments, and assignment information. DigiTrack is intended to help administrators review the distribution of training sessions and identify potential scheduling conflicts before confirming or modifying appointments. Final assignment decisions remain with authorized administrators because instructor qualifications, leave, student needs, and other operational circumstances may require human judgment. The study therefore focuses on scheduling support and workload visibility rather than claiming that an automated algorithm produces an optimal or perfectly equitable distribution.

To create a studentfacing feedback mechanism to track and store instructor performance metrics.

Another goal of the study is to design and implement a studentfacing feedback mechanism that will assist the school in collecting, storing, and analyzing instructor performance data systematically. For many driving

schools, gathering honest and timely student feedback can be difficult, especially when using paperbased forms or informal channels. The developed system will aim to assist Motosafe in collecting feedback in a structured and efficient manner immediately after each training session. Using the system, students will be able to rate their instructors and provide comments on their training experience through a simple, accessible web interface. The system will store all feedback data in a centralized database, allowing administrators to analyze performance metrics such as average ratings, common comments, and trend patterns over time. This objective also promotes transparency in instructor evaluation by giving students a voice in assessing service quality and providing management with actionable insights for improvement. The system will also help managers analyze instructor performance trends to identify highperforming instructors for recognition, detect instructors needing additional training, and improve overall service quality and efficiency. The feedback mechanism will include rating scales, optional comment fields, and immediate submission capabilities to ensure high response rates and timely data collection.

To provide an administrative dashboard for realtime visualization of instructor utilization and student outcomes.

This objective includes the efficient management of Motosafe Driving School operations through the development of an administrative dashboard. Through the dashboard, managers can monitor available instructor resources, track current training sessions, analyze utilization performance, and identify overall trends in operations. By monitoring operational processes and analyzing the information obtained, inefficiencies in service provision can be detected and addressed, thereby increasing operational efficiency. The development of the administrative dashboard is expected to contribute significantly to the efficient management of driving school operations. The dashboard will display important data on instructor scheduling, utilization rates, student progress, feedback scores, and session completion rates. It will provide structured reports, charts, graphs, and analyses of relevant information, allowing administrators to efficiently monitor driving school operations. Instead of checking all records and files one by one, administrators can obtain the needed information much faster using the dashboard

To test if the system is functional, accurate, and easy to use.

The research intends to assess and validate the functional efficiency, usability, and accuracy of the DigiTrack scheduling and performance monitoring system. System testing is significant in verifying whether the developed system satisfies the needs and objectives of the project. The system's ability to prevent scheduling conflicts effectively, distribute workloads fairly, collect and store feedback accurately, and generate reliable reports will be evaluated. Effective system functionality will ensure that Motosafe Driving School can operate smoothly without scheduling errors, workload imbalances, or data loss. The study will also evaluate the system's usability to determine whether administrators, instructors, and students find the system easy to navigate and use in performing their duties. Aspects such as interface design, responsiveness, accessibility, and learnability will be considered during testing and evaluation. Testing will include demonstration and evaluation by various users to obtain feedback on the effectiveness of the system. Proper evaluation through testing is critical to ensure that the proposed DigiTrack system functions efficiently, meets all specified requirements, and is ready for deployment at Motosafe Driving School.

1.4 SCOPE AND DELIMITATIONS OF THE STUDY

1.4.1 Scope

The proposed DigiTrack: Instructor Scheduling and Performance Monitoring System is a webbased application developed specifically for Motosafe Driving School. The system provides different functions according to three primary user roles: Administrator, Instructor, and Student.

The scope of the study includes the following major functions and components:

- Automated Scheduling and Booking Module

The system will provide an interface for managing training appointments and schedules. Authorized users can record appointment information and view existing schedules. The system will check relevant scheduling information to identify conflicting appointments and overlapping assignments.

The module is intended to reduce the need for manual crosschecking when creating or modifying appointments. However, administrators remain responsible for reviewing and confirming schedules.

- Instructor Management and Availability

The system will maintain instructor profiles and relevant information concerning instructor availability and assignments. Administrators can manage instructor information and review schedules associated with instructors.

The availability information recorded in the system will be used to support scheduling and assignment decisions.

- Instructor Workload Monitoring

DigiTrack will provide information concerning the number and distribution of training assignments among instructors. Administrators can use this information to review instructor workload and consider workload when making scheduling decisions.

The system does not claim to determine a universally optimal workload for every instructor. Workload decisions may depend on qualifications, availability, course requirements, vehicle considerations, and other operational factors.

- Student Feedback Mechanism

Students will be provided with a digital mechanism for submitting feedback about their training experiences and instructors. Feedback may include ratings and comments depending on the implemented system features.

The collected feedback will be stored in the centralized database and made available to authorized administrators for internal monitoring.

- Instructor Performance Monitoring

The system will organize information that may be used in monitoring instructor performance. This may include student ratings, feedback, completed training sessions, and other systemrecorded indicators.

The system does not represent these indicators as a complete professional assessment of instructor competence. Instead, they serve as information that can support the school's internal monitoring process.

- Administrative Dashboard

The administrator dashboard will provide a centralized view of selected operational information. Depending on the implemented features, this may include instructor schedules, instructor workload information, student records, session completion information, and student feedback.

The dashboard is intended to reduce the need to manually search through multiple records when reviewing operational information.

- Student and Instructor Records

The system will store relevant information concerning students and instructors. Authorized users will be able to access information according to their assigned roles.

Students will have access to functions relevant to their own training and account, while instructors will have access to information relevant to their assigned responsibilities.

- TrainingSession Monitoring

DigiTrack will record scheduled training sessions and their status. Authorized users can update the completion status of sessions after training activities have been conducted.

Session completion is not automatically detected by the system. The accuracy of sessioncompletion information therefore depends on authorized users entering the appropriate records.

- Centralized Database

The system will use a centralized database for storing information required by the application's modules. This includes student information, instructor information, appointments, training sessions, feedback, and other operational records necessary for the implemented features.

- User Management and RoleBased Access

DigiTrack will provide user authentication and rolebased access control. The primary roles are Administrator, Instructor, and Student. Each role will have access to functions appropriate to its responsibilities.

- WebBased Access

The system will be accessible through compatible web browsers on supported desktop, laptop, tablet, or smartphone devices. Actual accessibility depends on the availability of the system server, network connection, browser compatibility, and device capability.

- Theoretical Driving Course and Practical Driving Course

The system will support the recording and scheduling requirements associated with the training services offered by Motosafe Driving School, including theoretical and practical driving activities. Where applicable, the system may distinguish between TDC and PDC sessions and record the relevant schedule and completion information.

The study focuses on managing information associated with these activities rather than replacing the instructional content or official regulatory requirements governing them.

1.4.2 Delimitations

The study is delimited in the following ways:

- No Payment or Financial Module

DigiTrack does not include an online payment, billing, payroll, accounting, or financial management module. Payment transactions and financial records are outside the scope of the study.

The system focuses on scheduling, instructor management, training monitoring, feedback, and operational information.

- No Direct Integration with Government Databases

The system does not directly connect to the Land Transportation Office or other government databases. It does not automatically verify licenses, permits, government records, or official regulatory information.

LTO requirements remain governed by the applicable official regulations and procedures, including those established under LTO Memorandum Circular No. 2019-2176 [1].

- No GPS or VehicleSensor Integration

DigiTrack does not use GPS devices, vehicle sensors, telematics, or other hardwarebased tracking technologies to monitor the physical location or movement of vehicles.

The system therefore cannot automatically determine a vehicle's location, travel route, or actual driving activity.

- Internal Feedback Only

Student feedback collected through DigiTrack is intended for Motosafe Driving School's internal monitoring and qualityimprovement purposes. The feedback does not constitute an official government evaluation, certification, or licensing assessment.

- Manual SessionCompletion Updates

Although the system can organize training schedules and records, session completion must be entered or confirmed by an authorized instructor or administrator. DigiTrack does not automatically detect when a physical training session begins or ends.

- Limited to Motosafe Driving School

The system is designed according to the identified operational requirements of Motosafe Driving School in Miagao, Iloilo. Other driving schools may have different organizational structures, training procedures, user roles, scheduling policies, or regulatory requirements.

Consequently, the system may require modification before being used in another driving school.

- No Direct Measurement of RoadSafety Outcomes

The study does not measure whether DigiTrack directly reduces road accidents, traffic violations, or other communitylevel roadsafety indicators. Such outcomes would require a different research design, a longer observation period, and data beyond the scope of the current systemdevelopment study.

- No Guaranteed Equal Workload

DigiTrack provides workload information and supports scheduling decisions, but it does not guarantee mathematically equal workloads among instructors. Differences in instructor qualifications, availability, training types, student requirements, and other operational considerations may affect assignments.

The boundaries of the study also guide how the system should be evaluated. The research examines whether the implemented features support the stated operational tasks within the client environment; it does not attempt to determine the best staffing level for a driving school, calculate an optimal timetable across every possible assignment, or establish causal relationships between software use and road-safety outcomes. Those questions would require different data, models, and evaluation periods. Keeping the scope explicit prevents the expected benefits of the project from being presented as already proven results and helps readers understand the relationship between the system's functions and the evidence collected in the study.

1.5 SIGNIFICANCE OF THE STUDY

The development of DigiTrack may provide useful benefits to different stakeholders involved in the operation of Motosafe Driving School. The significance of the study is discussed in terms of the potential contribution of the system to each group.

- Administrators

DigiTrack can provide administrators with a centralized platform for managing instructor schedules, student information, training sessions, feedback, and operational records. Instead of relying entirely on separate paper records, spreadsheets, or informal communication, administrators can access relevant information through a common system.

The administrative dashboard can also provide summaries of selected operational information. This may assist administrators in reviewing instructor assignments, trainingsession status, feedback, and other records when making operational decisions.

The system does not replace administrative judgment. Instead, it provides organized information that can support the administrator's existing responsibilities.

- Instructors

Instructors can benefit from having a more organized view of their assigned training schedules and relevant student or session information. A centralized schedule can help instructors identify their assigned sessions and reduce uncertainty caused by inconsistent or outdated schedule information.

Workload information can also help administrators review how training assignments are distributed. This may support more informed scheduling decisions while recognizing that assignments must also consider qualifications, availability, course requirements, and other operational factors.

Student feedback may also provide instructors with information concerning learners' experiences. Such feedback can be considered as one source of information for identifying areas that may require improvement.

- Students

Students can benefit from more organized scheduling and easier access to their trainingrelated information. A centralized scheduling process may help reduce confusion concerning appointment dates, times, and assigned instructors.

The feedback mechanism also gives students an organized way to communicate their training experience to the school. Their feedback can provide information that administrators may use when reviewing the quality of services provided.

However, the system does not guarantee that all scheduling changes or administrative decisions will occur automatically. Students remain subject to the school's policies and procedures regarding training appointments and requirements.

- Motosafe Driving School

For Motosafe Driving School, DigiTrack can provide a centralized technological platform for managing operational information. The system can organize information concerning students, instructors, appointments, training sessions, feedback, and performancerelated indicators.

The system may also reduce dependence on scattered records and make frequently needed information easier to retrieve. This can support administrative coordination and improve the visibility of daytoday operations.

The system is intended as a tool for operational support rather than as a replacement for the school's management policies or regulatory responsibilities.

- Researchers and Future Developers

The study can serve as a reference for future researchers and developers interested in webbased management systems for driving schools, educational institutions, or other training organizations.

The system development process, database structure, user roles, scheduling functions, feedback mechanism, and evaluation procedures may provide ideas for future projects. Future researchers may also examine additional features such as advanced analytics, mobile applications, external system integration, automated notifications, or other technologies that are outside the current scope of DigiTrack.

- Academic Community

The study contributes to the academic application of information technology concepts in a real organizational setting. It demonstrates how software development, database management, web technologies, rolebased access control, scheduling, and system evaluation can be combined to address an operational problem.

The project may also serve as a case study of how a localized information system can be designed around the needs of a specific organization rather than adopting a generic system without considering its users and processes.

- Parents and Guardians

Parents and guardians may indirectly benefit when students have access to a more organized training schedule and when relevant training information is maintained systematically by the school.

However, the system is not designed as a dedicated parentmanagement platform. Any information available to parents or guardians remains subject to the school's policies regarding access to student information and privacy.

- Government and Regulatory Considerations

Although DigiTrack does not directly connect to government databases, centralized records may help the school organize information that may be needed for internal administrative purposes.

The system does not replace official LTO requirements, accreditation procedures, inspections, or regulatory reporting. Compliance with government regulations remains the responsibility of the driving school and must follow applicable official requirements [1].

- Local Community

The local community may benefit indirectly from the improved organization of a local training institution. However, this study does not claim that DigiTrack directly reduces road accidents or traffic violations.

Any broader community benefit would depend on many factors beyond the system itself, including the quality of driver education, student behavior, traffic enforcement, road conditions, and other roadsafety measures. Therefore, communitylevel roadsafety outcomes are outside the direct measurement of this study.

For school administrators, the main potential benefit is a more consolidated view of day-to-day operations. When schedules, instructor assignments, attendance, training records, and student feedback can be reviewed through related modules, administrators have a clearer basis for checking whether records are complete and whether follow-up is required. This can support more consistent supervision, but the system's usefulness still depends on staff updating information as events occur. A dashboard that contains outdated data may create a misleading picture, so responsibility for timely and accurate record maintenance remains important.

For instructors, the system is intended to make relevant assignment and training information easier to access. Organized schedules can reduce uncertainty about assigned sessions, while clear records can help instructors and administrators confirm whether training activities have been completed. The system should not be interpreted as a substitute for professional instruction or as a standalone measure of teaching quality. Performance-related indicators require context, and student feedback should be considered alongside workload, attendance, course assignments, and other relevant information rather than treated as a complete judgment of an instructor's ability [5].

For students, the value of DigiTrack lies in the availability of organized schedule and training information and in the use of feedback mechanisms where these are provided by the system. Students remain dependent on the driving school for official decisions, policy explanations, and changes that require staff approval. Clear information can support communication, but it cannot guarantee that every appointment will remain unchanged or that all requests will be resolved immediately. These boundaries are important because the system is designed to assist the existing organization, not to replace its responsibilities or promise outcomes beyond its implemented functions.

1.6 DEFINITION OF TERMS

To provide a consistent interpretation of important concepts used throughout the study, the following terms are operationally defined according to how they are used in DigiTrack.

- DigiTrack

A webbased Instructor Scheduling and Performance Monitoring System developed specifically for Motosafe Driving School. In this study, DigiTrack refers to the proposed and developed application used to manage scheduling, instructor information, student information, training sessions, feedback, workloadrelated information, and administrative monitoring.

- Manual Scheduling

The process of organizing appointments and training sessions using paper records, spreadsheets, verbal communication, or other methods that require users to manually check and update scheduling information.

- Workload Monitoring

The process of reviewing the number and distribution of training assignments or sessions associated with instructors during a specified period. In DigiTrack, workload monitoring provides information that administrators can use when reviewing instructor assignments.

- Performance Monitoring

The process of reviewing information that may indicate instructor performance, including student feedback, ratings, trainingsession records, and other systemrecorded indicators.

- Scheduling Conflict

A situation in which two or more appointments or training activities overlap in a manner that prevents the involved instructor, student, vehicle, or time slot from being properly used for the scheduled activities.

- Conflict Checking

A system process that checks existing schedule information when an appointment is created or modified to determine whether a conflicting appointment exists.

- Instructor Availability

The time periods recorded in the system during which an instructor is available for training assignments, subject to the school's scheduling rules and the instructor's qualifications.

- Instructor Utilization

The extent to which an instructor's available training time or schedule capacity is used for assigned training activities during a particular period.

- Student Feedback

Ratings, comments, or other responses submitted by students concerning their training experience or instructor.

- Automation

The use of software functions to perform specified tasks with reduced manual intervention. In DigiTrack, automation may include schedule validation, information processing, data storage, and report generation.

- Performance Metrics

Measurable information used to review aspects of instructor or system performance. Examples may include average student ratings, number of completed sessions, number of assigned sessions, or other indicators available within the system.

- Administrative Dashboard

A centralized interface that presents selected operational information to authorized administrators through tables, charts, summaries, or other visual elements.

- Feedback System

The DigiTrack feature that allows students to submit ratings and comments concerning their training experience and instructors.

- WebBased System

A software application that users access through a web browser over a network connection rather than through a locally installed desktop application.

- Centralized Database

A structured database that stores related system information in a common repository. In DigiTrack, the centralized database contains records such as student information, instructor information, schedules, training sessions, and feedback.

- RoleBased Access Control

A security mechanism in which system access is determined according to the user's assigned role. DigiTrack uses roles such as Administrator, Instructor, and Student to determine the functions and information available to each user.

- Theoretical Driving Course (TDC)

A theoretical drivereducation component involving instruction concerning traffic rules, road safety, and related driving knowledge. Within DigiTrack, TDCrelated information may be recorded and scheduled according to the operational requirements of Motosafe Driving School and applicable regulations.

- Practical Driving Course (PDC)

A practical drivingtraining component in which students receive handson driving instruction using an appropriate training vehicle. Within DigiTrack, PDCrelated appointments and trainingsession information may be scheduled and recorded.

- Training Session

A scheduled period during which a student receives theoretical or practical driving instruction. Training sessions are recorded in DigiTrack to support scheduling and progress monitoring.

- DoubleBooking

A scheduling situation in which the same instructor, student, vehicle, or time period is assigned to more than one activity at the same time.

- Utilization Rate

A measurement representing the proportion of an instructor's available scheduling capacity that is assigned to training activities during a specified period.

- Session Completion

The recorded status indicating that a scheduled training session has been completed and confirmed by an authorized user. In DigiTrack, session completion is updated manually by an instructor or administrator.

- Operational Efficiency

The effective use of available time, personnel, and organizational resources in performing the school's operational activities. In this study, operational efficiency is considered in relation to scheduling, information retrieval, record management, and administrative processes.

- Human Error

An incorrect action, omission, or inaccurate entry made during a manual or systemsupported process. In the context of DigiTrack, examples may include incorrect scheduling information, missed updates, or incorrect assignments.

- Workload Imbalance

A situation in which training assignments are distributed unevenly among instructors, resulting in differences in the number of sessions or workload assigned to individual instructors.

- Instructor Assignment

The process of assigning an instructor to a student's scheduled training activity based on relevant scheduling and operational considerations.

- Student Progress

Information concerning a student's recorded training activities, completed sessions, and other progressrelated records maintained by the system.

- Administrative Reporting

The process of generating or viewing organized information concerning school operations to support monitoring and decisionmaking.

- Scheduling Module

The component of DigiTrack responsible for managing appointments, schedules, instructor assignments, and conflict checking.

- Performance Monitoring Module

The component of DigiTrack that organizes feedback and other relevant information used by administrators to monitor instructorrelated performance indicators.

- User Authentication

The process of verifying the identity of a user before allowing access to the system.

- System Usability

The degree to which intended users can use the system to accomplish their required tasks effectively and conveniently.

- System Functionality

The degree to which the system provides and correctly performs the functions required by its intended users and specified requirements.

- System Accuracy

The degree to which information produced, stored, or processed by the system corresponds to the correct input and expected result.

- User Experience

The overall experience of administrators, instructors, and students when interacting with DigiTrack, including their perceptions of usability, accessibility, organization, and ease of use.

- Data Privacy

The appropriate handling and protection of personal information collected, stored, processed, and accessed through the system according to applicable policies, laws, and organizational requirements.

- Operational Records

Information generated or maintained as part of the school's daily activities, including student records, instructor records, schedules, appointments, training sessions, and feedback.

- Scheduling Availability

The recorded set of time periods that can be considered when creating or assigning training sessions.

- Training Completion Record

A stored record indicating that a scheduled training activity has been completed and confirmed by an authorized user.

- Centralized Information Management

The practice of storing and managing related organizational information through a common system so that authorized users can retrieve and use consistent records.

- Administrative DecisionMaking

The process through which school administrators review available information and make decisions concerning schedules, instructor assignments, training activities, records, and other operational matters.

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CHAPTER II
REVIEW OF RELATED LITERATURE

This chapter presents the review of related literature and related studies that provide the academic and practical foundation for the development of DigiTrack: An Instructor Scheduling and Performance Monitoring System for Driving School. The review examines concepts, frameworks, studies, and existing systems related to drivingschool operations, instructor scheduling, workload management, digital transformation in vocational training, information management, data security, and drivingschool management systems. The purpose of this review is not only to describe previous works but also to identify common findings, limitations, and research gaps that are relevant to the present study.

The review begins with the operational characteristics of driving schools and the role of structured driver education. It then discusses the regulatory environment affecting driving schools in the Philippines, particularly the requirements and administrative responsibilities associated with the Land Transportation Office (LTO). Understanding this environment is important because a drivingschool management system must operate within established training and documentation requirements rather than treating scheduling and record management as isolated administrative activities [1].

The chapter also examines manual scheduling and operational processes. Previous studies on scheduling and resource utilization indicate that manual approaches can make it more difficult to coordinate people, time, and other limited resources, particularly when several constraints must be considered simultaneously [2]. These concerns are relevant to driving schools because instructors, students, training schedules, courses, and available time slots must be coordinated within the school's operating conditions.

Another major area reviewed is the relationship between instructor workload and performance. Workload is an important consideration in educational environments because excessive or poorly distributed assignments may affect how instructors perform their responsibilities. Jiao and Hernandez found a relationship between workload and professional factors associated with teacher performance, although their study involved vocational college teachers rather than driving instructors specifically [5]. Therefore, the findings are used in this study as general educational evidence rather than as direct evidence about drivingschool instructors.

The chapter further examines digital transformation and Management Information Systems (MIS). Digital systems can centralize organizational information, support reporting, and make operational records more accessible to decisionmakers [9]. These capabilities are relevant to DigiTrack because the proposed system is designed to centralize information about instructors, students, courses, appointments, training sessions, and performance feedback.

Finally, related drivingschool systems and academic prototypes are reviewed. These systems demonstrate the usefulness of digital scheduling, booking, instructor management, student monitoring, and communication features. However, they also show that existing solutions vary considerably in scope, implementation context, and regulatory alignment. The review therefore identifies the specific gap that DigiTrack seeks to address: a localized webbased system designed around the scheduling, monitoring, and administrative requirements of a Philippine driving school.

2.1 DRIVINGSCHOOL OPERATIONS AND MANAGEMENT

Driving schools perform several interconnected administrative and instructional activities. These include student registration, course management, instructor assignment, scheduling, trainingsession recording, progress monitoring, certificationrelated records, and collection of student feedback. Because these activities involve different users and records, effective management requires accurate and accessible information.

A driving school therefore functions not only as a place where learners receive driving instruction but also as an organization that coordinates people, schedules, training resources, and records. The management of these resources becomes more challenging when several students and instructors must be accommodated within limited operating hours.

For DigiTrack, this operational perspective is important because the system is designed around the relationship among students, instructors, courses, schedules, appointments, training sessions, and performance records. Instead of treating each activity as a separate administrative task, the system provides a centralized environment in which related information can be stored and accessed.

2.1.1 Global Frameworks in Driver Education Logistics

Driver education forms part of a broader road-safety system. The International Transport Forum's Road Safety Annual Report 2023 reviews road-safety indicators and policy developments across countries [4]. This provides context for road-safety policy, but the report should not be treated as direct evidence that a specific driving-school management feature improves driver competence or road-safety outcomes.

The organization of driver education requires more than the delivery of lessons. Training institutions must coordinate instructional resources, maintain learner records, and ensure that required training activities are documented. These responsibilities make scheduling and information management important components of the operation of a driving school.

International developments in vocational education also demonstrate an increasing emphasis on digital transformation. Yang and Yang describe digital transformation in technical and vocational education and training as involving changes in institutional processes, instructional practices, information management, and organizational capabilities [10]. This perspective is relevant to driving schools because driver education contains both vocational training and administrative processes.

Digital transformation does not necessarily mean replacing instructors with technology. Instead, information systems can support instructors and administrators by providing better access to records, reducing repetitive administrative tasks, and organizing information required for daily operations. In this context, a digital management system can complement the instructional role of the driving instructor.

Driving education also requires coordination between theoretical and practical components. Learners may need to complete particular training activities before progressing to subsequent stages. Consequently, a management system should be capable of recording the status of training activities and making relevant information available to authorized users.

The global literature therefore establishes several important considerations for drivingschool management. First, driver education is connected to broader roadsafety objectives. Second, vocational training institutions increasingly depend on organized information systems. Third, training delivery involves the management of learners, instructors, schedules, and records. These considerations provide a foundation for examining how digital systems can support drivingschool operations.

For DigiTrack, these principles are translated into features that support course management, instructor assignment, appointment scheduling, trainingsession monitoring, student records, instructor ratings, and administrative reporting.

2.1.2 Local Regulatory Mandates and Compliance (LTO Frameworks)

The Philippine drivingschool environment is influenced by requirements established by the Land Transportation Office. These requirements are important to the present study because DigiTrack is intended for use by a driving school operating within the Philippine regulatory environment.

The LTO has established rules concerning driver education, drivingschool accreditation, and the training requirements associated with obtaining a driver's license. Memorandum Circular No. 2019-2176 provides requirements and guidelines related to driving schools and drivingschool accreditation [1]. These requirements demonstrate that drivingschool operations are subject to regulatory expectations and that schools need to maintain appropriate records and follow prescribed procedures.

The regulatory environment also distinguishes theoretical and practical driver education. Theoretical training provides learners with knowledge related to traffic rules, road safety, and responsible driving, while practical instruction provides opportunities to develop actual vehiclehandling and driving skills. The distinction between these components creates a need for accurate tracking of learner participation and training activities.

For a driving school, maintaining reliable records is therefore important. Records may include student information, course information, instructor assignments, schedules, training sessions, and other documentation required for administrative purposes. A centralized information system can help organize these records and make them easier to retrieve when needed.

However, it is important to distinguish between supporting compliance tracking and performing regulatory compliance itself. DigiTrack does not replace the LTO, does not issue driver's licenses, and does not independently determine whether a student satisfies every government requirement. Instead, the system provides

internal records and monitoring functions that can assist the driving school in organizing information relevant to its operations.

This distinction is particularly important when designing the system. Regulatory information should be represented accurately, while administrative decisions remain under the responsibility of authorized personnel. DigiTrack can therefore function as an internal management and monitoring tool rather than as a replacement for official government processes.

The Philippine regulatory environment also demonstrates why a localized system is useful. A generic scheduling application may provide appointment management but may not reflect the specific terminology, training structure, or administrative practices used by Philippine driving schools. A localized system can incorporate these contextual requirements into its data structures and workflows.

For DigiTrack, this localization is reflected in the inclusion of training related records, instructor assignments, course information, appointments, sessions, certification records, and administrative monitoring. These features are intended to support the school's internal organization while remaining separate from official LTO decisionmaking.

2.1.3 Manual Operational Vulnerabilities and Institutional Inefficiencies

Manual administrative processes remain common in organizations where resources, personnel, and technology are limited. These processes may include paper records, spreadsheets, handwritten schedules, verbal coordination, and separate files for different types of information.

Although manual methods can be functional for small operations, they become increasingly difficult to manage as the number of records and transactions increases. Scheduling is particularly sensitive because several pieces of information must be considered simultaneously. An administrator may need to determine whether an instructor is available, whether the student is available, whether the selected course is appropriate, and whether another appointment already occupies the same time slot.

Research on manual and automated timetabling has shown that scheduling methods affect resource utilization and the ability of institutions to coordinate available resources [2]. Nsulangi et al. examined manual and automated timetabling approaches in a highereducation setting and reported differences in resourceutilization considerations. Although the study was not conducted in a driving school, it provides relevant general evidence about the difficulty of coordinating limited resources through scheduling.

Driving school operations introduce additional scheduling considerations because instructors and students are both time dependent resources. A schedule may be affected by instructor availability, student availability, course requirements, cancellations, changes in appointments, and the operating schedule of the school.

Manual scheduling can also make information less centralized. When records are distributed across paper documents, spreadsheets, and communication channels, administrators may need to consult several sources before making a scheduling decision. This increases the amount of coordination required and can make it more difficult to maintain a consistent record.

Another concern is record updating. If a schedule is changed but the change is not immediately reflected in all relevant records, different users may have different versions of the same information. A centralized database can reduce this type of inconsistency by providing a common location for operational records.

However, digitalization should not be interpreted as automatically eliminating all operational problems. A digital system can still contain incorrect data if users enter incorrect information, if business rules are incomplete, or if the system is not maintained properly. Therefore, DigiTrack is designed to support administrators rather than remove the need for human review.

The primary value of DigiTrack in this context is the organization of information. Instead of depending primarily on separate records and verbal coordination, the system provides a centralized environment for storing instructor schedules, student appointments, training sessions, and performancerelated records.

2.1.4 Management Information System (MIS) Frameworks for Digital Firms

Management Information Systems provide a framework for understanding how organizations use information technology to support operations and decisionmaking. Laudon and Laudon describe information systems as systems that collect, process, store, and distribute information to support organizational activities and decision making [9].

The concept is directly relevant to DigiTrack because the system does not simply store information. It also organizes information into operational views that can assist administrators. Instructor records, student records, appointments, course schedules, session records, and ratings can be stored in a centralized database and retrieved through the application's interface.

The value of an MIS becomes particularly apparent when information originates from multiple activities. In a driving school, an appointment is related to a student and instructor, while the training session associated with that appointment may later be connected to completion information or feedback. Organizing these relationships allows administrators to retrieve related information without maintaining separate independent records.

Centralized information can also support reporting. Instead of manually compiling information from several sources, administrators can use stored records to review schedules, assignments, completed sessions, and ratings. This does not guarantee better decisions, but it provides decisionmakers with a more organized information base.

The MIS perspective also emphasizes that information systems should support organizational processes rather than exist independently of them. For DigiTrack, the system's features are therefore based on the operational activities of Motosafe Driving School. The system is designed around the roles of administrators, instructors, and students and the information required by each role.

Role-based access is another important aspect of an organizational information system. Different users should not necessarily have the same permissions. Administrators require management functions, instructors require access to their assigned activities, and students require access to their own appointments and trainingrelated information.

Consequently, the MIS framework supports the central design principle of DigiTrack: operational data should be collected and organized in a centralized system so that authorized users can access information appropriate to their responsibilities.

2.1.5 Synthesis

The literature on drivingschool operations shows that effective management requires coordination among students, instructors, courses, schedules, training activities, and records. Driver education also operates within a broader roadsafety and regulatory environment, making accurate administrative records important.

The Philippine context adds another layer of complexity because driving schools operate under LTO requirements. A management system must therefore consider local training and documentation practices while recognizing that official regulatory decisions remain outside the system.

Manual scheduling and recordkeeping can create difficulties when several resources and activities must be coordinated. Research on timetabling supports the importance of structured resource allocation, while MIS literature demonstrates how centralized information systems can organize operational data [2], [9].

These findings provide a foundation for DigiTrack. The system addresses the administrative side of drivingschool operations by centralizing records, supporting scheduling, monitoring training activities, and collecting performance feedback.

The literature on driving-school operations and general educational administration suggests that information organization is a foundational requirement for consistent service delivery. A management information system is useful when it connects records that are needed for routine decisions rather than merely converting paper forms into electronic copies [9]. For example, a student record becomes more useful when authorized staff can relate it to a course, scheduled appointment, assigned instructor, and completed training activity. The value comes from the relationships among records and the ability to review them consistently, not from digitization alone.

This distinction is important when interpreting the cited studies. Research on school information systems provides evidence that digital platforms can support administrative organization, but the specific workflows of a community college are not identical to those of a driving school [3]. Likewise, research on automated timetabling in higher education can inform the general challenge of allocating limited time and people, but it does not establish that the same scheduling rules apply without modification to driver training [2]. DigiTrack uses these sources as transferable design principles and adapts them to the client's requirements instead of claiming that prior studies directly validate the finished system.

The regulatory environment adds a further distinction between internal administration and official compliance. A driving school may use software to organize records associated with training and instructor assignments, but a local information system does not itself confer accreditation or certify that every legal obligation has been met. The system's records should therefore be viewed as operational support. Administrators remain responsible for verifying the accuracy and completeness of records and for following the applicable LTO procedures [1]. This boundary also helps define what should and should not be included in the system's claims.

2.2 INSTRUCTOR SCHEDULING AND WORKLOAD MANAGEMENT IN DRIVING SCHOOLS

Instructor scheduling is a major operational activity in a driving school because instructors are responsible for delivering practical and, depending on the school's structure, instructional activities to students. Their availability must be coordinated with student schedules and the school's available training periods.

Kannan et al. developed an Advanced Management System for Car Driving Schools that includes functions for student registration, scheduling, instructor or tutor management, and bookingrelated activities [6]. Their work demonstrates how a digital system can organize information involving students and instructors rather than relying entirely on manual communication.

Davanhipprage and Jagtap presented a web-based driving-school management project described as AI-driven, with functions related to scheduling, student management, and administrative processes [7]. The article describes these functions, but its claims about AI should not be treated as independently validated evidence.

These studies provide evidence that scheduling is a recurring concern in drivingschool systems. However, the proposed solutions differ in architecture, features, and level of automation. DigiTrack builds upon these concepts while focusing specifically on the operational requirements of Motosafe Driving School.

Instructor workload is another important consideration. Workload refers to the amount of instructional and related work assigned to an instructor during a particular period. When assignments are distributed unevenly, some instructors may have significantly more scheduled activities than others.

Jiao and Hernandez examined the relationship between workload and professional factors associated with teacher performance [5]. Their findings showed that workload can influence how professional factors, such as teaching practices and professional development, are reflected in teacher performance. They found that a moderate and properly distributed workload can support teachers' effectiveness and efficiency, while workload differences may affect how teachers perform. Although the study involved vocational college teachers rather than driving instructors, it provides a relevant educational perspective on why workload should be considered when designing scheduling and performance monitoring systems.

For DigiTrack, workload monitoring is therefore treated as an administrative support function. The system can present assignment and scheduling information that allows administrators to observe how training activities are distributed among instructors. The system does not independently determine whether an instructor is competent or whether a workload is psychologically appropriate.

2.2.1 The Mathematics of Multi Variable Resource Constraints

Scheduling can be understood as a resource allocation problem because multiple activities must be assigned to limited resources. In a driving school, instructors, students, time slots, courses, and training facilities may all affect whether an appointment can be scheduled.

At its simplest level, a scheduling problem involves matching an activity with an available instructor and an available time. However, actual scheduling may require several conditions to be satisfied simultaneously.

For example, an instructor should not normally be assigned to two appointments occupying the same time period. Similarly, a student cannot attend two different practical sessions at exactly the same time. The selected course and instructor may also need to be compatible.

These conditions can be represented conceptually as constraints:

- Instructor availability: the instructor must be available during the selected period.
- Student availability: the student must be available during the selected period.
- Time availability: the selected slot must fall within the school's operating schedule.
- Appointment conflict: the instructor and student should not already have another conflicting appointment.
- Course compatibility: the assigned instructor should be associated with the appropriate course or training category.
- Training sequence: activities should follow the school's required instructional process.

Scheduling literature commonly distinguishes between constraints that must be satisfied and preferences that are desirable. A required condition may be treated as a hard constraint, while a preferred time or balanced assignment may be treated as a soft constraint [2].

For example, an instructor's existing appointment at 9:00 A.M. represents a scheduling restriction because assigning another appointment to the same instructor and time would create a conflict. In contrast, a student's preference for a morning session may be treated as a preference rather than an absolute requirement.

Mathematically, instructor scheduling involves assigning students and instructors to available time slots while satisfying several requirements and constraints. These may include instructor availability, student availability, course requirements, and prevention of overlapping schedules. When multiple instructors, students, courses, and time slots are involved, manually creating schedules can become more difficult because several conditions must be considered at the same time.

More advanced scheduling systems may use optimization algorithms to minimize idle time, reduce scheduling conflicts, or distribute assignments according to predefined objectives. However, this mathematical concept should not be confused with the actual implementation of DigiTrack. DigiTrack is designed primarily as a web-based management and scheduling system rather than a full mathematical optimization engine. The discussion demonstrates why scheduling becomes complex when multiple constraints are involved, but it does not claim that DigiTrack implements integer programming or another optimization algorithm.

This distinction is important because DigiTrack provides administrative scheduling support and conflict checking rather than guaranteeing mathematically optimal instructor allocation.

2.2.2 Human Factors: Occupational Stress, Fatigue, and Pedagogical Quality

Scheduling affects not only the organization of appointments but also the people who perform the work. Instructors must manage teaching responsibilities, student interaction, vehicle operation, safety considerations, and administrative requirements.

An uneven workload may create practical difficulties for instructors. For example, an instructor who receives many consecutive lessons may have less time between sessions for preparation, documentation, and rest. Conversely, an instructor with relatively few assignments may have unused capacity.

Workload is therefore relevant to performance monitoring. Jiao and Hernandez reported that workload and professional factors were associated with teacher performance in their study of vocational college teachers [5]. Because the population of that study differs from driving instructors, the findings should be interpreted as related educational evidence rather than direct proof of the effects of driving instructor workload.

Driving instruction also involves psychomotor learning and continuous attention. Rehm et al. demonstrated that driving performance assessment can involve the evaluation of complex driving behaviors and that automated assessment approaches can be compared with human expert evaluation [11]. Their work illustrates the complexity of evaluating driving performance, although it is different from instructor performance monitoring.

Instructor performance in DigiTrack is therefore approached primarily through administrative records and student feedback rather than through automated assessment of actual driving behavior.

Student feedback can provide one source of information about the instructional experience. For example, students may evaluate an instructor after a completed session. The resulting rating can be stored and summarized for administrative review.

However, ratings should not be treated as a complete measure of instructor competency. A student's perception may reflect communication, punctuality, teaching style, or other aspects of the experience, but it may not measure every technical or safety related competency of the instructor.

DigiTrack consequently treats feedback as an additional performancemonitoring input rather than as an automatic determination of instructor quality.

Workload monitoring follows the same principle. The system can display assignments, schedules, and utilizationrelated information, but administrators remain responsible for interpreting the information and deciding whether changes to instructor assignments are appropriate.

2.2.3 Workload Monitoring and Administrative Scheduling Support

Digital scheduling systems can support workload management by making assignments more visible and by providing administrators with information about instructor availability and scheduled activities.

A centralized scheduling database is one important component. When schedules are stored in a common database, administrators can retrieve information about instructors, students, and appointments through the same system.

Conflict checking is another important function. A scheduling system can compare a requested appointment against existing records and identify conditions that may cause doublebooking. This is particularly useful when several users are creating or modifying appointments.

Kannan et al. demonstrated the use of centralized scheduling and tutor related functionality in a driving school management system [6]. Their system provides users with information about students and schedules, while administrators can manage assignments.

Davanhipprage and Jagtap describe booking and scheduling functions in a web-based driving-school management project [7]. This is an example of a proposed digital approach; the article should not be treated as independent evidence that it reduced coordination time in a measured deployment.

Commercial platforms such as Drive Scout advertise scheduling, booking, reminders, and instructor-management features [12]. These descriptions show the types of functions marketed to driving schools, but vendor descriptions should not be treated as independent evidence of effectiveness.

For DigiTrack, workload balancing is implemented as workload monitoring and administrative support, not as an unrestricted automatic optimization system. Administrators can review instructor schedules and assignments and use this information when making allocation decisions.

This distinction is necessary because workload fairness depends on factors that may not be represented completely in a database. An instructor's qualifications, availability, leave, vehicle assignment, student requirements, and other operational circumstances may affect whether an assignment is appropriate.

Therefore, DigiTrack supports the process by organizing relevant information rather than replacing the administrator's judgment.

2.2.4 Synthesis

The literature indicates that instructor scheduling involves multiple constraints and that digital systems can improve the organization and visibility of scheduling information. Existing drivingschool systems demonstrate that booking, instructor assignment, and schedule management can be incorporated into softwarebased solutions [6], [7].

Workload management is also relevant because instructor assignments can affect how instructional resources are distributed. Research involving vocational teachers provides general evidence that workload is

associated with professional performance, although the findings cannot be directly generalized to driving instructors [5].

The literature therefore supports the inclusion of scheduling, conflict checking, workload monitoring, and performance feedback functions in DigiTrack. At the same time, the reviewed evidence does not justify claiming that a digital system automatically produces perfectly balanced workloads or eliminates all scheduling conflicts.

DigiTrack consequently focuses on providing centralized information and administrative support. The system allows decisionmakers to see scheduling and workload information while retaining human control over instructor assignments.

A practical scheduling workflow can be understood as a sequence of checks. First, the administrator identifies the student, the required course or lesson, and the requested date and time. Second, the system checks whether the selected instructor is available and whether the student already has a conflicting appointment. Third, the administrator reviews the proposed assignment and confirms it according to the school's procedures. Finally, the confirmed appointment becomes part of the schedule and can be referenced by related training records. This sequence does not require an optimization algorithm; it requires consistent data, clear rules, and conflict detection at the point of assignment.

Workload information also needs to be interpreted carefully. A simple count of scheduled sessions can provide an initial view of assignment distribution, but session counts may not reflect differences in duration, course complexity, preparation requirements, or non-teaching responsibilities. If the system presents workload indicators, those indicators should be described according to the data they actually use. For instance, a count of appointments should not be presented as a complete measurement of effort unless the methodology defines and validates that interpretation. Jiao and Hernandez's discussion of workload and teacher performance supports considering workload as context, while also reminding researchers that findings from one educational setting should not be treated as direct proof about driving instructors [5].

Scheduling support is therefore best understood as decision support. The system can reduce the effort required to inspect records and can alert users to certain conflicts, but administrators may still need to consider exceptions, cancellations, instructor preferences, student needs, and school policies. A useful system makes these considerations visible and records the final administrative decision. It should not conceal the limits of its rules or imply that a schedule is fair simply because no time conflict exists. Workload distribution and conflict prevention are related but distinct concerns: one concerns how assignments are spread, while the other concerns whether simultaneous assignments are invalid.

2.3 DIGITAL TRANSFORMATION IN VOCATIONAL TRAINING

Digital transformation has become an important topic in technical and vocational education because training institutions increasingly use information technologies for administration, communication, learning support, and record management.

Yang and Yang discuss digital transformation in global technical and vocational education and training and emphasize the importance of digital capabilities within modern vocationaltraining institutions [10]. Digital transformation involves more than introducing individual software applications. It can include changes in organizational processes, information management, teaching practices, and institutional infrastructure.

Driving schools can be considered specialized vocational training environments because they provide structured training designed to develop practical competencies. Consequently, digital transformation can be applied not only to learning delivery but also to the administrative processes supporting training.

For a driving school, these processes include student registration, course selection, instructor assignment, scheduling, trainingsession tracking, certification records, and performance feedback.

A centralized webbased system can connect these activities. This allows the institution to move from isolated records toward an integrated information environment.

2.3.1 Paradigm Shifts in TechnicalVocational Information Systems (MIS)

Traditional administrative processes often depend on physical records and separate files. While these methods may be adequate for small numbers of transactions, they become increasingly difficult to maintain when the volume of information increases.

Digital MIS frameworks provide an alternative by storing records in structured databases and presenting information through user interfaces. Laudon and Laudon emphasize the role of information systems in collecting, processing, storing, and distributing information for organizational activities [9].

For vocationaltraining institutions, this can involve learner information, training records, instructor information, assessment results, and administrative reports.

Grepon et al. demonstrated the use of an information systems approach in designing and implementing an eschool system for a community college in Northern Mindanao, Philippines [3]. Their study provides a local example of how information systems can be used to centralize schoolmanagement processes.

The relevance to DigiTrack is the concept of centralized institutional information. Although a driving school differs from a college, both environments involve users, records, schedules, and administrative processes that benefit from organized information management.

A web based system also provides accessibility advantages. Authorized users can access the application using a browser rather than depending on a single physical workstation. This can be useful for administrators and instructors who need to view schedules or records from different devices.

However, accessibility must be accompanied by appropriate authentication and authorization. A system that makes information easier to access must also ensure that users are only permitted to access functions and records appropriate to their roles.

DigiTrack therefore uses rolebased access to separate administrative, instructor, and student functions. This structure supports the principle that information should be available to authorized users while limiting access to functions outside their responsibilities.

Digital transformation is more than the installation of a website or database. It involves reconsidering how information is captured, shared, maintained, and used in everyday work. In a vocational training environment, this may include replacing disconnected records with a common set of student, instructor, course, and training records; standardizing how appointments are entered; and making the status of a training activity visible to authorized users. The digital-transformation perspective described by Yang and Yang emphasizes changes in institutional processes and capabilities, making it relevant to the operational rather than purely technical goals of DigiTrack [10].

A web-based system can also support access from different workstations without requiring every user to maintain a separate local copy of the database. However, this advantage depends on stable hosting, appropriate account management, network access, and maintenance. A centralized system creates a common information source, but it also means that availability and security of the service become important operational concerns. The design must therefore balance convenient access with restrictions appropriate to each role. These requirements connect the digital-transformation discussion with the security and usability topics examined later in this chapter [13], [14], [18], [20].

For a small or localized organization, the appropriate level of digital transformation should be proportionate to its resources and needs. A system that requires expensive hardware, complex infrastructure, or specialist administration may be difficult to sustain even if it offers advanced features. DigiTrack is framed as a web-based management tool that uses commonly available web technologies and a relational database. Its contribution is the integration of selected operational functions in a client-specific workflow, rather than the introduction of a novel artificial-intelligence technique or a large-scale enterprise platform.

2.3.2 RealTime Feedback Mechanisms in Psychomotor Skill Mastery

Driving is a psychomotor activity requiring learners to coordinate physical actions, perception, judgment, and decisionmaking. Feedback is an important part of learning because learners need information about their performance and areas requiring improvement.

Rehm et al. investigated a virtual driving instructor capable of assessing driving performance and compared its assessment with human experts [11]. Their work demonstrates the potential of technology to support the assessment and feedback of drivingrelated performance.

However, an automated drivingperformance assessment system is substantially different from DigiTrack. DigiTrack does not analyze vehicle movement, steering, braking, road position, or other sensorbased driving behaviors.

Instead, DigiTrack provides administrative feedback mechanisms. Students can provide ratings and feedback concerning instructors after training activities. These records can then be reviewed by administrators.

This distinction is important because feedback about the instructor experience should not be represented as a direct measurement of the learner's driving competency.

The feedback feature can nevertheless support organizational monitoring. Repeated ratings or comments may help administrators identify areas that require attention or further review. The system therefore provides a structured location for information that might otherwise remain in informal communication.

The system's feedback mechanism can also support accountability. Because feedback is stored as part of the system's records, administrators have a more organized basis for reviewing instructorrelated concerns.

However, the system does not automatically determine disciplinary action or instructor competency based solely on student ratings. Administrative personnel remain responsible for interpreting the feedback within the school's policies.

2.3.3 Cybersecurity And Data Protection In Web-Based Training Systems

Security is particularly important in web-based educational management systems because these systems process personal information. DigiTrack stores information relating to students, instructors, administrators, schedules, feedback, and training-session records. Unauthorized access to such information could compromise confidentiality and organizational trust.

The security literature emphasizes that application security should be considered throughout system design and testing rather than added only after development. NIST's security testing guidance identifies activities such as vulnerability assessment, security testing, and evaluation of application controls as important components of information-system security [13].

2.3.3.1 OWASP Top 10 Web Application Risks

The OWASP Top 10 is a widely used awareness framework for identifying major web-application security risks. The current 2025 edition identifies risks including Broken Access Control, Security Misconfiguration, Software Supply Chain Failures, Cryptographic Failures, Injection, Insecure Design, Authentication Failures, Software or Data Integrity Failures, Security Logging and Alerting Failures, and Mishandling of Exceptional Conditions [14].

Broken access control is especially relevant to DigiTrack because the system has multiple user roles. A student should not be able to access administrative functions simply by changing a URL or manipulating a request. OWASP emphasizes that access control should be enforced on the server side and that applications should deny access by default unless a user has the necessary authorization [15].

Authentication is another important consideration. Web applications must properly manage login credentials, sessions, and authentication processes. OWASP identifies weak authentication and session-management practices as potential security risks [16].

Injection attacks are also relevant to database-driven systems. Applications that directly combine untrusted input with database queries can become vulnerable to SQL injection. Therefore, database queries should use appropriate parameterization and validation techniques.

Security logging is another concern because important actions may need to be traceable. OWASP identifies insufficient logging and monitoring as risks because organizations may be unable to detect or investigate suspicious activities when important events are not recorded [17].

The OWASP framework does not mean that DigiTrack automatically complies with every OWASP recommendation. Instead, it provides a security reference that can be used to identify areas that should be evaluated during development and testing.

2.3.3.2 Data Privacy Act of 2012

The Philippine Data Privacy Act of 2012, Republic Act No. 10173, establishes requirements for protecting personal information. Section 20 requires personal information controllers to implement reasonable and appropriate organizational, physical, and technical measures to protect personal information from accidental or unlawful destruction, alteration, disclosure, and other unlawful processing [18].

The implementing rules further state that security measures should maintain the availability, integrity, and confidentiality of personal data [19].

These principles are relevant to DigiTrack because the system processes personal information belonging to students, instructors, and administrators. Information should therefore only be collected and processed for legitimate purposes and should be protected against unauthorized access.

DigiTrack can support these requirements through authentication, role-based access, password protection, controlled database access, and security-related records. However, the presence of these technical controls does not by itself guarantee legal compliance. Compliance also depends on organizational policies, proper data handling, retention practices, privacy notices, access procedures, and other responsibilities of the organization.

2.3.3.3 Role-Based Access Control

Role-Based Access Control (RBAC) is a security model in which permissions are associated with organizational roles rather than individually assigned to every user. NIST explains that RBAC allows users to receive permissions through the roles to which they belong, simplifying authorization management [20].

RBAC is directly applicable to DigiTrack because the system contains three primary roles: administrator, instructor, and student.

The administrator requires broader permissions because the role involves managing instructors, students, schedules, courses, and monitoring information. Instructors require access to their assigned schedules and relevant training information but do not require unrestricted administrative permissions. Students require access to their own appointments, training information, and feedback functions.

This separation follows the principle that users should only receive permissions necessary for their responsibilities. NIST literature also discusses role hierarchies and constraints as mechanisms for managing organizational permissions [21].

For DigiTrack, RBAC therefore provides both a security mechanism and an organizational model. The roles reflect the responsibilities of users within Motosafe Driving School.

2.3.3.4 Authentication, Password Protection, and Input Validation

Authentication verifies a user's identity before the system provides access to protected resources. Secure authentication is especially important for administrative systems because unauthorized access can expose sensitive information.

Passwords should not be stored as plain text. Password hashing provides a mechanism for storing passwords in a form that is not directly reversible. DigiTrack uses password protection as part of its account-management process.

Input validation is another security mechanism. User-submitted information should be checked before being processed by the application. This is important for preventing invalid information and reducing risks associated with malicious input.

Server-side validation is particularly important because client-side validation can be bypassed. Therefore, the backend should independently verify submitted values before processing database operations.

2.3.3.5 Security Testing and Monitoring

Security testing should be integrated into the development and evaluation process. NIST SP 800-115 describes technical information-security testing and assessment as activities that can help organizations identify vulnerabilities and evaluate security controls [13].

For DigiTrack, security evaluation may include testing authentication, access control, input validation, database security, session management, and error handling. The purpose is to identify weaknesses before deployment or during maintenance.

Security testing is not a one-time activity. Web applications may change over time as new features, libraries, dependencies, and configurations are introduced. Therefore, security should remain part of system maintenance.

Security requirements should be translated into observable behaviors that can be checked. For authentication, testing can verify that valid credentials permit access, invalid credentials are rejected, and protected pages are not accessible merely by entering their URLs. For authorization, tests should verify that each role can reach the functions required for its responsibilities but cannot perform actions reserved for another role. For input validation, test cases can include missing values, unexpected characters, excessively long inputs, and invalid identifiers. These checks do not prove that a system is completely secure, but they provide a structured way to identify weaknesses relevant to the implemented features [13], [14].

Privacy protection is related to, but not identical with, application security. Security controls help prevent unauthorized access or alteration, while privacy practices also concern the purpose for which personal information is collected, how long it is retained, who may use it, and how requests concerning the information are handled. The Data Privacy Act and its implementing rules provide the applicable Philippine legal context for personal-information processing [18], [19]. For DigiTrack, practical safeguards include limiting access by role, collecting only information needed for the stated functions, protecting account credentials, and ensuring that operational records are visible only to authorized users.

A role-based permission structure is useful because the three main user groups do not need the same information or actions. Administrators may need to manage users and review school-wide records; instructors need access to their assignments and relevant training information; and students need access to their own schedules or progress information. This principle follows the logic of role-based access control, in which permissions are associated with roles and users obtain permissions through their assigned roles [20], [21]. It is important that the role structure be enforced on the server side as well as reflected in the interface, because hiding a menu item alone does not prevent a user from attempting to access a protected function directly.

Security monitoring and logging should be described according to what the implementation actually records. If the application stores authentication events or important administrative changes, those records may assist in reviewing suspicious activity or investigating mistakes. If such logging is not implemented, the manuscript should not imply that comprehensive monitoring is available. The same rule applies to encryption, backup, recovery, and vulnerability scanning: these should be reported as implemented controls only when they exist and have been checked. The OWASP Top 10 is a risk-awareness resource, not a certificate of security, so using it as a reference does not by itself establish that an application meets every security requirement [14].

2.3.4 Human-Computer Interaction And Usability Engineering

Human-computer interaction concerns the relationship between people and interactive computer systems. For DigiTrack, HCI is important because the system is used by people with different responsibilities, goals, and levels of technical experience.

An administrator may perform several tasks in a single session, such as checking appointments, assigning instructors, reviewing feedback, and examining records. An instructor may primarily need to view schedules and training assignments. A student may mainly need to check appointments and provide feedback.

These differences mean that the interface should present information according to the user's role and task.

2.3.4.1 Usability Engineering

Usability refers to the extent to which specified users can achieve specified goals effectively, efficiently, and satisfactorily within a particular context of use [22].

Effectiveness concerns whether users can successfully complete their intended tasks. Efficiency concerns the resources required to complete those tasks. Satisfaction concerns the user's response to the experience of using the system.

These dimensions are relevant to DigiTrack. For example, an administrator should be able to create or review an appointment without unnecessary steps. An instructor should be able to find assigned sessions without searching through unrelated records. A student should be able to view a schedule without needing extensive technical knowledge.

Usability therefore becomes part of system quality rather than simply an interface-design concern.

2.3.4.2 User Experience Frameworks

User experience includes users' perceptions and responses resulting from actual or anticipated use of a system [23].

A positive user experience depends on more than visual appearance. Users must understand what actions are available, what the system is currently doing, whether an action was successful, and how to recover from mistakes.

For DigiTrack, user experience can be improved through consistent navigation, clear labels, understandable messages, predictable interactions, and role-specific dashboards.

2.3.4.3 User Interface Design Principles

Nielsen's usability heuristics provide widely used principles for evaluating interfaces. These include visibility of system status, matching the system with the real world, user control, consistency, error prevention, recognition rather than recall, flexibility, minimalist design, error recovery, and help/documentation [24].

These principles can be directly applied to DigiTrack. For example, when an administrator creates an appointment, the system should provide clear feedback about whether the appointment was successfully saved or whether a conflict exists. This corresponds to visibility of system status.

Consistency is also important. Buttons, labels, navigation menus, and terminology should remain consistent across administrator, instructor, and student interfaces.

Error prevention is particularly relevant to scheduling. Instead of allowing users to create overlapping appointments and displaying an error only afterward, the system can check scheduling conditions before confirming an appointment.

2.3.4.4 Administrative Dashboard Design

Administrative dashboards are particularly important because administrators must process large amounts of operational information. A dashboard should prioritize information according to the decisions that users need to make.

DigiTrack's administrative dashboard includes information related to instructor performance ratings, lesson completion, attendance, and student feedback. These elements provide an overview of operational information without requiring administrators to open every individual record.

Effective dashboard design requires information hierarchy. Frequently needed information should be easy to locate, while secondary information can be placed in supporting views. Excessive visual elements can make dashboards difficult to interpret.

The dashboard should therefore focus on clarity rather than simply displaying as many statistics as possible.

2.3.4.5 Usability and DigiTrack

Usability is particularly relevant to DigiTrack because the system serves three different user groups. Each group interacts with the system differently, so the interface should reflect their responsibilities.

The administrator interface should emphasize management and monitoring. The instructor interface should emphasize schedules and training assignments. The student interface should emphasize appointments, training information, and feedback.

This role-oriented approach reduces unnecessary information and can make the system easier to understand.

The use of usability principles is also consistent with the evaluation of DigiTrack through the System Usability Scale in Chapter III. The usability evaluation provides a way to assess whether users perceive the system as understandable and manageable.

Usability evaluation should focus on representative tasks rather than asking only whether a user likes the interface. For an administrator, representative tasks may include locating an instructor, creating or modifying an appointment, checking a conflict, and reviewing a training record. For an instructor, tasks may include viewing an assigned schedule and finding relevant student or session information. For a student, tasks may include signing in and locating their own schedule or training information. Task-based evaluation helps identify where users hesitate, enter incorrect data, overlook a status message, or need assistance. The findings should be recorded in relation to the task and the interface element involved, rather than generalized from one comment.

A dashboard should prioritize information according to the decisions users need to make. A large number of charts does not necessarily make a dashboard useful. Indicators should have clear labels, defined meanings, and a reasonable connection to the underlying records. For example, a count of completed lessons should be based on records marked as completed under the school's process, while a feedback summary should indicate the records and period included. If the underlying data are incomplete or not updated, the dashboard should not be interpreted as a complete representation of school performance. Clear definitions and visible context help prevent users from drawing conclusions that the data cannot support.

Consistency across screens is also important. Similar actions should use familiar labels, related records should be presented in predictable locations, and validation messages should explain what needs to be corrected. A confirmation message should distinguish between an action that was saved and an action that still requires approval. Where a user can cancel, edit, or return to a previous screen, the interface should make these options

understandable. These principles are consistent with usability standards and established interface heuristics [22], [24]. They are particularly relevant to administrative systems because errors may affect several related records, not just the current screen.

Accessibility and practical use conditions should also be considered. Users may access the system from devices with different screen sizes, varying network conditions, or differing levels of digital confidence. A responsive layout, readable text, recognizable controls, and straightforward navigation can help reduce avoidable interaction difficulties. At the same time, the manuscript should avoid claiming that the system is fully accessible to all users unless it has been evaluated against a defined accessibility standard. In DigiTrack, usability claims should be limited to the design decisions made and the evaluation evidence actually collected.

2.3.5 Synthesis

Digital transformation can improve the organization of driving-school information, but digital systems also introduce responsibilities involving security, privacy, usability, and access control. The literature indicates that centralized systems should not only store information but should protect it and present it in ways that support users' tasks.

For DigiTrack, OWASP provides a reference for web-application security, the Data Privacy Act provides the Philippine legal context for personal-data protection, RBAC provides an access-control model, and HCI principles provide guidance for interface design.

Database design supports these requirements by defining the main entities, their attributes, and the relationships among them. Student information, instructor profiles, courses, appointments, training sessions, attendance, and feedback should be represented in a way that minimizes unnecessary duplication and supports accurate retrieval. If the same fact is stored independently in several places, updates can leave the records inconsistent. A relational database can reduce this risk by organizing related information into tables and connecting them through keys, although good outcomes still depend on appropriate schema design, validation, and transaction handling. The literature on information systems supports the role of structured data in organizational information management [9].

The system's reports and dashboards should be derived from clearly defined data. Before presenting a total, percentage, or status, the development team should identify which records are included and what event causes a record to be counted. For instance, a lesson should not be counted as completed merely because an appointment exists; the system must use the actual completion status or other defined confirmation. Similarly, a feedback summary should make clear whether it reflects all submitted responses or a selected period. Explicit definitions help administrators interpret indicators consistently and make it easier to test whether the displayed results match the underlying records.

2.4 RELATED SYSTEMS

Related systems provide practical evidence of how technology has been applied to driving-school management and educational administration. Reviewing these systems allows the study to identify existing approaches, common features, limitations, and areas that remain insufficiently addressed.

The systems reviewed in this section include academic prototypes and commercial or organizational platforms. Because commercial platforms do not always publicly disclose their internal architecture, technical characteristics are discussed only when supported by available documentation.

2.4.1 Advanced Management System for Car Driving Schools

Kannan et al. developed an Advanced Management System for Car Driving Schools [6]. The system focuses on managing driving-school activities through digital features for students, instructors, and administrators.

The system provides functions for student registration, package selection, instructor-related information, booking, and scheduling. The tutor module allows instructors to view enrolled students and booking information. Administrative functions support the management of schedules and user information.

From an operational perspective, the system demonstrates the usefulness of centralizing driving-school information. Instead of requiring instructors to coordinate all student information manually, the system gives users access to information through dedicated modules.

However, the available publication does not provide enough information to establish a complete modern technology-stack description for every component. Therefore, it would be inappropriate to assume a particular framework or database technology without explicit evidence.

The system also focuses strongly on scheduling and management functions. Its approach provides a useful foundation for DigiTrack but does not fully correspond to the localized regulatory, performance-monitoring, feedback, and administrative requirements of Motosafe.

Comparative Gap with DigiTrack

Kannan et al.'s system demonstrates that digital scheduling and instructor management are feasible in driving-school environments. DigiTrack extends the management perspective by incorporating instructor performance monitoring, student feedback, training-session monitoring, training-completion records, notifications, and role-specific functions.

The difference is therefore not simply the presence or absence of scheduling. It is the broader integration of scheduling with performance and operational monitoring.

2.4.2 AI- Driving School Management Systems

The article by Davanhipprage and Jagtap describes a web-based driving-school management project with functions related to registration, scheduling, payments, student records, and administration, and labels the project as AI-driven [7]. However, the paper does not provide strong independent validation that advanced AI methods were implemented or evaluated. It is therefore treated here as a proposed system description rather than evidence of demonstrated AI performance.

The system allows customers to select available time slots and provides notifications regarding upcoming sessions. It also provides instructors with access to assigned students and schedules.

One important contribution of this system is its emphasis on reducing manual coordination. Automated booking can reduce repeated communication between students, instructors, and administrative staff.

However, the presence of automated scheduling should not automatically be interpreted as mathematical optimization. The degree of optimization depends on the algorithms and constraints implemented by the system.

Comparative Gap with DigiTrack

Davanhipprage and Jagtap describe automation and scheduling functions. DigiTrack similarly supports scheduling but places additional emphasis on instructor performance monitoring, student feedback, training-session monitoring, and localized operational management.

The systems therefore overlap in scheduling functionality but differ in their overall management scope.

2.4.3 TrainMe: A Web-Based Management System for Driving School

Mithrasena developed TrainMe as a web-based management system for driving schools [8]. The system provides functions related to student enrollment, course selection, instructor assignment, and scheduling.

TrainMe demonstrates the value of browser-based access for driving-school operations. A web-based system can allow users to access information from different devices without requiring a separate desktop application.

The system also demonstrates the usefulness of role-based interaction because different users require different types of information.

However, TrainMe was developed within a different institutional and regulatory context. Its design therefore cannot automatically be assumed to satisfy the operational requirements of Philippine driving schools.

Comparative Gap with DigiTrack

DigiTrack applies the web-based management concept to Motosafe Driving School and incorporates functions relevant to the Philippine context. It also adds emphasis on instructor performance ratings, student feedback, training-session monitoring, training-completion records, and administrative reporting.

TrainMe therefore provides evidence supporting the feasibility of a web-based driving-school management system, while DigiTrack adapts the concept to a different operational environment.

2.4.4 DriveHub

DriveHub is an academic capstone project that proposes a multi-platform driving-school management system with a driving-school finder [25].

The system is relevant because it demonstrates an approach that combines driving-school management with student-facing discovery functions. Its multi-platform orientation illustrates how driving-school services can be made accessible to users through digital interfaces.

However, the driving-school finder component represents a different emphasis from DigiTrack. DigiTrack is designed specifically around the internal management of one driving-school organization rather than a broader marketplace or school-finder platform.

Comparative Gap with DigiTrack

The distinction between the two systems demonstrates the importance of defining system scope. DriveHub addresses discovery and multi-platform access, while DigiTrack concentrates on instructor scheduling, performance monitoring, training management, and administrative coordination within Motosafe.

2.4.5 Centralized Driving School Monitoring System

Kulshrestha et al. proposed a centralized driving-school monitoring system [26]. The work is relevant because it emphasizes centralized monitoring rather than relying entirely on separate records.

Centralized monitoring can provide administrators with improved access to information and can help organize operational activities. This is consistent with DigiTrack's objective of providing a central source of information for instructors, students, schedules, sessions, and feedback.

However, the exact technical architecture and implementation details should only be attributed to the study when explicitly documented. General assumptions about cloud architecture or audit logging should not be presented as facts without supporting evidence.

Comparative Gap with DigiTrack

DigiTrack combines centralized management with scheduling, instructor performance monitoring, student feedback, training-session tracking, training-completion records, and role-based access. Its primary

distinction is the integration of these functions around the specific operational needs of Motosafe Driving School.

2.4.6 Commercial Driving-School and Educational Management Platforms

Commercial platforms such as Drive Scout and other driving-school management products demonstrate that scheduling, student management, instructor management, and administrative functions can be delivered through software services [12].

Similarly, educational management systems demonstrate the broader use of centralized information systems in institutional administration.

However, commercial platforms may be designed for a broad international market and may include functions that are unnecessary for a smaller local driving school. They may also be configured around business models, regulations, and workflows different from those of Philippine driving schools.

DigiTrack is designed as a localized system for Motosafe. Its development therefore prioritizes the specific workflows identified during the study rather than attempting to reproduce every feature found in commercial enterprise platforms.

The systems reviewed differ in scope, intended users, and the level of evidence available. An academic prototype may demonstrate a proposed workflow without providing long-term evidence of operational performance. A commercial platform may advertise a broad feature set, but publicly available product descriptions do not independently establish how well each feature performs in a particular school. These differences mean that a comparison should consider both the features described and the type of source supporting the description. The review therefore treats system papers, theses, capstone projects, and commercial webpages as evidence of existing approaches, not as interchangeable proof of effectiveness.

A useful comparison for DigiTrack considers whether a system supports instructor and student records, scheduling, conflict checking, training-progress monitoring, feedback, role-specific access, and administrative reporting. It also considers whether the system is intended for a single school or supports discovery across multiple schools, whether specialized hardware is required, and whether the available source documents describe evaluation results. These dimensions are relevant because DigiTrack is designed for one client and focuses on internal scheduling and performance-monitoring support. A driving-school finder, for example, may be valuable for prospective customers but does not necessarily address the internal workload and record-management problem that motivated this study.

The review does not establish that DigiTrack is superior to every existing platform. Such a conclusion would require a systematic comparison using the same criteria, comparable data, and documented testing. Instead, the literature helps identify a specific design opportunity: combining core internal management functions in a localized web application, with attention to scheduling conflicts, role-specific information access, and instructor-related records. This narrower claim is more defensible than claiming that no comparable systems exist or that the proposed system resolves every operational problem.

A structured comparison also needs to distinguish feature availability from feature depth. A system may list scheduling as a feature without explaining whether it checks instructor and student conflicts, supports rescheduling, records cancellations, or maintains an audit trail of changes. Likewise, a platform may include feedback collection without describing how feedback is summarized or how access to feedback is controlled. Where the source does not provide these details, the review should mark them as unspecified rather than assume that the feature is equivalent to DigiTrack's intended workflow. This cautious comparison makes the research gap more transparent and avoids overstating differences between systems.

The selection of features for DigiTrack reflects the operational needs identified for the client and the resources available for the project. The system emphasizes the core functions that can be integrated and tested within the study: user-role management, instructor and student records, appointment scheduling, conflict checking, training

monitoring, feedback, and administrative summaries. Features that would require separate infrastructure, external data integration, or additional validation are outside the intended scope unless they are explicitly included in the implemented system. This bounded design helps keep the project manageable and provides a clearer basis for determining whether the system meets its requirements.

When interpreting the related systems, it is also necessary to distinguish between the stated objective of a project and its demonstrated result. A thesis or capstone report may describe intended functions and the development process, but the strength of its conclusions depends on whether it provides implementation details, test procedures, respondent information, and results that can be examined. A commercial website may describe a product's features but may not disclose the testing conditions or data behind performance claims. Therefore, the reviewed works are used primarily to understand design approaches and feature coverage. They are not treated as proof that a particular system eliminates scheduling errors, improves instructor performance, or guarantees better learning outcomes. This distinction is important for a fair comparison because the available evidence differs by source type. DigiTrack should be evaluated with the same caution: its functions can be described based on the implemented modules, while claims about usability or operational improvement should be tied to the test records and user feedback collected for this study. A feature comparison can identify how the system's scope differs from existing approaches, but a claim of superiority would require a direct evaluation using consistent criteria and comparable conditions.

2.5 RESEARCH GAP

The literature review identifies several areas where existing studies and systems do not completely address the requirements of the present study.

First, there is an operational integration gap. Existing systems demonstrate the feasibility of digital scheduling and driving-school management, but the reviewed systems differ in their combinations of scheduling, student management, instructor monitoring, feedback, and certification functions.

Second, there is a workload-visibility gap. Scheduling systems can show instructor assignments, but workload information is not always integrated with performance monitoring. DigiTrack addresses this by providing administrators with instructor-related information that can be reviewed alongside scheduling and performance records.

Third, there is a localization gap. Many reviewed systems were developed outside the specific operational and regulatory context of Motosafe Driving School. DigiTrack is designed around the requirements identified within the Philippine driving-school environment.

Fourth, there is a performance-monitoring gap. Some systems emphasize scheduling and booking, while DigiTrack incorporates instructor ratings and student feedback as part of the management process.

Fifth, there is a data-management gap. Manual processes may distribute information across separate records. DigiTrack provides centralized relational data storage to connect instructors, students, courses, schedules, appointments, sessions, and feedback.

Sixth, there is a usability gap. A system may provide many functions but still be difficult to use. DigiTrack therefore considers usability and user experience as part of system development and evaluation.

Finally, there is a security and privacy gap. A web-based system handling personal information must consider authentication, authorization, input validation, data protection, and security testing. DigiTrack incorporates these considerations as part of its system architecture and evaluation.

These gaps do not imply that existing systems are ineffective. Instead, they indicate that existing studies address different aspects of driving-school management. DigiTrack brings together selected functions into a localized system intended for Motosafe Driving School.

The gap is therefore practical and contextual rather than a claim that digital driving-school management has never been studied. Existing studies and products demonstrate that software can support enrollment, scheduling,

student records, and administrative workflows [6]–[8], [12], [25], [26]. However, the reviewed sources vary in focus, and not all give equal attention to internal workload visibility, conflict prevention, instructor performance-related records, and feedback within a single client-specific workflow. Some systems emphasize customer-facing discovery or booking, while others describe broader management functions without providing enough information to compare their implementation and evaluation in detail.

A second gap concerns the relationship between scheduling and performance monitoring. These functions are often connected operationally: schedules determine which activities are assigned, while completion records and feedback help administrators review what occurred. If the records are maintained separately, administrators may have to reconcile information manually before making a decision. DigiTrack addresses this relationship by organizing scheduling, training records, attendance, feedback, and performance-related information within a common application. The system does not assume that these data alone provide a complete or objective measure of instructor quality; rather, they provide organized inputs for administrative review.

A third gap is the need for a proportionate solution suited to a single driving school's workflow. Large or commercial platforms may provide functions beyond the immediate requirements of the client, while research prototypes may emphasize different priorities. DigiTrack focuses on the functions identified through the project requirements and uses a web-based architecture that does not depend on dedicated GPS or simulator hardware. This design choice is a scope decision, not evidence that hardware-supported systems are inherently less effective. It reflects the project's goal of supporting routine administrative work using the resources and information available to the client.

These gaps lead to the central contribution of the study: the design and development of a localized web-based system that integrates instructor scheduling, conflict checking, training-progress records, instructor attendance, student feedback, and administrative monitoring. Its contribution should be assessed through requirement coverage, correctness of implemented workflows, usability feedback, and the reliability of stored and displayed information. Claims about long-term productivity, improved road safety, or sustained workload fairness would require additional evidence beyond the implementation itself.

2.6 SYNTHESIS OF REVIEWED LITERATURE

The reviewed literature demonstrates that driving-school management involves several interconnected activities. Scheduling must coordinate students and instructors, while training records, feedback, and training-completion information must be maintained for administrative and instructional purposes.

Studies by Kannan et al. [6] and Davanhipprage and Jagtap [7] demonstrate the potential of digital scheduling and driving-school management systems. Mithrasena [8] demonstrates the feasibility of web-based driving-school management, while Grepon et al. [3] illustrates the use of information systems for educational administration.

The literature on workload indicates that workload should be considered when examining educational performance [5]. The literature on digital transformation demonstrates the growing role of information systems in vocational education [10]. Database architecture provides the technical foundation for organizing structured records, while HCI literature emphasizes the importance of effectiveness, efficiency, and satisfaction in system use [22].

Cybersecurity literature further demonstrates that web applications require protection against unauthorized access, authentication weaknesses, injection, and other risks [14]. Philippine privacy legislation establishes additional requirements for protecting personal information [18], while RBAC provides an approach for controlling access according to user roles [20].

Taken together, these findings support the development of a centralized web-based system for driving-school management. DigiTrack applies these concepts to Motosafe Driving School by integrating scheduling, instructor management, student management, training monitoring, feedback, performance monitoring, and centralized data management.

Across the reviewed literature, several principles converge. First, information systems can support administration when they centralize related records and make them available for appropriate decisions [3], [9],

[10]. Second, scheduling is a constrained coordination task because the availability of people and time slots must be considered together [2]. Third, workload-related information can be relevant to performance review, but it must be interpreted in context rather than reduced to one number [5]. Fourth, web-based applications need appropriate access controls, privacy safeguards, and security testing [13], [14], [18]–[21]. Finally, usability depends on whether intended users can complete their tasks effectively, efficiently, and satisfactorily in the context where the system is used [22], [24].

These principles are complementary. A technically functional system may still be difficult to use; a usable interface may still expose data to the wrong role; and a secure system may still provide little operational value if it does not match the client's workflow. DigiTrack therefore requires a combination of functional design, role-specific access, data organization, conflict checking, and user-oriented interface decisions. The literature supports treating these as related design requirements, while the system's actual success must be assessed using implementation and evaluation evidence rather than inferred from the presence of features alone.

The literature also establishes limits on the conclusions that can be drawn. Studies conducted in other educational settings provide useful concepts but may not directly predict outcomes in a driving school. Descriptions of existing platforms show that particular features are available or proposed, but they do not necessarily establish measurable improvements in efficiency. Security standards and guidelines identify risks and recommended practices, but citing them does not prove compliance. The present study uses the literature to justify design decisions and evaluation criteria; it does not treat the literature as a substitute for testing DigiTrack in its intended setting.

2.7 THEORETICAL FRAMEWORK

The theoretical framework provides the intellectual foundation for the development of DigiTrack. The study uses four major theoretical perspectives: Management Information Systems Theory, Resource Allocation Theory, Technology Acceptance Model, and Feedback and Performance Management concepts.

Each theory addresses a different aspect of the system. MIS Theory explains the role of information in organizational decision-making. Resource Allocation Theory explains the distribution of instructors and time. TAM explains user acceptance and the importance of usefulness and ease of use. Feedback and Performance Management concepts explain how performance information can be collected, reviewed, and used for organizational improvement.

2.7.1 Management Information Systems Theory

Management Information Systems Theory provides a foundation for understanding how organizations use information technology to support operations and managerial decisions. Laudon and Laudon describe information systems as systems that collect, process, store, and distribute information to support decision-making, coordination, control, analysis, and visualization [9].

The theory is directly relevant to DigiTrack because the system converts operational records into organized information. Raw records such as appointments, instructor assignments, student information, training sessions, and ratings have limited value if they cannot be easily retrieved or interpreted.

DigiTrack addresses this by centralizing records in a database. The system then provides interfaces through which users can interact with information according to their roles.

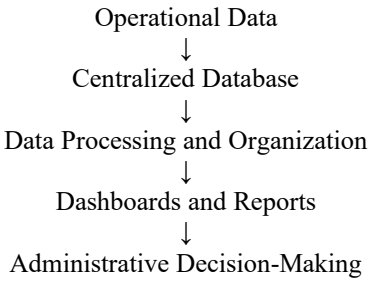
For administrators, MIS functions appear through dashboards, scheduling interfaces, instructor management, student management, and performance-monitoring functions. The administrator can use these features to review operational information without manually searching multiple records.

For instructors, the information system provides access to assigned training activities and schedules. This reduces the need to depend entirely on informal communication for schedule information.

For students, the system provides access to appointments, training information, and feedback functions.

The MIS perspective also emphasizes the relationship between information quality and decision-making. If incorrect or incomplete information is stored, the resulting reports may also be inaccurate. Therefore, DigiTrack requires appropriate data validation and organized database relationships.

MIS Theory and DigiTrack Mapping



The diagram demonstrates that DigiTrack functions as an information-processing system. Data generated from daily operations becomes information that can support administrative activities.

The theory therefore provides a foundation for the system's database, dashboards, reports, scheduling functions, and monitoring features.

2.7.2 Resource Allocation Theory

Resource Allocation Theory focuses on the distribution of limited resources according to organizational needs. Resources can include personnel, time, equipment, and other operational capacities.

In a driving school, instructors represent an important human resource. Their availability is limited, and each instructor can conduct only a certain number of training sessions during a given period. Students also have schedules and training requirements that must be accommodated.

Poor allocation may result in conflicts, underutilization, or excessive assignments. Digital scheduling can make resource information more visible and allow administrators to consider existing assignments when creating new schedules.

DigiTrack applies this perspective by maintaining instructor schedules and appointment records. Administrators can view existing assignments when coordinating new appointments.

The theory does not imply that DigiTrack produces a mathematically optimal schedule. Instead, it provides a conceptual explanation for why instructor availability and workload are relevant to scheduling.

The system's workload-related information can support administrative decisions by showing how assignments are distributed among instructors.

In DigiTrack, the resource-allocation perspective is applied through availability records and conflict rules rather than through a mathematical optimization engine. The relevant resources include instructor time and the time periods requested by students. A proposed assignment is valid only when the required participants are available and the assignment does not violate the scheduling rules implemented by the system. These checks can prevent certain overlapping bookings, but they do not guarantee a globally optimal timetable or mathematically equal workload. Administrators may still need to consider course requirements, individual circumstances, and school policies when deciding among valid alternatives.

Resource Allocation Theory and DigiTrack Mapping

Student Demand
+
Instructor Availability
+
Existing Assignments
+
Course Requirements
↓
Scheduling Decision
↓
Instructor Workload Visibility

This relationship shows how multiple operational factors influence scheduling.

Resource Allocation Theory therefore supports DigiTrack's scheduling and instructor-management functions.

2.7.3 Technology Acceptance Model

The Technology Acceptance Model (TAM), introduced by Davis, explains technology acceptance primarily through users' perceptions of usefulness and ease of use [27].

Perceived usefulness refers to the degree to which a person believes that using a system will improve job performance. Perceived ease of use refers to the degree to which a person believes that using the system will require little effort.

These concepts are highly relevant to DigiTrack because the system will only be useful if administrators, instructors, and students can understand and use its functions.

For administrators, perceived usefulness may be associated with easier scheduling, centralized records, performance monitoring, and reporting. For instructors, usefulness may involve easier access to schedules and assigned training activities. For students, usefulness may involve convenient access to appointments and feedback functions.

Ease of use is influenced by interface organization, navigation, labels, forms, messages, and consistency.

TAM and DigiTrack Mapping

Perceived Usefulness

- Centralized records
- Scheduling support
- Performance monitoring
- Feedback management
- Administrative reporting

Perceived Ease of Use

- Clear navigation
- Role-specific dashboards
- Simple forms
- Understandable labels
- Clear system feedback

↓

User Acceptance and Continued Use

The TAM perspective therefore explains why functionality alone is not sufficient. A system can contain many features but still experience poor acceptance if users find those features difficult to understand or operate.

This is one reason why usability evaluation is included in the study.

2.7.4 Feedback and Performance Management

Performance management involves monitoring performance-related information, providing feedback, and using the information to support improvement. In educational environments, performance information can include teaching activities, attendance, learner feedback, completed training activities, and other indicators.

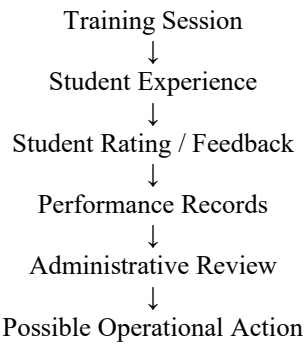
Jiao and Hernandez's study provides relevant evidence that workload and professional factors should be considered when examining teacher performance [5]. Although their participants were vocational college teachers rather than driving instructors, their findings support the broader idea that performance information should be interpreted within the context of workload.

DigiTrack applies this concept by collecting student ratings and feedback and organizing instructor-related information for administrative review.

The system does not automatically determine whether an instructor is competent. Instead, it provides information that administrators can use when reviewing instructor performance.

Feedback becomes more useful when the collection process is consistent and the interpretation is transparent. If students are asked to rate different instructors using different questions or at different stages of training, the resulting values may not be directly comparable. A system can help standardize the recording of feedback, but it cannot remove all response bias or ensure that every respondent represents the broader student population. Administrators should consider response counts, the period covered, and the context of comments before using feedback in a performance discussion. For DigiTrack, feedback is therefore an organized input for review rather than an automatic final score of instructor quality.

Feedback and Performance Management Mapping



This model emphasizes that feedback becomes useful when it is systematically collected and made available for review.

Feedback management also supports accountability. When feedback is stored consistently, administrators can examine records over time instead of relying entirely on informal comments.

However, feedback should not be treated as the only measure of instructor performance. Student ratings may reflect individual experiences and perceptions. Therefore, DigiTrack should present feedback as one

source of performance-related information alongside attendance, training-session records, workload information, and other available indicators.

2.7.5 Integration of the Supporting Theories

The four theoretical perspectives complement one another.

- MIS Theory explains how information is collected, processed, stored, and presented.
- Resource Allocation Theory explains why instructor availability, scheduling, and workload distribution are important.
- TAM explains why usefulness and ease of use influence the acceptance of the system.
- Feedback and Performance Management concepts explain how performance-related information can be collected and reviewed.

Together, the theories provide a comprehensive foundation for DigiTrack.

Integrated Theoretical Model

Driving-School Operational Data
↓
MIS Theory
Centralized collection and processing of information
↓
Resource Allocation Theory
Scheduling and workload considerations
↓
Feedback and Performance Management
Performance information and student feedback
↓
TAM
Usable and useful system for users
↓
DigiTrack
Instructor Scheduling and Performance Monitoring System

The integrated model demonstrates that DigiTrack is not based on one theoretical concept alone. Instead, each theory supports a particular dimension of the system.

2.8 CONCEPTUAL FRAMEWORK

The conceptual framework describes how the inputs of the study are transformed through system processes into expected outputs.

The framework follows the Input-Process-Output approach.

Input

The input consists of the information required by DigiTrack to perform its functions. This includes:

- Student information
- Instructor information
- Course information
- Instructor availability
- Student appointments
- Course schedules
- Training-session information
- Instructor attendance

- Student ratings and feedback
- Certification information
- User-account information
- Operational requirements of Motosafe Driving School

The quality of the output depends on the quality of the input. Inaccurate or incomplete information may produce inaccurate records or scheduling results.

Process

The process stage includes the activities performed by the system.

These include:

- User authentication
- Role-based access control
- Instructor management
- Student management
- Course management
- Schedule creation
- Appointment management
- Conflict checking
- Training-session recording
- Attendance monitoring
- Feedback collection
- Instructor performance monitoring
- Certification-record management
- Notification and messaging functions
- Database storage and retrieval
- Administrative reporting

These processes transform operational records into organized information that can be accessed by authorized users.

Output

The output consists of the information and services produced by the system.

These include:

- Organized instructor schedules
- Student appointment information
- Training-session records
- Instructor workload information
- Instructor performance records
- Student feedback records
- Certification records
- Administrative reports
- Notifications
- Centralized and accessible operational information

The output is intended to support the daily management of Motosafe Driving School.

2.8.1 Conceptual Relationship of the Study

The conceptual framework can be represented as:

INPUT

Student Data
Instructor Data
Course Data
Schedule Data
Appointment Data
Training Records
Feedback
Attendance
Certification Records
Operational Requirements

↓

PROCESS

Authentication
Role-Based Access
Scheduling
Conflict Checking
Training Monitoring
Feedback Collection
Performance Monitoring
Database Processing
Reporting

↓

OUTPUT

Organized Schedules
Instructor Workload Information
Training Records
Performance Information
Student Feedback
Certification Records
Administrative Reports
Centralized Information

The framework demonstrates how DigiTrack transforms operational data into information that can support driving-school management.

2.8.2 Overall Synthesis of the Theoretical and Conceptual Framework

The theoretical and conceptual frameworks collectively establish the foundation of DigiTrack. MIS Theory explains the importance of centralized information management, while Resource Allocation Theory provides a basis for considering instructor availability and workload during scheduling.

The Technology Acceptance Model emphasizes that users must perceive the system as useful and easy to use. This supports the study's attention to interface design and usability evaluation. Feedback and Performance Management concepts provide the foundation for recording student feedback and organizing instructor performance-related information.

The security perspective further establishes the importance of authentication, authorization, data protection, and role-based access for a web-based system handling personal information. HCI principles complement these security and functional requirements by ensuring that the system remains understandable and manageable for its intended users.

DigiTrack therefore brings together these concepts in one localized information system. The system does not claim to replace administrative judgment or guarantee optimal scheduling. Instead, it provides centralized information, scheduling support, monitoring functions, and feedback mechanisms that can assist Motosafe Driving School in managing its daily operations.

The conceptual framework ultimately reflects the relationship between operational information, digital processing, and management outputs. Accurate data are collected and stored, system processes organize the information, and the resulting outputs are presented to authorized users through role-specific interfaces. Through this structure, DigiTrack provides a technological foundation for improving the organization and accessibility of driving-school management information.

The framework also clarifies the difference between system outputs and organizational outcomes. Organized schedules, accessible records, conflict alerts, and summaries of feedback are outputs that can be observed within the application. Faster administration, fairer workload distribution, higher satisfaction, or improved instructional quality are broader outcomes that depend on user behavior, management practices, data accuracy, and conditions outside the software. The study can assess whether DigiTrack provides the intended outputs and can gather user feedback about its use, but stronger claims about sustained outcomes would require a longer evaluation and appropriate comparative evidence.

The relationship among inputs, processes, and outputs also depends on data quality. Incorrect availability information can lead to an inaccurate conflict check; incomplete training records can distort a completion summary; and missing or unrepresentative feedback can affect a performance report. For this reason, the conceptual framework assumes that users enter accurate information, that authorized staff update records when circumstances change, and that the system applies its validation and access rules consistently. These assumptions should be considered when interpreting the evaluation and when recommending future improvements to the system.

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CHAPTER III
METHODOLOGY

This chapter presents the methodology used in the design, development, testing, and evaluation of DigiTrack: An Instructor Scheduling and Performance Monitoring System for Motosafe Driving School. The chapter describes the research design, software development methodology, system architecture, technologies and development tools, data gathering techniques, testing procedures, and evaluation criteria used in the study.

The methodology was selected based on the nature and objectives of the study. DigiTrack is not limited to describing an existing operational problem. The study also involves the development of a functional web-based information system intended to address identified operational concerns of Motosafe Driving School. These concerns include instructor scheduling, instructor and student record management, training-session monitoring, performance feedback, certification-related record monitoring, and centralized access to operational information.

The study adopted a Research and Development (R&D) or design-and-development approach because the major output of the study is a technological product developed to address a specific organizational need. Design-and-development research focuses on systematic processes involving the design, development, and evaluation of products, tools, and related procedures [28]. In DigiTrack, the research component involved examining the existing processes and requirements of Motosafe Driving School, while the development component involved translating those requirements into a functional web-based system.

The software development component followed the Agile Scrum Model. Scrum was selected because the development of DigiTrack required repeated refinement of requirements, system features, interface designs, database structures, and user workflows. Instead of completing the entire system through one long development cycle, the team organized development into smaller iterations that allowed completed features to be reviewed and improved throughout the project [29].

The methodology also incorporated software engineering practices related to system architecture, database management, security, usability, testing, and evaluation. Security considerations were particularly important because DigiTrack handles user accounts, student information, instructor information, appointments, training records, ratings, and other operational data. The system therefore considered access control, authentication, input validation, data protection, and secure handling of personal information. These considerations are consistent with recognized application-security principles and the requirements of the Philippine Data Privacy Act of 2012 [18], [19].

The chapter is organized into seven major sections. Section 3.1 discusses the research design; Section 3.2 presents the software development methodology; Section 3.3 explains the system architecture; Section 3.4 identifies the tools and technologies used; Section 3.5 describes the data gathering techniques; Section 3.6 discusses the testing procedures; and Section 3.7 presents the evaluation criteria used to assess DigiTrack.

3.1 RESEARCH DESIGN

The study employed a Research and Development (R&D) design, specifically a design-and-development approach, because the primary purpose of the project was to develop and evaluate a functional information system for a specific organizational context. Design-and-development research involves the systematic study of design, development, and evaluation processes with the purpose of creating or improving products and tools [28].

The selected research design is appropriate for DigiTrack because the study begins with an identified operational problem and proceeds toward the creation of a technological solution. The researchers first examined the existing processes of Motosafe Driving School, identified operational requirements, analyzed the

difficulties encountered by users, and translated the findings into system requirements. These requirements then served as the basis for the design, implementation, testing, and evaluation of the proposed system.

The research design therefore combines two closely related components: descriptive analysis of the existing situation and development and evaluation of the proposed system. The descriptive component provides an understanding of the current processes, while the developmental component converts the identified requirements into a working software solution.

3.1.1 Understanding the Research and Development Design

Research and Development is suitable for studies where the researcher is expected not only to investigate a problem but also to develop a product, tool, process, or technological solution in response to that problem. Richey and Klein describe design-and-development research as a systematic study of design, development, and evaluation processes intended to establish an empirical basis for products, tools, and development models [28].

In the context of DigiTrack, the R&D design consists of several connected activities.

First, the researchers identified and examined the operational concerns of Motosafe Driving School. The study focused on activities involving administrators, instructors, and students. Particular attention was given to instructor scheduling, student appointments, training-session records, instructor monitoring, feedback, and the management of school-related information.

Second, the researchers gathered information from stakeholders and existing operational procedures. Interviews, observations, questionnaires, and discussions with stakeholders provided information about the current workflow and the requirements that the system needed to support.

Third, the information collected during the research stage was translated into functional and non-functional requirements. Functional requirements describe what the system should be able to perform, such as user authentication, instructor management, student management, scheduling, appointment management, session monitoring, ratings, and reporting. Non-functional requirements describe qualities expected from the system, such as usability, reliability, security, and maintainability.

Fourth, the researchers designed and developed DigiTrack based on the identified requirements. The system was organized as a web-based application with separate access privileges for administrators, instructors, and students.

Finally, the developed system was subjected to testing and evaluation. Testing focused on determining whether individual functions and interconnected modules worked according to their requirements. Evaluation focused on broader characteristics such as functionality, usability, reliability, and security.

The R&D design therefore establishes a connection between the original problem, the requirements identified from stakeholders, the developed system, and the evaluation of the resulting product.

3.1.2 Justification for Choosing the R&D Design

The R&D design was selected because the main objective of the study is to develop and evaluate DigiTrack, rather than merely describe or analyze the existing scheduling practices of Motosafe Driving School.

The first reason for selecting R&D is its alignment with the purpose of the study. The research begins with an actual organizational problem and produces a functional technological output. This relationship between problem identification, product development, and evaluation is consistent with the purpose of design-and-development research [28].

The second reason is that the study requires direct consideration of the actual users of the proposed system. Administrators, instructors, and students have different responsibilities and information requirements. The administrator needs access to management and monitoring functions, instructors need access to schedules

and training-related information, and students need access to appointments, training information, and feedback functions. The R&D approach allows these user requirements to be considered during the design and development process.

The third reason is the need for iterative refinement. System development rarely proceeds perfectly from the first design. During development, users may identify missing requirements, confusing interfaces, unnecessary steps, or changes in workflow. An R&D approach combined with Agile Scrum allows these findings to be incorporated into subsequent development activities.

The fourth reason is that the project requires evaluation of the developed system. The researchers do not consider system development complete simply because the software has been coded. The system must also be tested to determine whether its functions operate correctly and evaluated to determine whether it satisfies the intended requirements.

The fifth reason is the context-specific nature of the project. DigiTrack is designed specifically for Motosafe Driving School. Therefore, the study focuses on understanding the school's particular processes and developing a system that fits those processes instead of assuming that the same workflow applies to every driving school.

For these reasons, R&D provides an appropriate methodological structure for connecting investigation, system development, testing, and evaluation.

3.1.3 Strengths of the R&D Design

The R&D design provides several strengths for the development of DigiTrack.

1. Practical application.
The most direct strength is that the study produces a functional technological product. The research findings are translated into an actual system rather than remaining only as theoretical observations.

2. Alignment with user requirements.
The design allows system requirements to be derived from the actual operational environment of Motosafe Driving School. This reduces the likelihood that system features will be based solely on assumptions made by the developers.

3. Iterative improvement.
The R&D approach supports repeated refinement. Problems identified during development and testing can be addressed before the final version of the system is completed.

4. Integration of research and development.
The approach connects research activities with software engineering activities. Information obtained through interviews and observations can directly influence system requirements, interface design, database design, and workflow development.

5. Evaluation of the resulting product.
The developed system can be systematically tested and evaluated based on predetermined criteria. This allows the researchers to provide evidence regarding the system's functionality, usability, reliability, and security.

6. Context-specific development.
Because the system is developed for Motosafe Driving School, its functions can reflect the school's actual operational requirements. This makes the development process more focused and allows the researchers to address specific problems identified during the study.

3.1.4 Limitations of the R&D Design

Despite its usefulness, the R&D design also has limitations.

First, the findings are primarily context-specific. DigiTrack was designed for Motosafe Driving School, so the results may not automatically apply to other driving schools with different organizational structures, policies, staffing levels, or workflows.

Second, system development requires substantial time and resources. Requirements gathering, interface design, database development, programming, debugging, testing, documentation, and revision require coordinated effort from the development team.

Third, stakeholder availability may affect the development process. Administrators, instructors, and students have their own schedules and responsibilities, which may limit the time available for interviews, demonstrations, testing, and feedback.

Fourth, changes in requirements may affect the development schedule. Additional requests introduced during development may require changes to existing database structures, interfaces, workflows, or code.

Fifth, the study period limits the ability to observe long-term effects. The project can evaluate the developed system during the study and testing period, but long-term organizational effects such as sustained changes in workload distribution, operational efficiency, or user behavior may require longer observation.

These limitations were considered when planning the development and evaluation procedures.

The R&D approach also requires the researchers to distinguish between the product-development record and the evidence used to evaluate the product. A working feature demonstrates that the corresponding code and workflow exist, but it does not by itself demonstrate that the feature is reliable under all conditions or that users find it easy to use. For this reason, the study links each major requirement to a corresponding verification activity. Functional requirements are checked through test cases, interface-related requirements are examined through task-based use and feedback, and security-related requirements are reviewed through checks appropriate to the implemented controls. The resulting conclusions should remain within the limits of those procedures.

The unit of development is the DigiTrack application and its connected modules, while the organizational context is Motosafe Driving School. The study is not designed as a controlled experiment comparing multiple driving schools or measuring road-safety outcomes. It is a context-specific system-development study that examines whether a web-based application can support identified internal processes. This boundary is important when generalizing the findings: other schools may have different course structures, staffing arrangements, approval rules, and recordkeeping requirements, and would need their own requirements analysis before adopting a similar system.

3.2 SYSTEM DEVELOPMENT METHODOLOGY

The development of DigiTrack followed the Agile Scrum Model. Scrum provides a framework for developing and managing products through iterative work cycles. The official Scrum Guide describes a Sprint as a fixed-length event of one month or less in which a usable product increment is created [29].

Agile Scrum was selected because DigiTrack required continuous coordination between system developers and stakeholders. The system contained several interconnected modules, including user management, instructor management, student management, scheduling, appointments, training sessions, performance monitoring, feedback, certification-related records, messaging, and notifications.

Rather than attempting to complete all modules at once, the development team organized the work into smaller increments. Each increment could be designed, implemented, tested, reviewed, and refined before the team proceeded with additional functionality.

The general Scrum-based development cycle for DigiTrack consisted of:

1. Requirement identification and backlog preparation
2. Sprint planning
3. System and interface design

4. Development and implementation
5. Testing
6. Sprint review and stakeholder feedback
7. Revision and refinement and
8. Preparation of the next development cycle

The cycle continued until the required system functionality had been implemented and evaluated.

3.2.1 Core Principles of Agile Scrum

The following principles guided the development of DigiTrack.

1. Iterative Development

The system was developed through repeated development cycles. Instead of treating the entire system as one large task, features were divided into smaller deliverable components.

For example, user authentication could be developed and tested before additional administrative features were completed. Similarly, instructor management and scheduling functions could be developed and evaluated before being connected with performance-monitoring functions.

This approach allowed errors to be identified earlier and reduced the possibility of discovering major problems only after the entire system had been completed.

2. Collaboration

Development required communication among team members and stakeholders. Team members coordinated responsibilities related to requirements analysis, database design, interface development, backend development, testing, and documentation.

Stakeholder collaboration was also important because Motosafe administrators and users provided information about actual school procedures. Their feedback helped the researchers determine whether the proposed workflows reflected operational needs.

3. Stakeholder Feedback

Stakeholder feedback was incorporated throughout development. Demonstrations and reviews allowed users to examine completed features and identify concerns.

For example, if a scheduling interface contained unnecessary steps or did not clearly display instructor availability, the issue could be documented and addressed in a subsequent iteration.

4. Responding to Change

Requirements can change as users gain a better understanding of the system. Agile development allows changes to be considered during subsequent iterations rather than requiring the entire project to be redesigned.

For DigiTrack, this was relevant to scheduling workflows, user permissions, interface organization, reporting requirements, and other system functions.

5. Working Software

Working system functionality was treated as an important indicator of development progress. Each completed feature needed to function according to its intended requirements before it could be considered part of a usable system increment.

6. Continuous Improvement

The development team used testing results and stakeholder feedback to identify areas that required improvement. Improvements could involve correcting functional errors, simplifying interfaces, improving validation, revising database relationships, or strengthening access control.

3.2.2 Agile Scrum Artifacts

Several Scrum artifacts were used to organize the development of DigiTrack.

1. Product Backlog

The product backlog contained the features, requirements, improvements, and issues identified for DigiTrack. The backlog served as the central list of work that could be prioritized during development.

Examples of backlog items included:

- administrator login
- instructor account management;
- student account management;
- instructor-course assignment;
- schedule management;
- student appointments;
- training-session recording;
- instructor ratings;
- student feedback;
- certification-related records;
- notifications;
- messaging
- performance monitoring and
- reporting

2. Sprint Backlog

The sprint backlog contained the selected tasks to be completed during a particular development cycle. Items were selected based on priority, dependencies, technical requirements, and available development time.

3. Increment

An increment represented completed and integrated system functionality produced during development. An increment was considered usable when the implemented features satisfied their defined requirements and had undergone the necessary testing.

4. Definition of Done

The Definition of Done established the conditions used by the team to determine whether a development task was completed.

For DigiTrack, an item could be considered complete when:

- the required functionality had been implemented;
- the code had been reviewed;
- the feature had been tested;
- identified critical errors had been addressed;
- the feature was integrated with related modules when required;
- the user interface was functioning properly; and
- the corresponding documentation had been updated

5. Sprint Review

Sprint reviews were used to examine completed features and gather feedback. The review provided an opportunity for stakeholders to identify concerns before additional features were developed.

6. Sprint Retrospective

The development team reviewed its own development process to identify what worked well, what problems occurred, and what could be improved during the next iteration.

3.2.3 Development Phases

Although Scrum does not require a traditional linear sequence of software-development phases, the DigiTrack project organized development activities into the following recurring work areas:

1. Requirement Analysis

The team identified and analyzed the needs of Motosafe Driving School. Information from interviews, observations, questionnaires, and stakeholder discussions was converted into functional and non-functional requirements.

Functional requirements included operations such as user authentication, schedule management, appointments, training-session records, ratings, and monitoring.

Non-functional requirements included usability, reliability, security, responsiveness, and maintainability.

2. System Design

The team translated requirements into technical designs. This stage included:

- database structure
- user interface layouts
- navigation structure
- role permissions
- system workflows
- data relationships
- process diagrams
- system architecture and
- security considerations

3. Development

The development stage involved writing and integrating the source code. Frontend components were developed for user interaction, while backend components handled business rules, authentication, validation, database transactions, and system processing.

4. Testing

Developed functions were tested to identify defects and verify that system requirements were satisfied. Testing included individual functions, module interactions, complete workflows, and user acceptance.

5. Deployment

Completed system components were prepared for use in the intended environment. Deployment activities included configuring the application, database, user accounts, and required system settings.

6. Refinement

Testing and stakeholder feedback were used to identify improvements. Corrections were implemented and tested again before the corresponding features were considered complete.

During requirement analysis, the development team identifies the users, the operational problem, the records maintained by the school, and the actions each role must perform. Requirements should be written in observable terms. For example, a scheduling requirement should state which information must be entered, which conflicts must be checked, and what the user sees when a conflict is detected. A role-permission requirement should specify which role can perform an action and what happens when an unauthorized user attempts it. Clear requirements make it easier to design the database, implement the workflow, and prepare test cases that can be repeated.

During system design, the requirements are translated into screen layouts, navigation structures, database entities, relationships, validation rules, and role permissions. The design stage should consider both the normal path and common exceptions, such as a missing field, invalid identifier, duplicate assignment, cancelled appointment, or unavailable instructor. Considering these conditions early can reduce inconsistent behavior between modules. The design should also identify which actions require an administrator's confirmation and which records are visible to instructors or students, because these decisions affect both usability and data protection.

During implementation, related functions are developed and connected incrementally. A module may first be tested in isolation and then tested again after integration with other modules. For instance, a schedule entry may be saved correctly in the scheduling module but still fail to appear in a training-progress view if the database relationship or retrieval query is incorrect. Integration testing is therefore necessary in addition to checking individual screens. Changes made during a sprint should be reviewed for unintended effects on existing functions, especially when they alter database fields, permissions, or shared navigation components.

During review and refinement, the team compares the implemented behavior with the requirement and records defects or necessary changes. Stakeholder feedback can identify differences between the developers' assumptions and the school's actual workflow. However, feedback should be documented accurately: a suggestion, a reported difficulty, a confirmed defect, and an approved requirement change are not the same type of evidence. Maintaining this distinction supports traceability and makes the development history easier to explain during evaluation. The Scrum approach provides the iterative structure for this work, while the actual scope of each iteration depends on the team's project plan [29].

3.3 SYSTEM ARCHITECTURE

DigiTrack uses a web-based client-server architecture in which users access the system through a web browser while application processing and database operations are handled by the server.

The architecture separates the user interface from application processing and data storage. This separation allows different responsibilities to be managed independently and supports centralized management of system records.

The architecture consists primarily of:

Client side;

- Server side;
- Database
- Data flow between components and
- Security controls

The system supports three primary user roles:

- Administrator

- Instructor
- Student

Each role is provided with functions appropriate to its responsibilities.

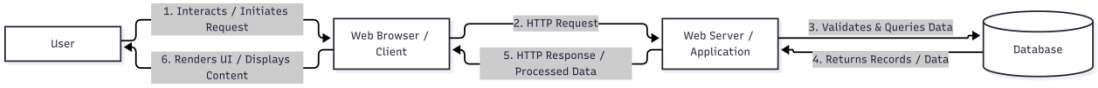
3.3.1 Overview of Client-Server Architecture

In the DigiTrack architecture, the client refers to the user's web browser and device. The client displays the user interface and sends requests to the server.

The server receives user requests, processes application logic, validates information, manages authentication and authorization, and communicates with the database.

The database provides centralized storage for system information.

The general interaction can be described as:



For example, when a student requests an available schedule, the browser sends a request to the application. The application checks the relevant records in the database and returns the appropriate schedule information to the browser.

When an administrator creates or modifies a schedule, the application validates the submitted information before storing the changes in the database.

This architecture allows the system to maintain a central source of information instead of requiring each user to maintain separate records.

3.3.2 Client Side

The client side represents the portion of DigiTrack that users directly interact with through a web browser.

The client-side interface provides:

- login forms;
- dashboards
- navigation menus
- forms for entering information
- schedule displays
- appointment interfaces
- feedback forms
- notifications
- tables and reports and
- other interactive interface components

The interface is designed according to the role of the logged-in user.

For example, an administrator has access to management and monitoring functions, while instructors are provided with functions related to their schedules and training activities. Students are provided with functions related to their accounts, appointments, training information, and feedback.

The separation of interfaces by role reduces unnecessary complexity and helps users access only the functions relevant to their responsibilities.

3.3.3 Server Side

The server side serves as the main processing component of DigiTrack.

The server handles:

- user authentication
- role verification
- authorization
- form processing
- input validation
- scheduling rules
- appointment processing
- training-session records
- ratings and feedback
- notifications
- database transactions and
- report generation

Database operations are handled by the server rather than allowing users to directly manipulate database records from their devices.

Where applicable, prepared statements and validated input are used to reduce the risk of injection attacks. Security controls are particularly important because injection and broken access control remain among the major categories identified in the OWASP Top 10 [14].

The server also applies role-based access restrictions. A user's ability to access a page or perform an operation should depend on the user's authenticated role and permissions rather than merely on whether the user knows the URL of a particular page.

3.3.4 Data Flow in the System

The general data flow in DigiTrack follows these steps:

Step 1: User interaction.
A user performs an action through the web interface, such as logging in, viewing a schedule, creating an appointment, recording a training session, or submitting feedback.

Step 2: Request transmission.
The browser sends the request to the web server.

Step 3: Authentication and authorization.
If the action requires an authenticated account, the server verifies the user's session and role.

Step 4: Input validation.
The system checks whether submitted information follows the required format and business rules.

Step 5: Business processing.
The application determines what operation should be performed.

Step 6: Database transaction.
The application retrieves, inserts, updates, or deletes the appropriate database records.

Step 7: Response generation.
The server prepares the result of the operation.

Step 8: Client display.
The browser displays the resulting information or an appropriate error or confirmation message.

This flow provides a centralized mechanism for controlling how information enters, moves through, and leaves the system.

3.3.5 Data Flow Diagram

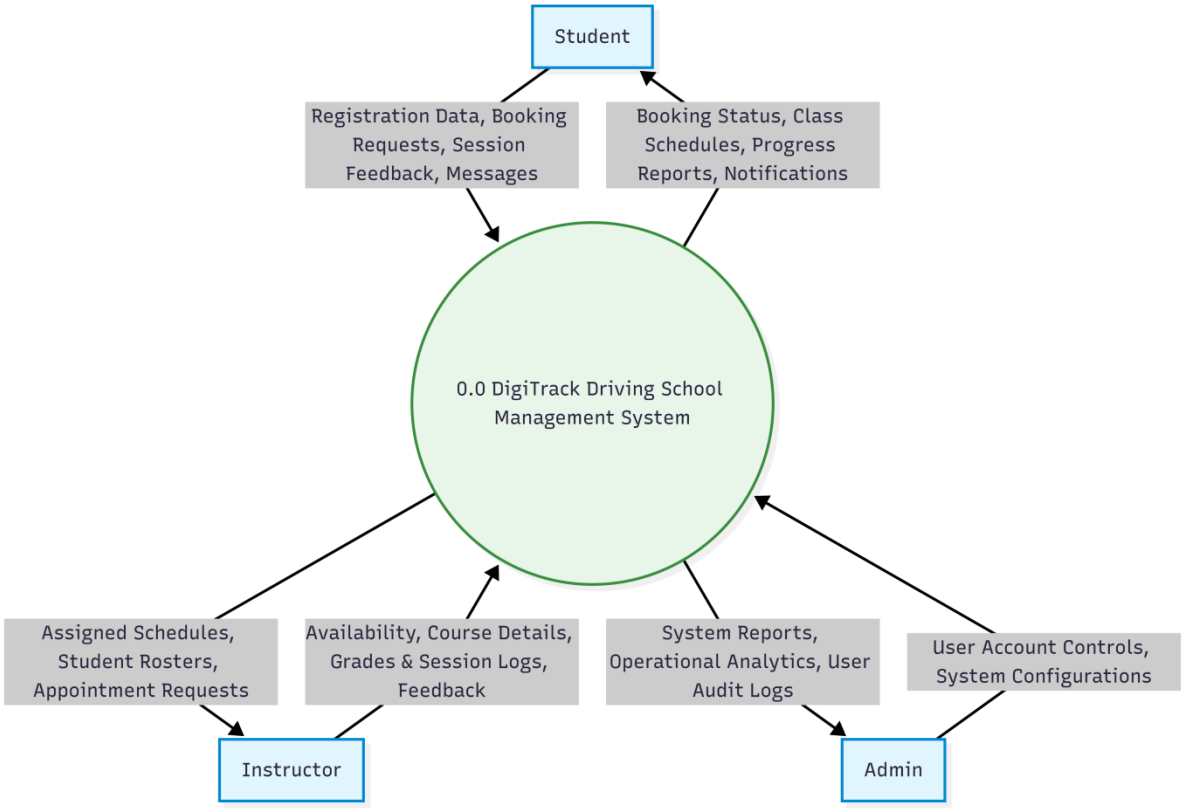
The Data Flow Diagram of DigiTrack, as shown in Figures 3.1 and 3.2, illustrates how information moves between the system's external entities, processes, and data storage components. It provides a clear representation of the system's functional flow and helps explain how each user role interacts with the application. In this diagram, the external entities consist of the Admin, Instructor, and Student, while the internal processes handle the main operations of the system. The diagram is important because it shows not only what data enters and leaves the system, but also how that data is processed and stored.

At the highest level, the diagram presents DigiTrack as a centralized system that receives information from users and produces appropriate outputs based on the type of request. The Student submits registration details, booking requests, session feedback, and messages to the system. The Instructor provides availability schedules, course details, grades, and session logs. The Admin manages user account controls and system configurations. These inputs are processed by the system so that each user receives the correct response, such as booking confirmations, class schedules, progress reports, notifications, and performance analytics.

The data flow also shows that DigiTrack relies on several internal processes to manage its operations efficiently. These processes include profile and account management, course and scheduling management, booking and appointment handling, lesson and session tracking, and evaluation and ratings. Each process interacts with the database to store, retrieve, or update information as needed. For example, student profiles are verified and saved, instructor course details and availability are updated, appointments are recorded, sessions are logged, and feedback and ratings are stored after completion of training requirements. This structured flow ensures that all transactions are properly organized and traceable.

Another important aspect of the DFD is the relationship between processes and data stores. The diagram emphasizes that information is not processed in isolation; instead, each function depends on accurate records stored in the database. This allows the system to validate user actions, avoid scheduling conflicts, track training progress, and maintain complete records of lessons, appointments, and performance. By linking processes with data stores, the DFD demonstrates how DigiTrack maintains data consistency and supports reliable decision-making.

Figure 3.1 Context Diagram (DFD Level 0)

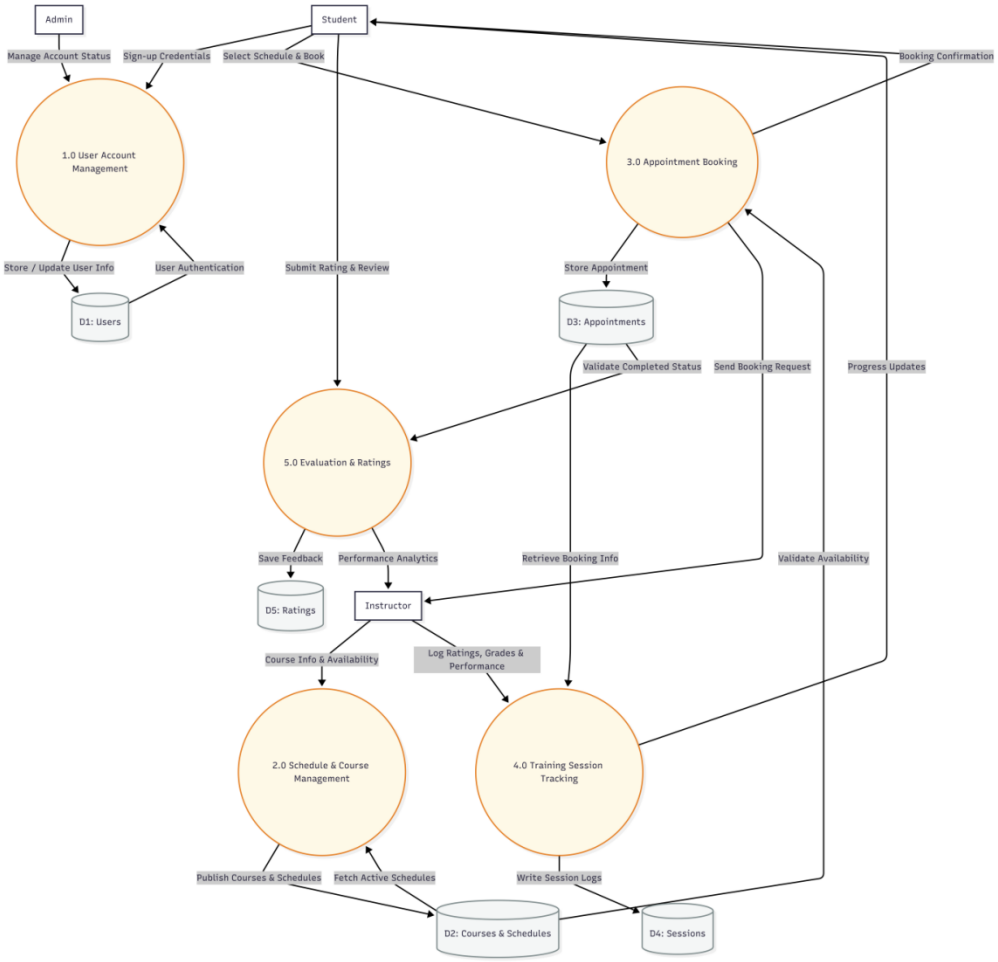


The Level 0 Data Flow Diagram presents the overall interaction between the external users of the system and the central process of DigiTrack Driving School Management System. At this level, the entire system is represented as a single major process called 0.0 DigiTrack Driving School Management System, which acts as the core that receives, processes, and distributes all essential data. The diagram shows the three main external entities of the system: the Student, Instructor, and Admin, each of whom exchanges information with the central system based on their specific roles and responsibilities.

The Student submits registration data, booking requests, session feedback, and messages to the Management System. In return, the system provides booking status, class schedules, progress reports, and notifications, allowing students to view their appointments and training progress in an organized and accessible way. This flow highlights how the system supports student enrollment, lesson scheduling, and progress monitoring in a centralized digital environment. It also reduces the need for manual coordination and helps students receive timely updates regarding their training status.

The Instructor sends availability information, course details, grades, and session logs to the system, while the Management System returns assigned schedules, student rosters, and appointment requests. This interaction allows instructors to update their schedules, record lesson activities, and receive clear training assignments from the system. On the other hand, the Admin submits user account controls and system configurations to the Management System and receives system reports, operational analytics, and user audit logs in return. This enables administrators to monitor overall system activity, verify records, and make informed decisions about school operations. Through these exchanges, the system ensures that operational data is properly managed and that each role has access only to the information relevant to its function.

Figure 3.2 Data Flow Diagram (DFD Level 1)



The Level 1 Data Flow Diagram breaks down the overall system into its major subprocesses and shows how each external entity interacts with specific functions of the DigiTrack Driving School Management System. Unlike Level 0, which presents the system as a single process, Level 1 expands it into five main processes: 1.0 User Account Management, 2.0 Schedule & Course Management, 3.0 Appointment Booking, 4.0 Training Session Tracking, and 5.0 Evaluation & Ratings. It also introduces the main data stores used to support these operations, namely D1: Users, D2: Courses & Schedules, D3: Appointments, D4: Sessions, and D5: Ratings.

The first process, 1.0 User Account Management, handles student registration and account verification. The Student submits sign-up credentials, which are stored in the Users database, while the Admin manages account status to ensure that only valid users are allowed to access the system. This process is important because it maintains accurate user records and helps secure the platform by confirming the identity and status of each student.

The second and third processes focus on course and booking management. In 2.0 Schedule & Course Management, the Instructor provides course information and availability, and the system publishes courses and schedules, which are saved in the Courses & Schedules database. In 3.0 Appointment Booking, the Student selects a schedule and submits a booking request. The system validates availability using the Courses & Schedules data store, then stores the appointment in the Appointments database and sends booking confirmations to the student and booking requests to the instructor. This ensures that bookings are properly linked to available courses and instructor schedules, making the reservation process organized and reliable.

The final two processes handle lesson tracking and performance evaluation. 4.0 Training Session Tracking records completed hours, grades, and performance data from the instructor and verifies appointment states using appointment data stored in the system. This process writes session logs to the Sessions database and provides progress updates to the student, ensuring that each driving session is properly logged and monitored. Meanwhile, 5.0 Evaluation & Ratings allows the student to submit ratings and reviews after completed sessions.

The system validates the completed status using the Appointments data store, then saves feedback to the Ratings database and provides performance analytics to the instructor.

This completes the training cycle by capturing structured feedback that can be used to evaluate instructor performance and improve service quality.

The Level 1 DFD shows how DigiTrack processes user requests through a structured set of operations and databases. Each subprocess has a clear responsibility, and the data stores support accurate record keeping across the system. This design improves efficiency, minimizes manual work, and ensures that student training, scheduling, session tracking, and performance evaluation are handled in a clear and organized way.

The data-flow representation should be interpreted as a description of information movement, not as proof that every possible data path has been tested. Each process should correspond to a function that exists in the application, and each data store should correspond to a database table or logical record group used by that function. Inputs should be checked before they change stored information, and outputs should be retrieved according to the current user's permissions. When the implementation changes, the diagram and related documentation should be reviewed so that they continue to represent the actual application rather than an earlier design assumption.

Traceability between requirements, database records, screens, and tests can help maintain consistency across modules. For example, the requirement to prevent conflicting appointments should map to the schedule-entry workflow, the data fields used to identify the instructor, student, date, and time, and the test cases that attempt both valid and conflicting assignments. The requirement for role-specific access should map to the account-role data, the server-side authorization checks, and tests using accounts with different permissions. This traceability makes it easier to locate the source of a defect and to demonstrate how each requirement was checked.

3.3.6 Security Considerations in the Architecture

Security is an important component of DigiTrack because the system stores personal and operational information.

The security architecture considers the following controls.

- Authentication

Authentication verifies that the person attempting to access the system has valid credentials.

DigiTrack requires users to log in before accessing protected functions. Authentication information should be processed securely and passwords should not be stored in plain text.

- Authorization and Role-Based Access Control

Authentication alone does not determine what a user is allowed to do. After authentication, the system must also determine whether the user's role has permission to perform the requested operation.

DigiTrack applies role-based access according to the three primary user groups:

Role	Main Access
Administrator	System management, records, scheduling, monitoring, reports
Instructor	Assigned schedules, training activities, student-related information
Student	Personal account, appointments, schedules, training information, feedback

Role-based access control provides a structured method for assigning permissions according to organizational roles [20], [21].

- Input Validation

User-submitted information should be validated before it is processed or stored. Validation reduces the possibility of malformed or malicious input being accepted by the application.

- Protection Against Injection

Database queries should use secure query practices, including prepared statements where supported. Injection is identified as one of the major application-security risks in the OWASP Top 10:2025 [14].

- Session Security

User sessions should be protected against unauthorized use. Sessions should be invalidated appropriately during logout, and protected pages should verify that a valid authenticated session exists.

- Password Protection

Passwords should be stored using appropriate password-hashing mechanisms rather than plain-text storage. Password verification should be performed through secure password-hashing functions.

- Access Control

Protected pages should not rely solely on hiding links from users. The server must verify authorization whenever a protected operation is requested.

- Data Privacy

DigiTrack may process personal information belonging to students, instructors, and administrators. The Philippine Data Privacy Act of 2012 establishes protections for personal information processed through information and communications systems and requires appropriate safeguards for personal information [18], [19].

Therefore, the project considers:

- collection of only necessary information
- restricted access to personal information
- confidentiality of participant information
- secure storage
- controlled disclosure
- appropriate retention and
- responsible disposal of research data

The architecture should maintain a clear separation between presentation, application logic, and data storage. The client interface is responsible for presenting forms and results and for collecting user input. Server-side processing is responsible for checking sessions, validating submitted values, enforcing role permissions, applying business rules, and communicating with the database. The database stores the structured records required by the application. This separation does not automatically make the application secure, but it creates a clear place to enforce checks that must not depend solely on client-side behavior.

Data validation should occur at the point where input is received by the server, even when the interface already performs browser-side validation. Client-side validation can improve usability by helping users correct mistakes

quickly, but it can be bypassed. Server-side checks should confirm that required fields are present, submitted identifiers refer to valid records, values follow the expected format, and the current user is permitted to perform the requested action. Database queries should use safe parameter handling where supported, and error messages shown to users should avoid exposing sensitive implementation details. These are practical controls related to common web-application risks identified in security guidance [13], [14].

Session handling is another architectural concern. Protected pages should require an authenticated session, and the server should verify that the session corresponds to a valid user and role. Signing out should invalidate the active session according to the application's implementation. The application should not rely on a role name submitted by the browser as proof of authorization; permissions should be determined from trusted server-side records. These controls are especially important when administrator, instructor, and student accounts share the same application but have different access requirements [20], [21].

The data layer should preserve relationships among students, instructors, courses, appointments, training sessions, attendance, and feedback. Primary keys identify records, while foreign-key relationships or equivalent application-level checks help ensure that a record refers to the intended entity. Validation should prevent impossible or inconsistent states where the implemented rules permit such checks. For example, an appointment should not reference a nonexistent student or instructor. Where a business rule is enforced only in application code, the implementation and testing documentation should state that limitation rather than implying that the database independently guarantees it.

The architecture also has operational dependencies. The application requires a functioning web server, database service, configuration, and network access. A database error or unavailable server can prevent users from accessing otherwise correctly implemented features. Backup and recovery arrangements, if implemented, should be documented and tested separately from normal functional testing. Similarly, availability should not be claimed merely because the application works during a demonstration. These operational considerations help define what a security and reliability evaluation can reasonably conclude.

3.3.7 Entity Relationship Diagram

The table relationships in DigiTrack are illustrated in Figure 3.3, which presents the Entity Relationship Diagram (ERD) of the system. The ERD shows how the different database tables are connected to one another and how data is organized within the system. It provides a structural view of the database design by identifying the main entities, their attributes, and the relationships that link them together. This diagram is important because it helps explain how DigiTrack stores and manages information in a logical and organized manner.

The ERD includes the major entities used in the system, such as users, courses, appointments, sessions, certificates, and ratings. Each entity represents a specific part of the driving school workflow and contains fields that store relevant details. For example, student and instructor tables store personal and account information, while appointment and session tables record booking and lesson activities. Certificate and rating tables store outcome-based records that reflect training completion and feedback. Through these entities, the database supports the full lifecycle of student training, from registration to certification.

The relationships between the tables show how information flows across the system. For instance, one instructor can offer many courses, one student can create many appointments, and one appointment can generate multiple sessions. In addition, certificates and ratings are linked to appointments so that training outcomes can be traced back to the specific booking and lesson records that produced them. These relationships help ensure data consistency and reduce duplication because shared information is stored only once and referenced where needed. This structure also makes it easier to retrieve related records when generating reports, certificates, or performance summaries.

Figure 3.3 Entity Relationship Diagram (ERD)

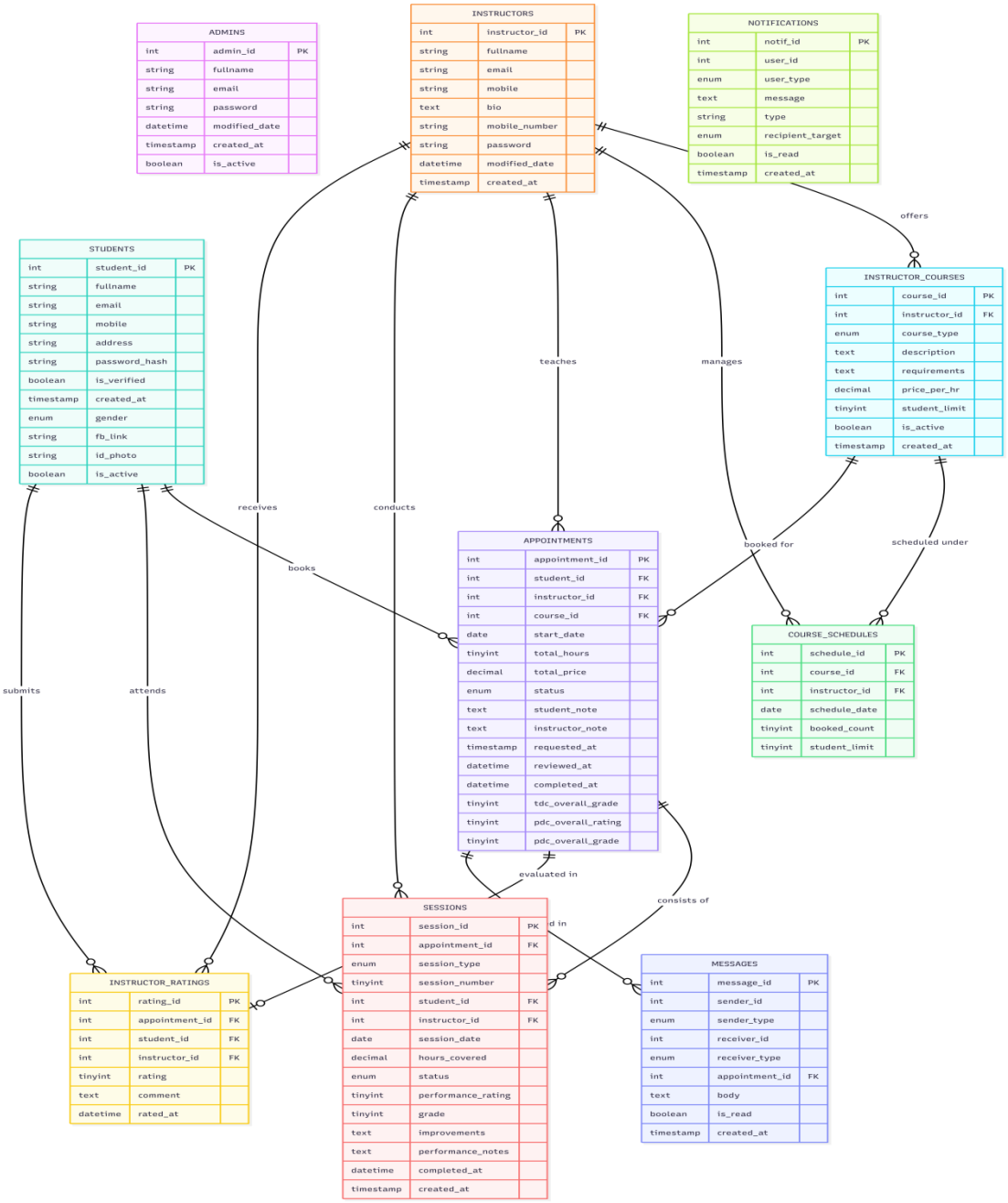


Figure 3.3 represents the Entity–Relationship Diagram (ERD) of DigiTrack: An Instructor Scheduling and Performance Monitoring System for Driving Schools. The diagram illustrates the logical structure of the system database and shows how the major entities are related to one another. It serves as a blueprint for organizing, storing, and retrieving the data required for instructor scheduling, student appointments, training-session monitoring, completion tracking, and performance evaluation.

The ERD consists of the main entities used by the system, including administrators, instructors, students, instructor courses, course schedules, appointments, sessions, and instructor ratings. These entities represent the major users, activities, and records involved in the operation of Motosafe Driving School. Each entity contains a primary key that uniquely identifies its records, while foreign keys are used to establish relationships between related entities.

The database structure allows students to book appointments for available courses and schedules. Instructors may offer courses, manage schedules, handle appointments, and conduct training sessions. An appointment is divided into one or more sessions, which are used to track completed training hours, session

performance, and progress toward course requirements. Official certificate issuance remains an external, manual process handled by the driving school or regulatory authorities; DigiTrack focuses on verifying and logging completion rather than generating certificates. Students may also submit ratings and feedback regarding the instructors assigned to their appointments.

The relationships shown in the ERD support the core business processes of DigiTrack. A student may have multiple appointments, sessions, and ratings, while an instructor may be connected to several courses, schedules, appointments, sessions, and ratings. The relationships between course schedules and appointments allow the system to monitor available slots, booking status, and assigned instructors.

The ERD also demonstrates the use of cardinality to define the number of records that may be associated with each entity. The relationships include one-to-many relationships, such as one student having multiple appointments and sessions, and optional one-to-one relationships, such as an appointment having zero or one instructor rating. These relationships help maintain data consistency and ensure that records are properly connected throughout the system.

3.4 TOOLS AND TECHNOLOGY USED

The development of DigiTrack relies on a carefully selected technology stack that supports reliability, compatibility, ease of use, and cost-effectiveness. The chosen tools and technologies are grouped into four main categories: frontend technologies, backend technology, database technology, and development tools. Each category plays a specific role in ensuring that the system functions properly as a web-based Management Information System. Together, they provide the foundation needed to build a system that is responsive, secure, and maintainable.

3.4.1 Frontend Technologies

The frontend of DigiTrack is built using standard web technologies, namely HTML, CSS, and JavaScript. These technologies are widely supported by modern web browsers and are suitable for creating a user interface that is both interactive and accessible across different devices. Since DigiTrack is intended for use by students, instructors, and administrators, the frontend must be easy to navigate and responsive to different screen sizes.

HTML (HyperText Markup Language) is used to define the structure and content of the web pages. It provides the basic layout of the system, including forms, tables, buttons, headings, and navigation elements. HTML5 is used in DigiTrack to take advantage of modern features such as semantic tags, improved form handling, and multimedia support. By using HTML5, the system maintains compatibility with current browser standards while keeping the interface organized and easy to understand.

CSS (Cascading Style Sheets) is used to control the appearance and layout of the interface. It defines the visual presentation of the pages, including colors, fonts, spacing, alignment, and responsiveness. CSS3 is particularly useful in DigiTrack because it supports media queries, which allow the interface to adapt to desktops, tablets, and mobile phones. It also enables visual enhancements such as transitions and animations that improve the overall user experience.

JavaScript adds interactivity and dynamic behavior to the frontend. It is used for form validation, dynamic content updates, AJAX communication, and other client-side functions that improve usability. For example, JavaScript can check whether required fields are completed before submission, update dashboard content without reloading the page, and display available booking slots in real time. The combination of HTML, CSS, and JavaScript ensures that DigiTrack is responsive, interactive, and compatible with major browsers such as Chrome, Firefox, Safari, and Edge.

3.4.2 Backend Technology

The backend of DigiTrack is developed using PHP, a server-side scripting language commonly used for web applications. PHP is well suited for the system because it can process form submissions, handle authentication, manage business rules, and communicate with the database. It serves as the core of the server-side logic, ensuring that data is processed correctly before being stored or returned to the user.

One advantage of PHP is its ease of use. Its syntax is relatively simple, which makes it accessible to developers with different skill levels. PHP can also be embedded directly into HTML, which simplifies the creation of dynamic web pages. This makes development faster and allows the system to combine layout and logic efficiently when needed.

PHP is also widely chosen because of its compatibility. It works well with MySQL databases and is supported by most web servers, including Apache and Nginx. This makes integration between the application and the database straightforward. In addition, PHP has a large developer community and extensive documentation, which makes it easier to troubleshoot problems and find reusable solutions.

Another important reason for selecting PHP is its built-in security support. It provides functions for password hashing, input validation, session handling, and prepared statements, all of which help protect the system from common attacks such as SQL injection. Since DigiTrack handles sensitive user and academic information, these security features are essential. PHP is also open-source and cost-effective, which makes it appropriate for a project that needs a practical and sustainable development environment.

3.4.3 Database Technology

DigiTrack uses MySQL as its relational database management system. MySQL is used to store all system records, including user accounts, student profiles, instructor information, bookings, sessions, certificates, ratings, and audit logs. Because DigiTrack depends on accurate and organized data storage, MySQL provides a reliable foundation for managing relationships between different parts of the system.

One of the main strengths of MySQL is its reliability. It is a mature and stable database system that has been used in many production environments for years. This makes it a dependable choice for storing important records that must remain accurate and accessible. In addition, MySQL supports ACID transactions, which help maintain data integrity when records are created, updated, or deleted.

MySQL is also known for its scalability. It can handle growing amounts of data and users while maintaining good performance through indexing and query optimization. This is important for DigiTrack because the system may expand as more students, instructors, and courses are added. Its compatibility with PHP also makes it convenient for backend integration, especially when using MySQLi or PDO for database access.

Another benefit of MySQL is its security features, which include authentication, access control, and support for secure data handling. These features help protect sensitive records from unauthorized access. Since the system stores personal details, schedules, and certification data, MySQL is an appropriate choice for ensuring that information remains centralized, organized, and protected.

Database testing should verify both the values stored and the relationships between records. A successful form submission does not necessarily mean that the resulting data are correct: the system could save an incorrect foreign-key value, omit a related record, or display a stale value in another module. Test cases should therefore compare the input, the stored record, and the value displayed after retrieval. Where a record is linked to another entity, the test should confirm that the relationship points to the intended student, instructor, course, or appointment. Duplicate submissions, missing required fields, and invalid references should also be considered when they are relevant to the implemented workflow.

Consistency checks are particularly important when multiple modules read or update the same operational data. For example, a change in appointment time should be reflected in the schedule view and should be used by the conflict-checking process during subsequent assignments. If completion status is used in a dashboard summary, the summary should change only when the underlying record reaches the status defined by the system's rules. These checks help detect discrepancies between database storage and the information presented to users. The manuscript should describe the database engine and the checks actually performed, while avoiding claims that all integrity problems have been eliminated unless the testing evidence supports that conclusion.

3.4.4 Development Tools

The development of DigiTrack uses Visual Studio Code and XAMPP/Apache as primary tools. These tools support coding, testing, and local deployment during the development phase. They are widely used in web development because they are practical, cost-effective, and easy to configure.

Visual Studio Code is the main code editor used for building the application. It supports extensions for HTML, CSS, JavaScript, PHP, and other languages, making it suitable for full-stack web development. Its integrated terminal, debugging tools, syntax highlighting, and code completion features help improve productivity and reduce coding errors. It also supports Git integration, which makes version tracking and collaboration easier during development.

XAMPP is used as the local server environment for testing the system. It includes Apache, MySQL, and PHP in one package, allowing the application to run locally before being deployed to a live server. This

makes it easier to test forms, database connections, and server-side logic without depending on an online hosting environment. Apache, the web server component of XAMPP, handles HTTP requests and serves the PHP files needed by the system.

These tools are valuable because they simplify installation, testing, and troubleshooting during development. They also allow the project to be built in a controlled environment, which reduces the risk of errors before deployment. Since DigiTrack is a web-based system, having a local development setup is essential for efficient programming and debugging.

3.5 DATA GATHERING TECHNIQUES

Data gathering was conducted to obtain information necessary for understanding the existing operations of Motosafe Driving School and for evaluating DigiTrack.

The study used multiple data-gathering techniques to obtain information from different perspectives.

The major techniques included:

- interviews
- observation
- questionnaires
- document and record review
- usability testing and
- system-generated technical information, where available and actually collected

The use of multiple sources allowed the researchers to compare information obtained through different methods.

3.5.1 Interviews

Semi-structured interviews were used to obtain detailed information from stakeholders.

The interview questions focused on:

- current scheduling practices
- instructor availability
- student appointments
- training-session recording
- instructor monitoring
- student feedback
- record management
- reporting needs
- problems encountered during daily operations and
- expectations for the proposed system

Semi-structured interviews allowed the researchers to use prepared questions while also asking follow-up questions when participants provided information that required clarification.

Interview responses were used to identify operational requirements and system features.

3.5.2 Participants and Sampling

The study involved stakeholders directly related to the operation and use of DigiTrack.

The primary stakeholder groups were:

- Motosafe Driving School administrators

- Instructors and
- students

Administrators provided information concerning management processes and system requirements. Instructors provided information concerning schedules, training activities, student handling, and instructor-related workflows. Students provided information concerning appointments, training-related information, and feedback requirements.

Participants were selected based on their relevance to the system and their ability to provide information related to the research objectives.

Participant selection should reflect the roles that are expected to use the application. Administrators are appropriate respondents for questions about record management, schedule coordination, reporting, and administrative control. Instructors can provide feedback about access to assignments and training information, while students can comment on the clarity and accessibility of student-facing functions. If the number of available users is limited, the study may use purposive participation based on role and relevance to the system, but it should describe the actual participants and should not imply that the group statistically represents all driving schools. The selection method should be reported as it was carried out, not merely as an ideal procedure.

The researchers should also distinguish between participants who contribute to requirements gathering and participants who evaluate the finished system. Some individuals may take part in both activities, but their roles and the timing of their input should be recorded. Early interviews help identify workflow requirements; later task-based testing examines whether the implementation supports those requirements. Feedback given during development may influence revisions, while feedback gathered after a stable build can help evaluate the resulting interface. Keeping these stages separate reduces the risk of presenting informal development conversations as a formal acceptance evaluation.

3.5.3 Data Collection Instruments

Several instruments were used to collect information.

- Interview Guide

The interview guide contained questions related to existing processes, problems, requirements, and expectations.

- Observation Guide

The observation guide was used to document actual operational procedures. The researchers observed how schedules, appointments, records, and other school-related activities were handled.

- Questionnaire

A questionnaire was used to collect structured responses from participants where appropriate. The questionnaire may contain items concerning current problems, desired features, system usability, and user satisfaction.

- Usability Testing Form

A usability-testing form was used to record participant performance during task-based system testing.

Possible observations included:

- whether the participant completed the task
- whether assistance was required
- errors encountered
- time required to complete the task and
- participant comments
- Evaluation Questionnaire

The evaluation questionnaire was used to collect user perceptions concerning system characteristics such as functionality, usability, reliability, and security.

3.5.4 Data Collection Procedure and Timeline

The data collection and development activities were organized into several stages.

- Stage 1: Initial Requirement Gathering

The researchers gathered information regarding the existing processes of Motosafe Driving School.

Interviews and observations were used to understand how the school manages instructors, students, schedules, appointments, training activities, and feedback.

The information collected during this stage was organized into initial requirements.

- Stage 2: Requirement Analysis

The researchers analyzed the information collected during the initial stage.

Similar problems and requirements were grouped together. These were then translated into system features and user stories.

For example, difficulties in determining instructor availability contributed to the requirement for an instructor scheduling component.

Problems associated with scattered records contributed to the requirement for centralized information management.

- Stage 3: Iterative Development

During development, completed system features were demonstrated and reviewed.

Stakeholder comments were considered when appropriate. Features requiring changes were returned to the development backlog for refinement.

- Stage 4: Testing

After the major features were implemented, the system underwent functional and integration testing.

Users were also involved in usability and acceptance testing where applicable.

- Stage 5: Evaluation

Evaluation data were collected using the established evaluation criteria.

The collected information was analyzed to determine whether DigiTrack satisfied its intended requirements.

3.5.5 Data Analysis Methods

The collected data were analyzed using both qualitative and quantitative methods. Qualitative data from interviews and observations were transcribed and reviewed using thematic analysis. Recurring issues, needs, constraints, and feature requests were identified and grouped into themes. These themes were then translated into user stories and acceptance criteria for the product backlog, ensuring that the system was aligned with the needs of the users.

Quantitative data from surveys and usability tests were analyzed using descriptive statistics such as means, medians, and standard deviations. Usability metrics, including task success rates and time on task, were computed and summarized, and SUS scores were generated from the standardized questionnaire. Where applicable, pre- and post-deployment results were compared, such as the frequency of scheduling conflicts or the amount of time required to complete a booking task.

System logs and performance data were also analyzed to assess technical efficiency. Server logs were aggregated to determine response time distributions, error frequencies, and peak load behavior. The results were used to identify areas for optimization, such as database queries, caching strategies, or server configuration improvements. This combination of qualitative and quantitative analysis provided a balanced view of both user experience and system performance.

For quantitative data, the analysis should match the level of measurement and the purpose of the instrument. Counts and percentages can describe how many test cases passed or how many respondents selected each response option. A mean score may be reported when it is appropriate to the rating scale and when the method of interpretation is clearly stated. The researchers should identify the denominator used for percentages and avoid treating missing answers as negative responses unless the instrument defines that treatment. For qualitative comments, responses can be grouped by recurring issue, such as schedule clarity, navigation, access restrictions, or suggestions for improvement. The coding process should be described briefly so that readers can understand how comments were summarized.

Data analysis should not manufacture precision that the study design cannot support. If only a small number of users participate, descriptive results can still identify practical problems, but they should not be used to claim broad statistical significance. If the test cases are designed by the development team, passing them demonstrates conformance to those defined cases, not independent proof that all user needs have been met. The interpretation should state the conditions under which testing occurred, the build that was tested, and any known limitations. This approach provides a transparent connection between collected evidence and the conclusions drawn from it.

3.5.6 Validity, Reliability, and Bias Mitigation

Several measures were considered to improve the credibility of the gathered data.

- Triangulation

Information from different sources was compared where possible. For example, information obtained from interviews could be compared with observations of actual workflows.

Using multiple sources reduces dependence on a single participant's perspective.

- Instrument Review

Research instruments were reviewed before their use to ensure that questions were understandable and relevant to the research objectives.

- Consistent Procedures

The researchers used consistent instructions during interviews, questionnaires, and usability sessions.

- Participant Confidentiality

Participants were informed that their responses would be treated confidentially and used for research purposes.

- Bias Reduction

The researchers avoided presenting expected answers to participants and allowed participants to provide their own observations and opinions.

Participants were also informed that their responses should reflect their actual experience with the current process or system.

To improve validity, each question in a data-gathering instrument should correspond to a defined research objective or system requirement. Questions about scheduling should focus on how appointments are currently created, changed, checked, and communicated. Questions about performance monitoring should distinguish between the records administrators need and the judgments they make using those records. Questions about usability should refer to tasks that respondents can reasonably assess, such as locating a schedule or understanding a status message. Avoiding unrelated or leading questions reduces the risk of collecting opinions that do not answer the study's research questions.

Reliability can be supported by using consistent instructions, the same task definitions, and a documented method for recording responses. If more than one researcher observes a test, the team should agree on what counts as task completion, an error, a request for help, or a successful correction. Where a rating scale is used, its labels and interpretation should remain consistent across the questionnaire and the analysis. The manuscript should report the instrument that was actually used and should not introduce a different scale after data collection merely because it appears easier to interpret.

Potential sources of bias include selecting only users who are comfortable with technology, asking questions that suggest a preferred answer, observing a user only after extensive coaching, and treating developer expectations as user feedback. These risks can be reduced by inviting appropriate representatives of the intended roles, giving neutral task instructions, recording difficulties as well as successes, and distinguishing direct observations from interpretations. If the number of participants is small or the evaluation is limited to one school, those limitations should be stated explicitly rather than hidden by broad claims.

The researchers should also maintain a clear record of which data were collected and how they were analyzed. Interview notes, observation records, completed questionnaires, test logs, and issue lists serve different purposes and should not be combined without explanation. Quantitative responses may be summarized using the descriptive statistics appropriate to the instrument, while open-ended comments may be grouped by recurring topic. Any reported result should be traceable to the relevant record, and missing responses or incomplete test runs should be acknowledged where they affect interpretation.

3.5.7 Ethical Considerations

Ethical considerations were observed throughout the study.

1. Informed Consent

Participants were informed about:

- the purpose of the study
- the nature of their participation
- the information being collected
- how the information would be used and
- their right to participate voluntarily

2. Voluntary Participation

Participation was voluntary. Participants were not required to provide information unrelated to the objectives of the study.

3. Confidentiality

Personal information collected from participants was treated confidentially. Information that could identify individual participants should not be unnecessarily disclosed in the research report.

4. Privacy

DigiTrack processes information that may identify individuals. The Philippine Data Privacy Act of 2012 establishes protections for personal information processed through information and communications systems [18].

Accordingly, the researchers considered privacy principles when gathering, storing, processing, and reporting research information.

5. Responsible Use of Data

Collected data were used only for purposes related to the study. Access to research data was limited to authorized individuals.

3.5.8 Data Management and Storage

All collected data were stored using a secure data management plan. The researcher kept files on an encrypted local machine and maintained backups in a restricted cloud repository. Code and project documents were version-controlled using Git to ensure that changes could be tracked and previous versions restored if needed. Access to raw data was limited to authorized project members and supervisors, while anonymized datasets were used for analysis and reporting.

This approach helped protect the integrity, privacy, and availability of the research data throughout the study. It also supported organized development and made it easier to manage revisions, analysis, and reporting tasks.

Ethical handling of information is particularly important because the system may contain identifying information about students, instructors, and administrators. Data gathering should collect only the information necessary to understand the workflow or evaluate the system. Respondents should be informed of the purpose of the activity and how their responses will be used. Personal comments should not be attributed to named individuals in the manuscript unless this is authorized and necessary. Research records should be stored securely and made available only to team members who need them for the study, consistent with applicable privacy requirements [18], [19].

The development team should avoid using real personal information in screenshots, demonstrations, or sample records when dummy data can serve the same purpose. If real records are required for a legitimate test, access should be limited and the data should not be copied into public repositories or shared materials. Test accounts should use controlled credentials, and screenshots included in the manuscript should be reviewed for exposed names, contact details, account identifiers, or other unnecessary personal information. These practices reduce avoidable exposure while allowing the system's workflow to be documented.

The ethical responsibilities also apply to performance information. Student feedback and instructor-related records should be treated as information for authorized administrative review, not as a basis for unsupported personal conclusions. Feedback may reflect an individual experience, a specific lesson, or a temporary circumstance and should be interpreted within its context. The manuscript should not present feedback ratings as an objective or comprehensive measure of an instructor's competence unless a suitable instrument and validation process support that claim.

3.5.9 Limitations of Data Gathering

The data-gathering process was subject to several limitations.

First, the study focused on one driving school. Therefore, the collected information primarily represents the operational context of Motosafe Driving School.

Second, stakeholder availability may have limited the duration of interviews and observations.

Third, participants may have provided responses based on their personal experiences, which may differ from the experiences of other users.

Fourth, the study period limited the amount of time available for long-term observation.

Finally, some operational information may not have been available to the researchers because of confidentiality or organizational restrictions.

These limitations were considered when interpreting the results.

3.6 TESTING PROCEDURE

Testing was conducted to determine whether DigiTrack operated according to its requirements and whether the different system components worked together correctly.

Testing was performed at multiple levels because a system may contain individual functions that work correctly while still experiencing problems when those functions interact.

The major testing activities were:

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing

Testing focused on the major system functions, including authentication, role-based access, instructor management, student management, course management, scheduling, appointments, training sessions, ratings, feedback, notifications, and reporting.

3.6.1 Unit Testing

Unit testing focused on individual functions or components.

The purpose was to verify that a specific function produced the expected result under defined conditions.

Examples include:

- login validation
- account creation
- instructor record creation
- student record creation
- schedule creation
- appointment submission
- appointment status update
- rating submission
- training-session recording
- notification generation

A typical unit-test procedure involved:

- identifying the function to be tested

- defining the expected result
- preparing test data
- executing the function
- comparing the actual result with the expected result
- recording any error
- correcting and retesting the function when necessary

3.6.2 System Testing

System testing evaluated DigiTrack as a complete application.

Instead of examining only individual functions, system testing considered whether the entire application could support complete workflows.

Example scenarios included:

Student Appointment Workflow

- Student logs in
- Student views available schedule
- Student submits an appointment
- System validates the appointment
- Appointment is stored
- Appropriate schedule information is updated
- Relevant user receives the corresponding status or notification

Instructor Scheduling Workflow

- Administrator logs in
- Administrator views instructor information
- Administrator creates or modifies a schedule
- System checks relevant scheduling conditions
- Schedule is stored
- Authorized users can view the updated schedule

Training Session Workflow

- Instructor accesses an assigned appointment
- Instructor records the completed training activity
- System stores the session information
- Appropriate student progress information is updated

Feedback Workflow

- Student submits a rating or feedback
- System validates the submission
- Feedback is stored
- Authorized administrators can view the feedback
- Information can contribute to instructor performance monitoring

System testing helped determine whether these workflows functioned correctly from beginning to end.

System-level test cases should cover complete user workflows rather than isolated controls alone. A typical case may begin with a user signing in, opening the relevant module, entering required information, submitting the form, and verifying the saved record in the appropriate view. Negative cases should be included where

applicable, such as submitting a form with missing required information, attempting to save an invalid date or identifier, or opening a protected page without the required role. For each case, the test record should state the preconditions, steps, expected result, observed result, and pass/fail decision. This structure helps another team member repeat the test and understand why it was considered successful or unsuccessful.

The system test set should also account for changes to existing records. Creating a record is only one part of a management workflow; users may need to edit an appointment, cancel a session, correct a student's details, or review a completed training activity. Tests should confirm that updates are reflected in the related views and that cancelled or modified records are not mistakenly presented as unchanged active assignments. Where the application preserves historical records, the expected behavior should be stated clearly. Where it overwrites or deletes information, the team should confirm that this behavior matches the client's requirements and does not remove information needed for routine administration. The purpose of these tests is to verify the implemented behavior under defined conditions, not to assume that every possible situation has been covered.

3.6.3 Integration Testing

Integration testing examined interactions between different system modules.

This was necessary because DigiTrack contains interconnected features.

Examples include:

- Authentication and Role Management

The system was tested to verify that users were directed to the appropriate interface after successful authentication and that access permissions corresponded to their roles.

- Scheduling and Appointments

The scheduling module was tested together with appointments to verify that appointment information corresponded to available schedules and that relevant records were updated correctly.

- Appointments and Training Sessions

The system was tested to verify that training-session records could be associated with the appropriate appointment, student, and instructor.

- Ratings and Instructor Records

The rating module was tested to verify that student feedback was associated with the correct instructor and relevant training transaction.

Integration testing was used to identify problems that might not appear when individual modules were tested separately.

An integration test should examine the transfer of information between connected modules, including whether the same record is interpreted consistently by each module. A schedule created by an administrator should appear in the appropriate schedule view; a change to an appointment should be reflected where that appointment is displayed; and a completed training activity should be reflected in the relevant progress record when the implemented workflow requires it. Tests should also include invalid or incomplete data to verify that the application responds appropriately instead of saving a misleading or disconnected record. The exact test results should be recorded from the actual system build.

Regression testing is useful after a defect is corrected or a shared component is changed. A fix to appointment validation, for example, should not prevent valid appointments from being saved or cause unrelated schedule views to fail. Changes to role permissions should be checked against all affected user roles, not only the account used during development. A concise regression checklist can identify high-priority workflows that should be

repeated after updates. This process supports confidence in the current build but does not guarantee the absence of all defects; its value depends on the coverage and quality of the selected test cases.

3.6.4 User Acceptance Testing

User Acceptance Testing (UAT) involved intended users evaluating whether DigiTrack supported their operational needs.

The primary user groups were:

- administrators;
- instructors
- students

Participants were provided with representative tasks corresponding to their system roles.

Examples include:

Administrator tasks :

- log in
- manage instructor records
- manage student records
- create or modify schedules
- monitor appointments
- view instructor performance information

Instructor tasks ;

- log in
- view assigned schedule
- view appointments
- record training information
- access appropriate student information

Student tasks

- log in
- view courses
- view available schedules;
- submit an appointment
- view appointment information
- submit instructor feedback

Participants' comments and observed difficulties were documented.

If users encountered errors, unclear instructions, confusing navigation, or missing functionality, the development team reviewed the issue and determined whether modification was required.

UAT therefore considered not only whether the system technically worked but also whether the system supported actual user tasks.

User Acceptance Testing should be organized around realistic workflows that reflect the responsibilities of the intended user roles. An administrator scenario may begin with signing in, selecting a student and instructor, creating an appointment, checking for conflicts, saving the assignment, and confirming that the resulting record

appears in the relevant schedule. An instructor scenario may involve signing in, reviewing assigned sessions, and locating relevant training information. A student scenario may involve signing in and reviewing the schedule or progress information available to that account. Each scenario should specify the expected result before the test begins.

During a test, the observer should record whether the user completed the task, whether the expected result appeared, whether an error occurred, and whether assistance was needed. If a task fails, the record should identify the point of failure and the observed behavior rather than merely stating that the system did not work. A defect caused by an implementation error should be distinguished from a misunderstanding caused by unclear labels or unfamiliar navigation. Both may matter, but they require different corrective actions. After a correction, the affected task should be repeated to verify that the problem was addressed and that the change did not introduce another issue.

Acceptance criteria should be defined before results are interpreted. For example, a scheduling test may pass when the system saves a valid appointment and prevents a conflicting assignment under the specified conditions. A role-access test may pass when a user can access permitted pages and is denied access to restricted actions. A usability task may be considered successful when the user completes it without assistance, according to the study's predefined rule. The criteria must match the actual application behavior and should not be changed after observing results simply to make the system appear successful.

User feedback is useful for identifying issues that functional testing may not reveal, such as confusing wording, an unclear confirmation, or a workflow that requires unnecessary navigation. Nevertheless, satisfaction and acceptance should be reported only when feedback has actually been collected through the stated procedure. A planned UAT procedure is not evidence that users have approved the system, and a demonstration is not equivalent to a completed evaluation. The final manuscript should distinguish between tests completed, issues corrected, feedback received, and evaluation activities that remain future work.

3.7 EVALUATION CRITERIA

The evaluation criteria were established to determine the overall quality and suitability of DigiTrack.

The primary criteria were:

1. Functionality
2. Usability
3. Reliability
4. Security

These criteria correspond to important qualities of a web-based information system and to the objectives of DigiTrack.

3.7.1 Functionality

Functionality refers to whether the system provides the required functions and produces correct results.

For DigiTrack, functionality includes:

- user authentication
- role-based access
- instructor management
- student management
- course management
- schedule management
- appointment management
- training-session monitoring
- instructor ratings
- student feedback

- notifications
- messaging
- certification-related record management
- reporting.

Functionality was assessed through test cases and requirements verification.

A feature was considered functionally acceptable when it performed the expected operation under the defined test conditions.

Special attention was given to business rules such as:

- restricting functions according to user roles
- preventing invalid scheduling conditions
- maintaining relationships among records
- storing submitted information correctly
- producing appropriate results.

3.7.2 Usability

Usability refers to the extent to which specified users can achieve specified goals with effectiveness, efficiency, and satisfaction within a specified context of use [22].

For DigiTrack, usability is important because the system will be used by people with different responsibilities and levels of technical familiarity.

Usability evaluation considered:

- Learnability

  The degree to which new users can understand the system and learn how to complete common tasks.

- Efficiency

  The amount of effort and time required to complete tasks.

- Error Handling

  The ability of the system to prevent, identify, and recover from user errors.

- Navigation

  The clarity and consistency of menus, buttons, forms, and page relationships.

- Satisfaction

  The user's overall perception of the interface and interaction with the system.

Task-based usability testing was used to observe whether participants could complete representative activities.

The researchers may record:

- task success
- task completion time
- errors

- assistance required
- user comments

Where the System Usability Scale (SUS) is used, the final manuscript should report the actual SUS results obtained from the study rather than inserting a score that has not been collected.

3.7.3 Reliability

Reliability refers to the ability of the system to perform its intended functions consistently under specified conditions.

For DigiTrack, reliability is important because the system stores operational records and supports activities such as scheduling, appointments, training records, and feedback.

Reliability evaluation may consider:

- Availability

  Whether the system is accessible when required.

- Consistency

  Whether the system produces consistent results when the same valid operation is performed under the same conditions.

- Error Handling

  Whether the system responds appropriately when invalid or unexpected inputs occur.

- Data Integrity

  Whether records remain accurate and consistent during system operations.

- Recoverability

  Whether data and system functions can be restored following an identified failure when backup and recovery procedures are available.

Testing for reliability may include repeated execution of important functions, error handling tests, database consistency checks, and backup/recovery verification where those procedures are actually implemented.

3.7.4 Security

Security evaluates the system's ability to protect information and restrict access to authorized users.

Security evaluation considered:

- authentication
- authorization
- role-based access
- password protection
- input validation
- protection against injection
- session management
- data protection
- access restrictions
- security-related logging where implemented

The OWASP Top 10:2025 identifies major web application risks including Broken Access Control, Security Misconfiguration, Cryptographic Failures, Injection, Insecure Design, Authentication Failures, Software and Data Integrity Failures, and Logging and Alerting Failures [14].

Therefore, security testing should consider whether DigiTrack contains weaknesses related to these areas that are applicable to the actual implementation.

Security evaluation may involve:

- source-code review
- authentication testing
- authorization testing
- input-validation testing
- database query review
- configuration review
- controlled vulnerability testing

The study should not claim that DigiTrack is completely secure or immune to all vulnerabilities. Instead, the evaluation should report the security controls that were implemented and the vulnerabilities that were tested.

3.7.5 Evaluation Instrument

The evaluation instrument was designed to collect structured feedback concerning the quality of DigiTrack.

The instrument may contain statements corresponding to the major evaluation criteria:

Criterion	Areas Evaluated
Functionality	Completeness, correctness, feature operation
Usability	Ease of use, navigation, learnability, satisfaction
Reliability	Consistency, stability, error handling
Security	Authentication, access control, data protection

Respondents may evaluate each statement using the rating scale approved for the study.

The exact scale used in the final manuscript must match the questionnaire actually administered.

For example, if a five-point Likert scale was used, the researchers should present the exact scale and interpretation adopted by the research instrument.

The evaluation results should then be summarized using the appropriate descriptive statistics.

The final results should report;

- number of respondents
- responses per criterion
- computed values
- interpretation of the values
- relevant qualitative comments

The evaluation results presented in Chapter IV should correspond directly to the criteria and procedures described in this chapter.

The evaluation instrument should define each criterion in terms that respondents can understand and assess. Functionality items should address whether required tasks work as expected. Usability items should address clarity, navigation, ease of learning, and the effort needed to complete tasks. Reliability items should address consistency of results and behavior during repeated use or defined error conditions. Security items should focus on visible and testable safeguards, such as role restrictions and handling of invalid input, rather than asking respondents to judge technical properties they cannot observe. This alignment helps keep the questionnaire appropriate to the respondents' experience.

If a rating scale is used, the number of response options, the meaning of each option, and the method of computing the summary must be stated consistently. The researchers should explain how missing answers are handled and whether each criterion is summarized separately. If open-ended comments are included, the analysis should identify recurring concerns and suggestions without presenting an isolated comment as representative of every user. The instrument should be attached or described in sufficient detail for readers to understand what was measured.

The evaluation should also separate evidence of conformance from evidence of perceived quality. A test log can show whether a particular function met its expected result under the test conditions. A user questionnaire can indicate how respondents perceived the interface or workflow. These forms of evidence complement one another but are not interchangeable. A high satisfaction rating does not prove that the application is free of defects, and a passing functional test does not prove that the interface is intuitive. Reporting both types of evidence, when available, provides a more balanced assessment of DigiTrack.

Finally, the evaluation results should be interpreted in light of the study's setting and limitations. Results from one driving school may be informative for that client but may not represent all driving schools. A small respondent group may provide useful formative feedback but may not support broad statistical generalization. If long-term operational data are unavailable, the study should avoid claiming sustained improvements in workload distribution or efficiency. These limitations do not invalidate the development work; they clarify what the collected evidence can reasonably establish.

A final methodological safeguard is the use of requirement-to-test traceability. Each major system requirement should be linked to the module that implements it, the data used by the module, and at least one test or evaluation activity that checks the expected behavior. For instance, a requirement concerning instructor availability should be linked to the availability data, the scheduling interface, the server-side conflict rule, and a test that attempts to assign an instructor during an occupied time. A requirement concerning role-based access should be linked to the permissions assigned to each role and to tests that attempt both permitted and restricted actions. This mapping helps the team identify requirements that have not been tested and makes it easier to explain how the evaluation relates to the objectives of the study. Traceability should be based on the actual implemented system; if a requirement was changed or deferred during development, the documentation should record that status rather than imply that the original requirement was fully satisfied. The resulting evidence can then be organized into a concise summary showing the requirement, test procedure, expected result, observed result, and status. Such a summary supports transparent reporting and makes it possible for readers to distinguish between a feature that exists, a feature that was tested, and a feature whose broader operational benefit has yet to be established.

3.8 SUMMARY OF THE METHODOLOGY

The methodology of the DigiTrack study combines Research and Development with Agile Scrum to support the systematic development and evaluation of a web-based instructor scheduling and performance monitoring system.

The R&D design provides the overall research structure by connecting the identification of the operational problem, requirements gathering, system development, and evaluation. The Agile Scrum Model provides the software development structure by organizing implementation into iterative development cycles that allow testing, feedback, and refinement.

The system architecture follows a web-based client-server structure consisting of the client interface, server-side processing, and centralized relational database. The architecture supports the three major user roles of DigiTrack: administrator, instructor, and student.

The system was developed using web technologies and tools appropriate for the project, including HTML, CSS, JavaScript, PHP, MySQL, Visual Studio Code, XAMPP, phpMyAdmin, Git, and GitHub.

Data gathering involved interviews, observations, questionnaires, and system evaluation activities. The information gathered was used to identify requirements, guide development, and assess the completed system.

Testing was conducted through unit testing, system testing, integration testing, and User Acceptance Testing. These procedures were intended to verify that individual functions, interconnected modules, complete workflows, and user requirements were properly addressed.

Finally, DigiTrack was evaluated using functionality, usability, reliability, and security criteria. These criteria provided a structured basis for determining whether the developed system satisfied its intended requirements and provided an appropriate foundation for the findings presented in the succeeding chapter.

The methodological sequence is intended to maintain a clear connection between the original operational problem and the evidence reported at the end of the study. Requirements gathering identifies the needs and constraints of the client; system design converts those needs into workflows, data structures, and role permissions; implementation produces the working application; and testing checks whether the specified functions behave as expected. User evaluation, where completed, adds evidence about how intended users experience the workflows. These stages are related, but one stage cannot substitute for another. A requirement documented during an interview is not yet an implemented feature, and an implemented feature is not yet evidence that the requirement has been met reliably. The manuscript should therefore report each stage according to the records available for it and should identify any requirement that changed during development.

The final interpretation should bring together the functional test results, usability observations, and relevant security checks without treating them as a single undifferentiated measure of success. A system may pass its scheduling tests but still require interface improvements; users may find a dashboard helpful while a particular permission check needs correction. These findings can coexist and should be reported separately before the researchers draw an overall conclusion. If additional testing or client feedback remains outstanding, that status should be made explicit. The purpose of the methodology is to make the development and evaluation process understandable and repeatable, not to guarantee a positive result in advance.

The methodological documentation should also make the system version and test conditions clear. A test result applies to the build, data, accounts, and configuration used during the test; a later code change may alter the result and should be followed by appropriate retesting. Where test data are created specifically for evaluation, the records should be identified as test data and should not be confused with actual school transactions. If the system is accessed through a hosted environment, the manuscript should distinguish between local development testing and tests performed on the deployed version. These details matter because database configuration, permissions, and deployment settings can affect system behavior even when the source code is unchanged. The researchers should preserve test notes and record any known limitations so that the reported findings can be traced to the conditions under which they were obtained. This practice supports transparency and helps future developers reproduce important checks when the system is maintained or extended.

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CHAPTER IV

PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the results of the development, implementation, and evaluation of DigiTrack: An Instructor Scheduling and Performance Monitoring System for Motosafe Driving School. It describes the developed system, its major modules, development results, system architecture, hardware and software requirements, database design, and testing activities.

The presentation of the results follows the major objectives of the study. The developed system is presented according to its major functions, including user authentication, instructor management, student management, course and schedule management, appointment booking, training-session monitoring, certification monitoring, and instructor performance evaluation. The chapter also presents the relationships among the system components and the database structures that support the application.

The results presented in this chapter are based primarily on the actual system developed by the researchers. Therefore, descriptions of the system's modules, workflows, database structures, and testing activities represent the implementation of DigiTrack. Where recognized software-development, usability, or security concepts are discussed, appropriate references are used to distinguish established principles from the researchers' own implementation.

4.1 OVERVIEW OF THE DEVELOPED SYSTEM

DigiTrack is a web-based developed to support instructor scheduling, student appointments, training-session monitoring, and instructor performance monitoring at Motosafe Driving School. The system was designed in response to operational concerns involving schedule management, instructor availability, student records, training progress, and feedback monitoring.

The system centralizes important information that would otherwise need to be maintained through separate records or communication channels. Through a centralized application, authorized users can access information according to their assigned roles and perform activities appropriate to their responsibilities.

The completed system provides three primary user roles: Administrator, Instructor, and Student. Each role has a different set of functions. The Administrator is responsible for managing users, courses, instructors, schedules, appointments, and monitoring information. The Instructor can view assigned schedules, monitor training activities, and record relevant session and performance information. The Student can access training-related information, manage appointments, view schedules, and provide instructor feedback.

The major problems addressed by the system include scheduling conflicts and overlapping appointments, difficulty in monitoring instructor workload, difficulty in tracking student training progress, and limited visibility of instructor performance and student feedback. These concerns were among the operational issues identified as part of the development of the system.

The major objectives reflected in the completed system are:

- To automate the recording and management of lesson schedules and bookings
- To provide a centralized view of instructor availability and workload
- To record student feedback and instructor performance information
- To support monitoring of TDC and PDC-related training requirements
- To provide organized summaries and reports for administrative use
- To provide role-specific access to system information and functions

The completed system consists of the following major modules:

1. Login and Role-Based Access Control
2. Administrator Dashboard
3. Instructor Management and Availability
4. Student Management and Enrollment
5. Lesson Scheduling and Booking
6. Training-Session Monitoring
7. Compliance and Certification Monitoring
8. Feedback and Instructor Performance Monitoring
9. Database and Record Management

These modules are interconnected so that information entered in one part of the system can support related processes in another part. For example, instructor and course information can be used during scheduling, while appointment information can subsequently be associated with training sessions and instructor ratings.

4.1.1 Overall System Workflow

The overall workflow of DigiTrack begins with student onboarding and the submission of required information. Administrative personnel verify the submitted information before the student proceeds to the available courses and instructors.

After approval, the student can view available course schedules and submit an appointment request. The system processes the booking according to the available schedule and provides schedule information to the appropriate users.

Once a training appointment has been established, the instructor can view the assigned lesson and record the appropriate training information. Training-session records can then be used to monitor student progress and completion of required training activities.

After a completed training activity, the student can provide a rating and feedback regarding the instructor. The collected feedback can then be accessed by authorized users for performance monitoring.

Figure 4.1 presents the overall workflow of the developed DigiTrack system. The existing figure shows the progression from student registration and verification through course and instructor selection, booking, instructor scheduling, training records, and completion-related activities.

Figure 4.1
Overall Workflow of the DigiTrack System

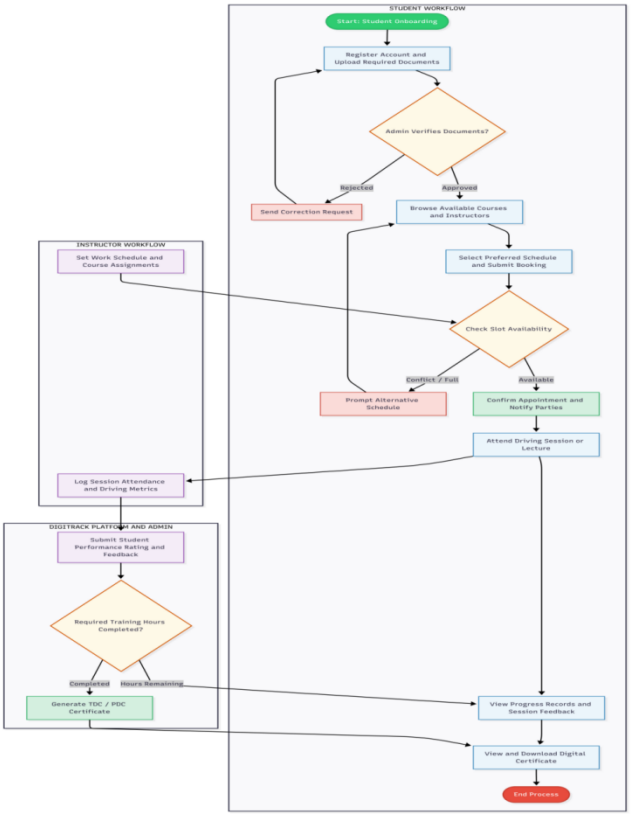


Figure 4.1 presents the overall workflow of DigiTrack. The process begins with student onboarding and administrative verification. After approval, the student can select available courses and instructors and submit a booking request. The system checks schedule availability before confirming the appointment. The instructor can then access the assigned schedule and record training information. The resulting records support monitoring of student progress and training completion. The workflow also provides a process for recording completion-related information and presenting the corresponding records to the student.

4.2 SYSTEM DEVELOPMENT RESULTS

The development of DigiTrack followed the Agile Software Development Methodology. The development process was divided into manageable stages so that requirements, design, development, testing, and deployment could be addressed progressively.

The development process consisted of requirements gathering, system planning, system design, software development, testing, deployment, and maintenance planning. These activities correspond with the Agile approach described in Chapter III and were used to organize the development of the system.

4.2.1 Requirements Gathering

The first stage involved identifying the operational requirements of Motosafe Driving School. Information was obtained through interviews, observations, and document analysis involving relevant stakeholders.

The requirements focused on the major activities that the system needed to support. These included instructor scheduling, student management, course management, appointment booking, training monitoring, compliance-related recordkeeping, and feedback collection.

The information gathered during this stage served as the basis for determining the functions and information requirements of DigiTrack. The requirements were subsequently translated into system features and development tasks.

4.2.2 System Planning

During the planning stage, the researchers established the scope, objectives, system specifications, development schedule, and required resources.

The scope of the system focused on scheduling, booking, instructor workload monitoring, training-session monitoring, feedback collection, and related administrative functions. Planning also considered the hardware and software requirements necessary to develop and operate the application.

The planning stage helped the researchers organize the development activities and determine the sequence in which the system components would be implemented.

4.2.3 System Design

The system-design stage involved preparing the architecture, database structures, process flows, interface layouts, and system modules.

The researchers prepared system models such as process diagrams, Entity Relationship Diagrams, and interface designs to serve as references during implementation. The design emphasized separation of system responsibilities and the organization of data into related database entities.

The resulting design was used as the blueprint for the development of the application.

4.2.4 Software Development

During software development, the researchers implemented the planned system modules incrementally. The major functionalities developed included authentication, instructor management, student registration, course management, schedule management, appointment booking, training-session recording, certification monitoring, and feedback recording.

The modules were progressively integrated so that the researchers could test the interaction between different system functions during development.

The development process resulted in an integrated web-based application that connects user interfaces, application logic, and a centralized relational database.

4.2.5 Testing

Testing was performed after individual modules were developed and integrated. Unit testing was used to check individual functions, while integration and system testing were used to verify the interaction of different components.

The testing activities focused on important system functions such as authentication, appointment booking, scheduling, database transactions, role-based access, training-session recording, and feedback submission.

User evaluation was also planned to determine whether the completed system met the expectations and operational needs of its intended users.

The actual numerical results of the user evaluation should be presented in a separate results subsection once the completed questionnaires and computed scores are available.

4.2.6 Deployment

After testing and necessary corrections, the application was prepared for deployment in the target environment.

Deployment involved database configuration, server configuration, and installation of the application components required to make DigiTrack accessible to authorized users.

The deployed application was then made available for controlled demonstration and evaluation.

4.2.7 Maintenance

Maintenance involves monitoring the system, correcting identified issues, updating system components, and implementing improvements based on user feedback.

Because software requirements may change after implementation, maintenance provides an opportunity to improve the system according to the operational needs of Motosafe Driving School.

Figure 4.2
Agile Software Development Life Cycle Adopted in the Study

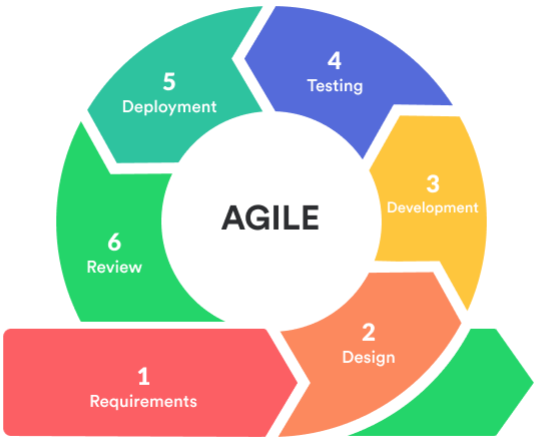


Figure 4.2 illustrates the Agile development cycle used by the researchers. The existing figure presents the major development activities as Requirements, Design, Development, Testing, Deployment, and Review. These activities represent the iterative nature of the development process described in the methodology.

4.3 SYSTEM ARCHITECTURE

DigiTrack uses a three-tier client-server architecture consisting of the Presentation Layer, Application Layer, and Data Layer.

The separation of these layers organizes the application according to different responsibilities. The Presentation Layer handles interaction with users, the Application Layer processes requests and applies system rules, and the Data Layer stores and retrieves system information.

This architectural organization also makes it easier to describe how information moves between users, application processes, and the database.

Figure 4.3
Three-Tier Architecture of DigiTrack

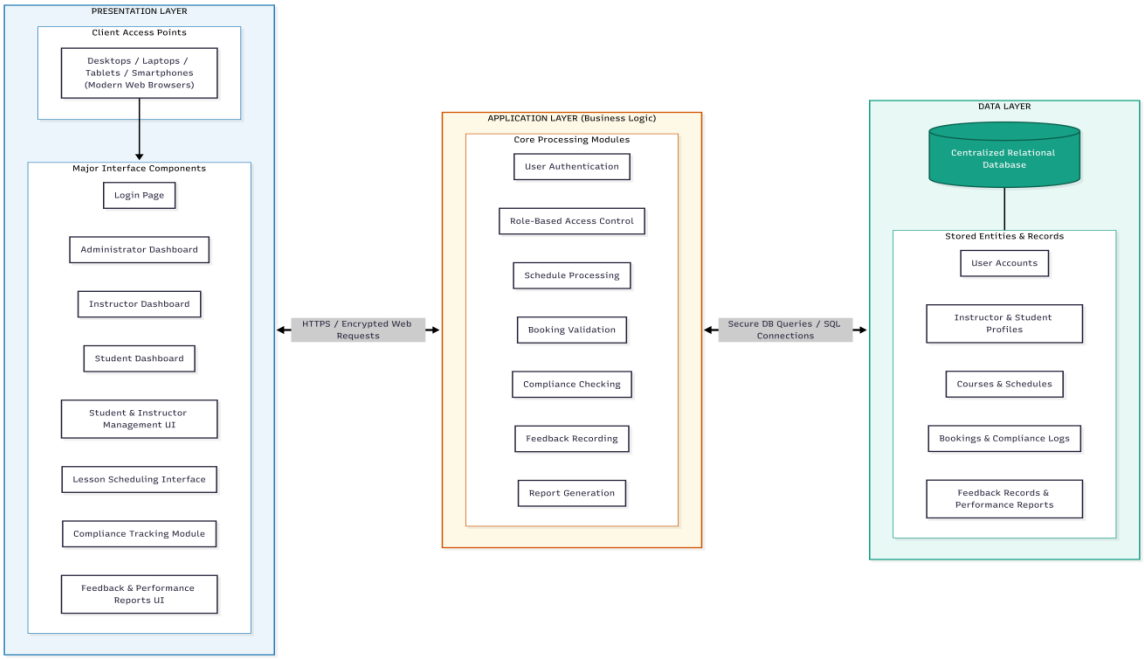


Figure 4.3 presents the architecture of DigiTrack and illustrates the relationship among the presentation, application, and data layers.

4.3.1 Presentation Layer

The Presentation Layer serves as the interface between the users and the application. It displays information and receives input from users through the system's web interfaces.

The major interface components include:

- Login Page
- Administrator Dashboard
- Instructor Dashboard
- Student Dashboard

- Student Management
- Instructor Management
- Lesson Scheduling
- Compliance Tracking;
- Feedback and Performance Reports

The system is designed to be accessible through commonly used web-enabled devices such as desktop computers, laptops, tablets, and smartphones, subject to the capabilities of the device and browser used.

4.3.2 Application Layer

The Application Layer contains the system's processing logic. It acts as an intermediary between the user interface and the database.

Its major responsibilities include:

- User authentication
- Role-based access control
- Schedule processing
- Appointment validation
- Compliance-related checking
- Training-session processing
- Feedback recording
- Report generation

The application layer processes information received from users before storing or retrieving the corresponding records from the database.

4.3.3 Data Layer

The Data Layer stores the records required by DigiTrack. The application uses a centralized relational database to organize and retrieve system information.

The major categories of information stored include:

- User accounts
- Instructor profiles
- Student profiles
- Courses
- Course schedules
- Appointments
- Training sessions
- Instructor ratings and feedback

The database relationships allow the system to associate records across different processes. For example, a student appointment can be associated with a particular instructor, course, schedule, and subsequent training session.

4.4 HARDWARE AND SOFTWARE REQUIREMENTS

The implementation of DigiTrack requires appropriate hardware and software resources. The requirements presented in this section are based on the specifications contained in the existing project document.

4.4.1 Hardware Requirements

Table 4.1
Recommended Hardware Specifications

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD
Monitor	1366 × 768 resolution	Full HD display
Internet Connection	10 Mbps	25–50 Mbps
Mobile Device	Android 10 / iOS equivalent	Latest stable Android/iOS version

The existing document identifies these specifications as the recommended hardware configuration for typical web-based operations and graphical dashboard use.

4.4.2 Software Requirements

Table 4.2
Software Requirements

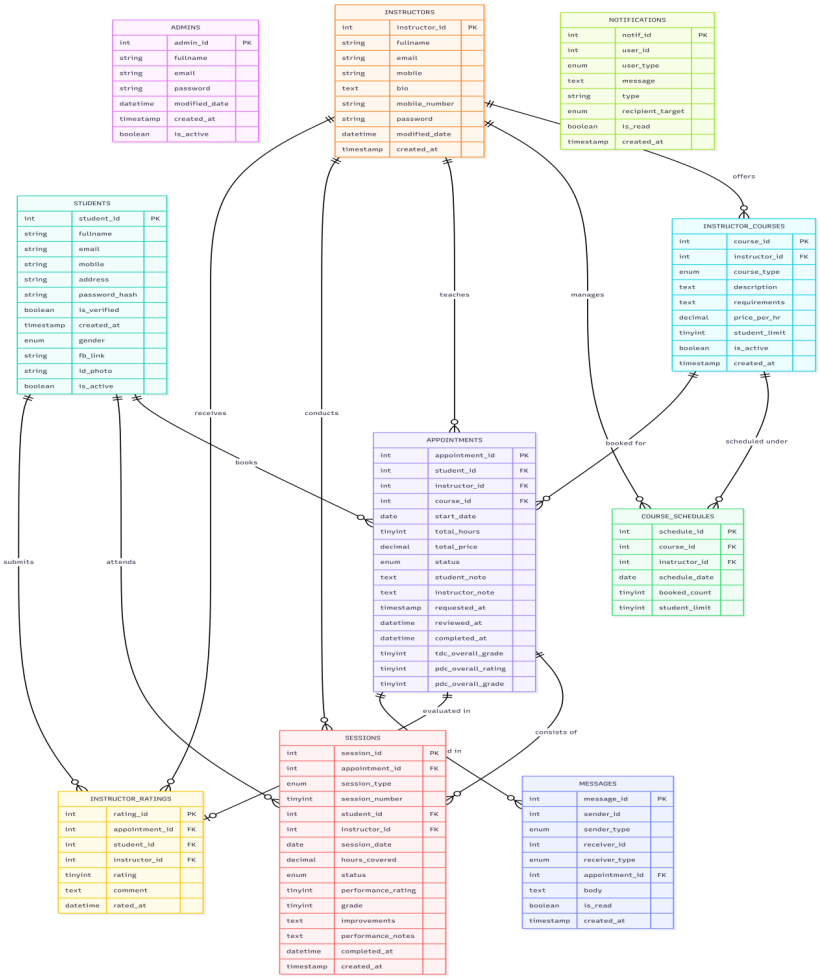
Software	Purpose
Windows / Linux OS	Operating system
Visual Studio Code	Source-code editor
MySQL	Database management system
PHP	Backend development
HTML / CSS / JavaScript	Frontend development
Google Chrome	Web browser
Git	Version control
XAMPP / WAMP	Local development server

4.5 DATABASE DESIGN

The database serves as the foundation for DigiTrack because it stores the information required for scheduling, student management, instructor management, appointments, training sessions, certification records, and instructor evaluations.

The system uses a relational database model through MySQL. The database is organized into related tables, with primary and foreign keys used to establish relationships among records.

Figure 4.4
Entity Relationship Diagram of DigiTrack



The major entities identified in the existing database design are:

- Admins
- Instructors
- Students
- Instructor Courses
- Course Schedules
- Appointments
- Sessions
- Instructor Ratings

4.5.1 Admin Table

The Admin table stores the account and profile information of system administrators.

Field Name	Description
admin_id	Primary key and unique administrator identifier
first_name	Administrator's first name
last_name	Administrator's last name
email	Administrator's email address
password_hash	Stored password hash

Field Name	Description
role	Administrative access level
created_at	Date and time the account was created
updated_at	Date and time the account was updated

The table supports administrator authentication and administrative access to system functions.

4.5.2 Instructor Table

The Instructor table stores the personal, contact, authentication, and account-status information of instructors.

Field Name	Description
instructor_id	Primary key and unique instructor identifier
first_name	Instructor's first name
last_name	Instructor's last name
email	Instructor's email address
phone_number	Instructor's contact number
password_hash	Stored password hash
status	Account or employment status
created_at	Date and time the record was created
updated_at	Date and time the record was updated

The table supports instructor profile management, course assignment, schedule management, appointments, and performance monitoring.

4.5.3 Student Table

The Student table stores the personal, contact, authentication, and enrollment-related information of students.

Field Name	Description
student_id	Primary key and unique student identifier
first_name	Student's first name
last_name	Student's last name
email	Student's email address
phone_number	Student's contact number
password_hash	Stored password hash
date_of_birth	Student's date of birth

Field Name	Description
address	Student's residential address
status	Account or enrollment status
created_at	Date and time the record was created
updated_at	Date and time the record was updated

The Student table supports appointment booking, training monitoring, certification-related records, and instructor evaluation.

4.5.4 Instructor Courses Table

The Instructor Courses table connects instructors with the courses they offer.

Field Name	Description
course_id	Primary key and unique course identifier
instructor_id	Foreign key linked to the instructor
course_name	Name of the driving course
description	Description of the course
price	Course cost
duration_hours	Required training duration
created_at	Date and time the course was created

An instructor may be associated with one or more courses.

4.5.5 Course Schedule Table

The Course Schedule table stores available schedules for courses offered by instructors.

Field Name	Description
schedule_id	Primary key and unique schedule identifier
course_id	Foreign key linked to the course
instructor_id	Foreign key linked to the instructor
start_datetime	Starting date and time
end_datetime	Ending date and time
max_slots	Maximum number of students

Field Name	Description
slots_available	Number of remaining available slots
status	Current schedule status

The table allows the system to organize instructor availability and available course schedules.

4.5.6 Appointment Table

The Appointment table stores booking information submitted by students.

Field Name	Description
appointment_id	Primary key and unique appointment identifier
student_id	Foreign key linked to the student
instructor_id	Foreign key linked to the instructor
course_id	Foreign key linked to the course
schedule_id	Foreign key linked to the schedule
booking_status	Current appointment status
total_amount	Total amount associated with the appointment
created_at	Date and time the appointment was created

The appointment record connects the student, instructor, course, and schedule involved in a booking.

4.5.7 Sessions Table

The Sessions table records training sessions associated with student appointments.

Field Name	Description
session_id	Primary key and unique session identifier
appointment_id	Foreign key linked to the appointment
student_id	Foreign key linked to the student
instructor_id	Foreign key linked to the instructor
session_number	Training-session sequence
scheduled_date	Date of the training session
hours_rendered	Number of completed training hours
instructor_notes	Instructor observations or notes
status	Current session status

The table allows the system to monitor completed training sessions and the number of hours rendered by students.

4.5.8 Instructor Rating Table

The Instructor Rating table stores student evaluations of instructors.

Field Name	Description
rating_id	Primary key and unique rating identifier
appointment_id	Foreign key linked to the appointment
student_id	Foreign key linked to the student
instructor_id	Foreign key linked to the instructor
rating	Numerical rating provided by the student
feedback	Written feedback
created_at	Date and time the rating was submitted

The relationship between the rating and the appointment allows the feedback to be associated with a specific training transaction.

4.5.9 Relationship Among Tables

The relationships among the database tables support the major operational processes of DigiTrack.

A student may have multiple appointments and training sessions. An instructor may be associated with multiple courses and schedules. Appointments connect students with instructors and courses, while sessions record the actual training activities associated with appointments.

Instructor ratings are associated with the relevant student, instructor, and appointment. This allows the system to maintain feedback records that can be reviewed for performance-monitoring purposes.

The use of primary and foreign keys provides a structured relationship among records and supports consistent retrieval of related information. The database structure presented in the existing document is intended to support appointment management, session monitoring, certification records, and instructor performance evaluation.

4.6 SYSTEM FUNCTIONAL RESULTS

The developed system provides the principal functions identified during the requirements and design stages.

4.6.1 User Authentication and Role Management

The authentication function provides the entry point to the system. Users are required to log in before accessing role-specific functions.

After successful authentication, users are directed to the appropriate interface according to their assigned role. This separates administrator, instructor, and student functions and prevents users from being presented with unrelated system operations.

4.6.2 Administrator Functions

The Administrator is provided with functions for managing the major records and operations of the system.

The administrator can manage:

- instructor records
- student records
- courses
- instructor availability
- schedules
- appointments
- training records
- certification-related information
- instructor ratings
- administrative reports

The dashboard provides organized information that assists the administrator in monitoring the school's activities.

4.6.3 Instructor Functions

The Instructor interface provides access to assigned schedules and training-related records.

The instructor can:

- view assigned schedules
- view student assignments
- access training information
- record training-session information
- monitor relevant student progress
- review available performance-related information

These functions provide instructors with a centralized location for managing their assigned training activities.

4.6.4 Student Functions

The Student interface provides access to training and appointment information.

Students can:

- access their account
- view available courses
- view instructor schedules
- submit appointments
- view training schedules
- monitor training-related records
- submit instructor ratings and feedback

These functions provide students with access to information related to their training activities.

CHAPTER V

SUMMARY, CONCLUSIONS, LIMITATIONS, AND RECOMMENDATIONS

This chapter presents the summary of the study, major findings, conclusions, limitations, and recommendations for the continuous improvement and implementation of DigiTrack: An Instructor Scheduling and Performance Monitoring System for Motosafe Driving School.

The study focused on developing a centralized web-based system to support the management of instructors, students, courses, schedules, appointments, training sessions, certifications, instructor performance, communication, and account security. The system was developed in response to operational concerns associated

with manual scheduling, record management, instructor workload monitoring, student training monitoring, and performance feedback.

DigiTrack was developed using an Agile development approach and was organized using a layered system architecture consisting of the presentation, application, and data layers. The completed system provides separate functions and access privileges for administrators, instructors, and students. The system was also deployed as part of the implementation of the developed solution.

The chapter summarizes the major results obtained from the development and testing of the system, discusses the conclusions derived from the completed system, identifies the limitations encountered during the study, and provides recommendations for future improvement, maintenance, and expansion.

5.1 SUMMARY OF THE STUDY

The study focused on the development of DigiTrack for Motosafe Driving School. The system was designed to provide a centralized platform for organizing instructor schedules, managing student appointments, monitoring completed training sessions, recording certification information, and supporting instructor performance monitoring.

The study was conducted in response to operational concerns related to manual scheduling and record management. These concerns included difficulty in identifying instructor availability, the possibility of overlapping appointments, uneven distribution of instructor workload, difficulty in monitoring student training progress, and limited visibility of instructor performance information.

The main objective of the study was to develop an Instructor Scheduling and Performance Monitoring System that would improve the organization, accessibility, and management of driving school records.

Specifically, the system was developed to:

- Automate the recording and management of instructor schedules and student appointments
- Provide a centralized platform for managing administrator, instructor, and student records
- Support the assignment of instructors to courses and appointments
- Monitor scheduled and completed training sessions
- Record student ratings and feedback regarding instructor performance
- Provide notifications and messages related to appointments and system activities
- Improve the security of user accounts through authentication and account-history features
- Provide organized information that can assist administrators in operational decision-making.

An Agile development methodology was used in developing DigiTrack. The development process included requirements gathering, planning, system design, software development, testing, implementation, and maintenance planning. The iterative approach allowed the researchers to develop the system in manageable components and integrate the major functions according to the identified requirements.

The system was designed using a layered architecture composed of the Presentation Layer, Application Layer, and Data Layer. The Presentation Layer provides the interfaces accessed by users through supported devices. The Application Layer contains the business logic responsible for authentication, role-based access, appointment validation, scheduling, certification monitoring, ratings, messaging, notifications, and other system processes. The Data Layer contains the centralized relational database used to store and manage the system records.

The database contains interconnected records for administrators, students, instructors, courses, instructor-course assignments, schedules, appointments, training sessions, PDC certificates, instructor ratings, messages, notifications, and security-related history. Primary and foreign key relationships were used to organize the relationships among the different records.

The completed system was deployed to provide a working implementation of the proposed solution. At the time of documentation, formal client feedback following deployment had not yet been obtained. Therefore,

the conclusions of this study focus primarily on the developed system, its implemented functions, and the documented development and testing results rather than on claims of client satisfaction.

5.2 SUMMARY OF FINDINGS

The development of DigiTrack resulted in a completed web-based system that integrates several operational functions required for the management of driving school activities. The findings below summarize the major functions and system capabilities demonstrated through the development and implementation of DigiTrack.

5.2.1 Instructor Scheduling

DigiTrack provides an organized platform for managing course schedules, instructor availability, and student appointments. Administrators can assign instructors to courses and appointments, while instructors can access their assigned schedules.

Centralizing scheduling information reduces the need to depend entirely on paper records, separate files, spreadsheets, or informal communication when managing appointments. The system provides a common source of scheduling information that can be accessed according to user permissions.

The scheduling functions also organize information involving dates, instructors, students, and courses. This structure helps administrators identify scheduling information more efficiently and provides a more organized method of managing appointments. However, administrator supervision remains necessary when reviewing and approving schedules.

5.2.2 Instructor and Student Management

The system provides administrators with a structured method of managing instructor and student records. Information can be stored in the centralized database and accessed according to the user's assigned role.

For instructors, the system maintains information related to profiles, course assignments, schedules, appointments, training sessions, ratings, and certification-related activities. For students, the system manages information related to appointments, training sessions, course participation, and feedback.

Centralizing these records reduces the need to search through multiple files and provides a more organized method of retrieving and updating information. It also establishes relationships between users and their corresponding courses, schedules, appointments, sessions, and feedback records.

5.2.3 Course and Schedule Management

DigiTrack supports the management of driving courses and instructor-course assignments. Administrators can identify instructors assigned to particular courses and manage available schedules associated with those courses.

The course structure supports different training requirements, including theoretical and practical driving activities. This allows course information to be associated with students, instructors, schedules, and appointments.

The integration of these records provides a more organized representation of the relationships among the major entities involved in driving school operations.

5.2.4 Appointment and Session Management

The system provides a digital method for recording and managing student appointments. Each appointment can be associated with a student, instructor, and course and can contain information such as the scheduled date and appointment status.

Appointment statuses allow the school to distinguish between different stages of an appointment, such as pending, approved, completed, or cancelled, depending on the operational process implemented in the system.

DigiTrack also provides training-session records associated with appointments. This allows the system to distinguish between a planned appointment and a training activity that has already been conducted. The relationship between appointments and sessions provides administrators and instructors with better visibility of student training activities.

5.2.5 Performance Monitoring

DigiTrack supports instructor performance monitoring through its rating and feedback functionality. Students can provide ratings and feedback associated with completed appointments or training sessions.

The rating record can be connected to the student, instructor, and appointment involved in the evaluation. This creates a structured record that administrators can review when monitoring instructor performance and service quality.

The feedback feature does not replace formal performance evaluation procedures. Instead, it provides an additional source of information that can help administrators identify strengths, possible areas for improvement, and service concerns.

5.2.6 Communication and Notifications

The messaging and notification features support communication among administrators, instructors, and students.

Messages may be associated with appointments or other system activities, allowing communication to remain related to the relevant transaction or training activity. Notifications may provide information concerning appointment updates, messages, schedule changes, and other system activities.

These functions provide a centralized method of communicating system-related information and can help reduce dependence on informal communication methods.

5.2.7 Account Security

DigiTrack includes security-related functions and records designed to support account protection. These include login history, OTP history, and student lockout history.

Login-history records provide information about account access. OTP records support identity-verification processes, while account lockout records help protect accounts following repeated unsuccessful login attempts.

These functions demonstrate that security considerations were incorporated into the development of the system. However, security requires continuous monitoring, maintenance, and improvement as the system is used in an actual operational environment.

5.2.8 Centralized Information Management

One of the major findings of the study is the integration of the system's major records into a centralized relational database. Instead of managing instructor, student, course, schedule, appointment, session, certificate, and rating information separately, DigiTrack connects these records through database relationships.

This centralized structure provides a more organized method of storing and retrieving information. It also supports the different system modules because related information can be accessed by the appropriate functions according to user permissions.

5.3 CONCLUSIONS

Based on the development, implementation, and documented testing of DigiTrack, the following conclusions were reached.

5.3.1 DigiTrack Addresses the Identified Operational Problems

The completed system addresses several operational concerns associated with manual scheduling and record management at Motosafe Driving School.

DigiTrack provides a centralized platform where information about instructors, students, courses, schedules, appointments, training sessions, and ratings can be stored and managed. By organizing these records within one system, the application reduces dependence on separate files and informal methods of managing operational information.

The system therefore provides a structured technological solution to the operational problems identified during the study.

5.3.2 The System Supports More Organized Scheduling

DigiTrack provides administrators with tools for organizing course schedules and assigning instructors to appointments. It also allows users to access appointment and training-session information according to their roles.

By associating dates, instructors, students, and courses within centralized records, the system provides a more organized approach to scheduling. Although administrator review remains necessary, DigiTrack provides structured information that can assist in identifying and managing scheduling conflicts.

5.3.3 The System Supports Training Activity Monitoring

The system supports monitoring of student appointments and completed training sessions. The relationship between appointments and sessions allows administrators to distinguish scheduled activities from activities that have already been conducted.

This provides better visibility of student training progress and allows relevant users to access records concerning completed, pending, or cancelled activities.

5.3.4 The System Supports Instructor Performance Monitoring

The rating and feedback feature provides a structured method for recording student evaluations of instructor performance.

The system connects ratings and feedback with the appropriate instructor and appointment, allowing administrators to review performance-related information in an organized manner. While these records should not be used as the sole basis for formal instructor evaluation, they provide an additional source of information that can support performance monitoring.

5.3.5 The System Supports Centralized Information Management

DigiTrack integrates different operational records into a centralized relational database. The relationships among students, instructors, courses, schedules, appointments, sessions, ratings, messages, and notifications allow the system to manage related information within one environment.

This centralized structure supports more organized information retrieval and provides administrators with a consolidated view of relevant school records.

5.3.6 The System Provides Role-Based Access and Security Controls

The system provides different functions according to the roles of administrators, instructors, and students. This allows users to access functions appropriate to their responsibilities.

Security-related features, including authentication, OTP-related records, account lockout controls, login history, and password-history records, were incorporated into the system. These controls provide a foundation for protecting user accounts and restricting access to authorized functions.

5.3.7 The Study Objectives Were Addressed Through System Development

The development of DigiTrack addressed the major objectives of the study by providing functions for instructor scheduling, appointment management, instructor and student management, training-session monitoring, instructor ratings, communication, and account security.

The completed system demonstrates how a centralized web-based application can organize the major information and processes involved in driving school operations. However, long-term operational effectiveness and client satisfaction require continued use, monitoring, and formal evaluation after deployment.

5.4 LIMITATIONS OF THE STUDY

Although DigiTrack was completed and deployed, several limitations should be considered when interpreting the results of the study.

First, the system was developed specifically for the operational requirements of Motosafe Driving School. Other driving schools may have different scheduling procedures, course structures, user roles, and reporting needs. Therefore, additional configuration or modification may be necessary before the system can be applied to another driving school.

Second, the current version of DigiTrack does not include some advanced features. These may include online payment processing, SMS integration, automated email delivery, advanced analytics, digital certificate verification, and integration with external government or institutional systems.

Third, the effectiveness of the system depends on accurate and consistent data entry. Incorrect or incomplete information entered by users may affect the accuracy of schedules, records, reports, and other system outputs.

Fourth, continued system effectiveness depends on proper user training, regular maintenance, database backups, security monitoring, and appropriate management of user accounts and permissions.

Fifth, formal client feedback following deployment had not yet been obtained at the time of documentation. Consequently, the study does not claim a specific level of client satisfaction or long-term operational impact based on client survey results.

Finally, the current study focuses on the implementation of DigiTrack within the context of Motosafe Driving School. Its results should therefore be interpreted within the operational setting for which the system was designed.

5.5 RECOMMENDATIONS

Based on the findings, conclusions, and limitations of the study, the following recommendations are proposed to support the continued improvement and implementation of DigiTrack.

5.5.1 System Improvements

- Provide workload summaries showing the number of appointments and completed sessions assigned to each instructor.
- Add advanced search and filtering options for students, instructors, appointments, courses, sessions, certificates, and ratings.
- Add an appointment-rescheduling feature that records the original schedule, revised schedule, reason for the change, and user responsible for the update.
- Add a waiting-list feature for students when preferred schedules are unavailable.
- Improve dashboard summaries to provide administrators with clearer information about schedules, workloads, appointments, and training progress.

5.5.2 Notifications and Communication

- Add automated reminders for upcoming appointments and training sessions.
- Provide email or SMS notifications for important school announcements and appointment changes.
- Improve the messaging feature by including message status, timestamps, attachments, and conversation history.
- Provide notification settings that allow users to select the types of alerts they wish to receive.
- Maintain an organized notification history so users can review previously received system notifications.

5.5.3 Performance Monitoring

- Allow administrators to generate periodic instructor performance reports.
- Include rating categories such as punctuality, communication, teaching ability, professionalism, and vehicle or equipment preparedness.
- Apply appropriate privacy controls to ensure that instructor feedback is viewed only by authorized users.
- Use student feedback as a supporting source of information rather than the only basis for instructor evaluation.

5.5.4 Security and Data Protection

- Maintain strong password requirements and encourage periodic password updates.
- Continue using OTP verification, account lockout controls, and login-history monitoring where applicable.
- Protect sensitive information during transmission and storage through appropriate security mechanisms.
- Conduct periodic security testing to identify vulnerabilities and security weaknesses.
- Maintain regular database backups and periodically test the restoration process.
- Establish procedures for handling lost accounts, suspicious login activities, and unauthorized access.
- Apply appropriate privacy practices when storing student, instructor, appointment, rating, and certification information.
- Review user roles and access permissions regularly to ensure that users only have access to functions required for their responsibilities.

5.5.5 Usability and Accessibility

- Provide user orientation and training before full operational implementation.
- Maintain compatibility with desktop and mobile devices.
- Use clear labels, readable text, and consistent navigation to make the system easier to use.

- Consider language options if future users require additional language support.
- Conduct periodic usability evaluation after the system has been used in actual school operations.

5.5.6 Implementation and Maintenance

- Assign an authorized system administrator to manage accounts, courses, schedules, and system settings.
- Establish a regular schedule for software updates, database maintenance, security reviews, and performance monitoring.
- Maintain a record of system issues, corrections, and enhancement requests.
- Establish a recovery plan for technical problems, server failure, data loss, or extended internet interruption.
- Review the system periodically to determine whether its functions continue to meet the operational needs of the school.
- Obtain formal feedback from administrators, instructors, and students after sufficient actual use of the system.
- Use the collected feedback to identify features that require modification or improvement.

5.6 PROPOSED FUTURE ENHANCEMENTS

The following features may be considered for future versions of DigiTrack.

5.6.1 Mobile Application

A dedicated mobile application or progressive web application may be developed to provide instructors and students with easier access to schedules, appointments, notifications, messages, and feedback features.

A mobile-focused version could provide more convenient access for users who primarily use smartphones during daily school activities.

5.6.2 Online Payment Integration

An online payment feature may be added to allow students to pay course or appointment fees through supported digital payment platforms.

If implemented, the feature should include transaction records, payment status, receipts, and appropriate security controls. Any financial functionality should also be carefully tested before actual deployment.

5.6.3 Training Completion and Certificate Release

Future versions of DigiTrack may include a feature for managing the release of training completion certificates to students who have successfully completed the required training.

The feature may allow administrators to record the completion status of students and manage the release of their corresponding training certificates through the system.

This feature is not implemented in the current version of DigiTrack.

5.6.4 Multi-Branch Support

If Motosafe Driving School expands to additional branches, DigiTrack may be enhanced to support multiple locations.

This would require branch-based user access, instructor assignment, course scheduling, reporting, and data management. Such functionality would allow administrators to manage multiple branches within a centralized system.

5.6.5 External System Integration

Future versions may be integrated with external services such as email platforms, SMS gateways, payment systems, cloud storage, or other systems used by driving schools.

Any external integration should be carefully tested to ensure compatibility, privacy, security, and reliable data exchange.

5.7 FINAL STATEMENT

DigiTrack was developed and deployed as a centralized web-based Instructor Scheduling and Performance Monitoring System for Motosafe Driving School. The completed system integrates functions for managing instructors, students, courses, schedules, appointments, training sessions, instructor ratings, messages, notifications, and account-security information.

The development of DigiTrack provides a structured technological approach to managing information that was previously dependent on separate records and manual processes. By connecting the major operational records through a centralized database and providing role-based functions, the system supports more organized scheduling, training-session monitoring, instructor performance monitoring, and information management.

The system also establishes a foundation for future improvements. Continued maintenance, user training, security monitoring, database management, and formal user evaluation are necessary to determine the system's effectiveness during sustained operational use.

Since formal client feedback had not yet been obtained at the time of documentation, further evaluation should be conducted after the system has been used by the intended users. Their feedback can provide additional evidence regarding usability, functionality, satisfaction, and areas requiring improvement.

Overall, DigiTrack provides Motosafe Driving School with a developed and deployed platform that addresses the identified need for a more centralized and organized approach to driving school scheduling and performance monitoring. With continued improvement and appropriate implementation, the system can serve as a foundation for the school's ongoing digital management of its operational activities.